V. B. Berestetskii   E. M. Lifshitz   L. P. Pitaevskii

Quantum Electrodynamics

Course of Theoretical Physics
Volume IV

New English translation from the fourth corrected Russian edition

Contents

Introduction

§ 1. Uncertainty Relations in the Relativistic Domain

The quantum theory presented in Vol. III of this course is non-relativistic in its very substance, and it fails for phenomena in which motion occurs at speeds no longer small next to the speed of light. One might at first imagine that a relativistic theory could be reached by generalizing the formalism of non-relativistic quantum mechanics in a more or less straightforward manner. A closer look shows, however, that a logically complete relativistic theory cannot be built without invoking fresh physical principles.

We begin by recalling several physical notions on which non-relativistic quantum mechanics rests (Vol. III, § 1). A fundamental part was played there, as we saw, by the concept of measurement — the process in which a quantum system interacts with a "classical object" (an "apparatus") and thereby comes to possess definite values of particular dynamical variables (coordinates, velocities, and so forth). We saw, too, that quantum mechanics places severe limits on the extent to which different dynamical variables of an electron[2] can exist together. Thus the uncertainties Δq and Δp with which coordinate and momentum can coexist obey the relation $\Delta q \Delta p \sim \hbar$;[3] the more precisely either of these quantities is measured, the less precisely the other can be known at the same moment.

What matters, however, is that any single dynamical variable of the electron, on its own, admitted measurement of unlimited precision, and within a time interval that could be taken as short as desired. On this circumstance the whole of non-relativistic quantum mechanics depends in a fundamental way. It alone makes it possible to introduce the wave function, the central object of the theory's formalism. The physical content of the wave function $\psi(q)$ is, after all, that its squared modulus gives the probability that a measurement performed at a given instant will yield one or another value of the electron's coordinate. Evidently the very notion of such a probability presupposes that a coordinate measurement of unlimited precision and speed can be realized in principle; failing that, the notion would be empty of content and would forfeit its physical meaning.

The existence of a limiting speed (the speed of light c) leads to new limitations, of a fundamental kind, on the possibilities of measuring various physical quantities (*L. D. Landau, R. Peierls*, 1930).

In Vol. III, § 44 the relation

$$(v' - v)\,\Delta p\,\Delta t \sim \hbar \tag{1.1}$$

was obtained; it relates the uncertainty Δp of a momentum measurement on the electron to the length Δt of the measuring process itself, v and v' denoting the electron's velocity before and after that process. It follows that momentum can be measured accurately within a short time (small Δp together with small Δt) only if the measurement itself

[2] As in Vol. III, § 1, we say "electron" for brevity, understanding by it any quantum system.

[3] Ordinary units are used in this section.

alters the velocity strongly enough. In the non-relativistic theory this expresses the fact that a momentum measurement repeated after a short interval will not reproduce its result, but it places no restriction of principle on how accurate a single measurement of the momentum may be, the difference $v' - v$ being capable of unlimited increase.

The presence of a limiting speed, however, changes the state of affairs radically. The difference $v' - v$, like the velocities themselves, can now not exceed c (more precisely, $2c$). Replacing $v' - v$ in (1.1) by c, we obtain the relation

$$\Delta p \, \Delta t \sim \hbar / c, \tag{1.2}$$

which determines the best accuracy of momentum measurement attainable in principle for a given duration Δt of the measurement. Thus in the relativistic theory an arbitrarily accurate and arbitrarily rapid measurement of the momentum turns out to be impossible in principle. An exact measurement of the momentum ($\Delta p \to 0$) is possible only in the limit of an infinitely long duration of the measurement.

One has reason to expect that the measurability of the electron's coordinate as such is likewise affected. Within the mathematical formalism of the theory this appears as a conflict between an exact coordinate measurement and the statement that a free particle's energy is positive. As we shall see, the full set of eigenfunctions of a free particle's relativistic wave equation contains, besides the solutions whose time dependence is the "proper" one, solutions of "negative frequency" as well. These will in general appear also in the expansion of a wave packet describing an electron confined to a small region of space.

As will be shown, the "negative-frequency" wave functions are tied to the existence of antiparticles — the positrons. Their presence in the expansion of the wave packet reflects the creation of electron–positron pairs, unavoidable in the general case, during a measurement of the electron's coordinates. Since the emergence of these new particles cannot be controlled by the measuring process itself, coordinate measurement for the electron is drained of meaning.

In the rest frame of the electron the minimum error in a measurement of its coordinates is

$$\Delta q \sim \hbar / mc. \tag{1.3}$$

This value (the only one that dimensional considerations would permit) answers to a momentum uncertainty $\Delta p \sim mc$, which corresponds in turn to the minimum threshold energy needed to create a pair.

In a reference frame where the electron moves with energy ε, (1.3) is replaced by

$$\Delta q \sim c\hbar / \varepsilon. \tag{1.4}$$

In the extreme ultra-relativistic limit, in particular, energy and momentum are related by $\varepsilon \approx cp$, so that

$$\Delta q \sim \hbar / p, \tag{1.5}$$

the error Δq then coinciding with the particle's de Broglie wavelength.[4]

[4]What is meant here are measurements any outcome of which permits an inference about the state of

For photons the ultra-relativistic case always holds, so that expression (1.5) is valid. This means that it makes sense to speak of the coordinates of a photon only in those cases where the characteristic dimensions are large compared with the wavelength. But this is nothing other than the "classical" limiting case, corresponding to geometrical optics, in which one may speak of the propagation of light along definite trajectories — rays. In the quantum case, however, when the wavelength cannot be regarded as small, the concept of the coordinates of a photon becomes vacuous. We shall see later (see § 4) that in the mathematical formalism of the theory the non-measurability of the photon's coordinates already shows itself in the impossibility of forming from its wave function a quantity that could play the part of a probability density satisfying the requirements which relativistic invariance demands.

On the basis of all that has been said, it is natural to suppose that the future theory will renounce altogether the consideration of the time development of the processes of interaction of particles. It will show that in these processes no exactly definable characteristics exist (even within the limits of the usual quantum-mechanical accuracy), so that the description of a process in time will turn out to be just as illusory as classical trajectories proved to be in non-relativistic quantum mechanics. The only observable quantities will be the characteristics (momenta, polarizations) of free particles — the initial particles that enter into the interaction and the final particles that have arisen as a result of the process (*L. D. Landau, R. Peierls*, 1930).

The characteristic formulation of a problem in relativistic quantum theory consists in the determination of the probability amplitudes of transitions connecting given initial and final states (that is, at $t \to \mp\infty$) of a system of particles. The totality of the transition amplitudes between all possible states constitutes the *scattering matrix*, or *S-matrix*. This matrix will be the carrier of all the information about the processes of interaction of particles that possesses an observable physical meaning (*W. Heisenberg*, 1938).

At the present time a complete, logically closed relativistic quantum theory does not yet exist. We shall see that the existing theory introduces new physical aspects into the manner of describing the states of particles, which acquires certain features of field theory (see § 10). The theory is constructed, however, to a considerable degree on the model of, and with the help of the concepts of, ordinary quantum mechanics. This way of building the theory has led to success in the domain of quantum electrodynamics. The absence of complete logical closure shows itself in this theory in the existence of divergent expressions when its mathematical apparatus is applied directly; but for the removal of these divergences there exist entirely unambiguous procedures. Nevertheless these procedures retain to a considerable degree the character of semi-empirical recipes, and our confidence in the correctness of the results obtained in this way rests, in the end, on their excellent agreement with experiment, and not on the internal consistency and logical coherence of the fundamental principles of the theory.

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the electron; we thus set aside coordinate measurements by collision, in which over the time of observation the result occurs with a probability short of unity. In such a case a deflection of the particle does allow the electron's position to be inferred, but from the absence of a deflection nothing at all can be concluded.

CHAPTER I

Photon

§ 2. Quantization of the Free Electromagnetic Field

In order to treat the electromagnetic field as a quantum object, it is convenient to start from a classical description in which the field is characterized by a discrete, albeit infinite, set of variables; such a description will allow one to apply the standard formalism of quantum mechanics directly. A description by means of potentials specified at every point of space is, in essence, a description in terms of a continuous set of variables.

Let $\mathbf{A}(\mathbf{r}, t)$ be the vector potential of the free electromagnetic field, satisfying the "transversality condition"

$$\nabla \cdot \mathbf{A} = 0. \tag{2.1}$$

The scalar potential then vanishes, $\Phi = 0$, and the fields $\mathbf{E}$ and $\mathbf{H}$ are

$$\mathbf{E} = -\dot{\mathbf{A}}, \qquad \mathbf{H} = \nabla \times \mathbf{A}. \tag{2.2}$$

Maxwell's equations reduce to the wave equation for $\mathbf{A}$:

$$\Delta \mathbf{A} - \frac{\partial^2 \mathbf{A}}{\partial t^2} = 0. \tag{2.3}$$

As is well known (see Vol. II, § 52), in classical electrodynamics the passage to a description in terms of a discrete set of variables is effected by considering the field inside a large but finite volume V.[1] Let us recall, omitting the details of the calculation, how this is done.

A field in a finite volume can be expanded in travelling plane waves, so that its potential takes the form of a series

$$\mathbf{A} = \sum_{\mathbf{k}} (\mathbf{a}_{\mathbf{k}} e^{i\mathbf{kr}} + \mathbf{a}_{\mathbf{k}}^* e^{-i\mathbf{kr}}), \tag{2.4}$$

where the coefficients $\mathbf{a}_{\mathbf{k}}$ depend on time according to

$$\mathbf{a}_{\mathbf{k}} \sim e^{-i\omega t}, \quad \omega = |\mathbf{k}|. \tag{2.5}$$

By virtue of condition (2.1) the complex vectors $\mathbf{a}_{\mathbf{k}}$ are orthogonal to the corresponding wave vectors: $\mathbf{a}_{\mathbf{k}}\mathbf{k} = 0$.

The summation in (2.4) runs over an infinite discrete set of values of the wave vector (its three components k_x, k_y, k_z). The transition to an integration over a continuous distribution is made with the aid of the expression

$$\frac{d^3k}{(2\pi)^3}$$

[1]To avoid burdening formulae with superfluous factors, we set $V = 1$.

for the number of allowed values of $\mathbf{k}$ falling within a volume element $d^3k = dk_x\,dk_y\,dk_z$ of $\mathbf{k}$-space.

Once the vectors $\mathbf{a}_\mathbf{k}$ are given, the field inside the volume in question is completely determined. One may therefore view these quantities as a discrete set of classical "field variables". In order to find the passage to a quantum theory, however, a further transformation of these quantities is needed, one that brings the field equations into a form analogous to the canonical (Hamiltonian) equations of classical mechanics. The canonical variables of the field are introduced by means of

$$\mathbf{Q}_\mathbf{k} = \frac{1}{\sqrt{4\pi}}(\mathbf{a}_\mathbf{k} + \mathbf{a}_\mathbf{k}^*), \qquad \mathbf{P}_\mathbf{k} = -\frac{i\omega}{\sqrt{4\pi}}(\mathbf{a}_\mathbf{k} - \mathbf{a}_\mathbf{k}^*) = \dot{\mathbf{Q}}_\mathbf{k} \tag{2.6}$$

(they are evidently real). The vector potential is expressed through the canonical variables as

$$\mathbf{A} = \sqrt{4\pi}\sum_\mathbf{k}\left(\mathbf{Q}_\mathbf{k}\cos\mathbf{kr} - \frac{1}{\omega}\mathbf{P}_\mathbf{k}\sin\mathbf{kr}\right). \tag{2.7}$$

To find the Hamiltonian H one must evaluate the total field energy

$$\frac{1}{8\pi}\int(\mathbf{E}^2 + \mathbf{H}^2)\,d^3x,$$

expressed in terms of the quantities $\mathbf{Q}_\mathbf{k}$, $\mathbf{P}_\mathbf{k}$. Substituting the expansion (2.7) for $\mathbf{A}$, computing $\mathbf{E}$ and $\mathbf{H}$ by means of (2.2), and integrating, one obtains

$$H = \frac{1}{2}\sum_\mathbf{k}(\mathbf{P}_\mathbf{k}^2 + \omega^2\mathbf{Q}_\mathbf{k}^2).$$

Each of the vectors $\mathbf{P}_\mathbf{k}$ and $\mathbf{Q}_\mathbf{k}$ is perpendicular to the wave vector $\mathbf{k}$ and therefore has two independent components. The direction of these vectors fixes the polarization direction of the corresponding wave. Denoting the two components of $\mathbf{Q}_\mathbf{k}$, $\mathbf{P}_\mathbf{k}$ (in the plane perpendicular to $\mathbf{k}$) by $Q_{\mathbf{k}\alpha}$, $P_{\mathbf{k}\alpha}$ ($\alpha = 1, 2$), we rewrite the Hamiltonian as

$$H = \sum_{\mathbf{k}\alpha}\frac{1}{2}(P_{\mathbf{k}\alpha}^2 + \omega^2 Q_{\mathbf{k}\alpha}^2). \tag{2.8}$$

The Hamiltonian has thus broken up into a sum of independent terms, each involving just one pair $Q_{\mathbf{k}\alpha}$, $P_{\mathbf{k}\alpha}$. Every term answers to a travelling wave of a given wave vector and polarization and reproduces the structure of a single harmonic-oscillator Hamiltonian. One accordingly speaks of this decomposition as the expansion of the field in *oscillators*.

We now turn to the quantization of the free electromagnetic field. The method of classical description just set out makes the route to quantum theory obvious. The canonical variables — the generalized coordinates $Q_{\mathbf{k}\alpha}$ and generalized momenta $P_{\mathbf{k}\alpha}$ — are now to be treated as operators obeying the commutation rule

$$\hat{P}_{\mathbf{k}\alpha}\hat{Q}_{\mathbf{k}\alpha} - \hat{Q}_{\mathbf{k}\alpha}\hat{P}_{\mathbf{k}\alpha} = -i \tag{2.9}$$

(operators with different $\mathbf{k}\alpha$ all commute with one another). Together with them the potential $\mathbf{A}$ and, by (2.2), the field strengths $\mathbf{E}$ and $\mathbf{H}$ also become (Hermitian) operators.

A consistent determination of the field Hamiltonian requires evaluation of the integral

$$\hat{H} = \frac{1}{8\pi} \int (\hat{\mathbf{E}}^2 + \hat{\mathbf{H}}^2)\, d^3x, \tag{2.10}$$

in which $\hat{\mathbf{E}}$ and $\hat{\mathbf{H}}$ are expressed through the operators $\hat{P}_{\mathbf{k}\alpha}$, $\hat{Q}_{\mathbf{k}\alpha}$. In practice, however, the non-commutativity of the latter does not make itself felt here, since the products $Q_{\mathbf{k}\alpha}P_{\mathbf{k}\alpha}$ enter with the factor $\cos \mathbf{kr} \cdot \sin \mathbf{kr}$, which vanishes upon integration over the full volume. Consequently the Hamiltonian takes the form

$$\hat{H} = \sum_{\mathbf{k}\alpha} \frac{1}{2}(\hat{P}^2_{\mathbf{k}\alpha} + \omega^2 \hat{Q}^2_{\mathbf{k}\alpha}), \tag{2.11}$$

exactly reproducing the classical Hamiltonian, as one would naturally expect.

Finding the eigenvalues of this Hamiltonian requires no special computation, since the problem reduces to the familiar one of the energy levels of linear oscillators (see Vol. III, § 23). We can therefore write down at once the energy levels of the field:

$$E = \sum_{\mathbf{k}\alpha} \left(N_{\mathbf{k}\alpha} + \frac{1}{2} \right) \omega, \tag{2.12}$$

where $N_{\mathbf{k}\alpha}$ are non-negative integers.

Discussion of this formula will be taken up in the next section; here we write out the matrix elements of the quantities $Q_{\mathbf{k}\alpha}$, which can be obtained directly from the known formulae for the matrix elements of the oscillator coordinate (see Vol. III, § 23). The non-vanishing matrix elements are

$$\langle N_{\mathbf{k}\alpha} | Q_{\mathbf{k}\alpha} | N_{\mathbf{k}\alpha} - 1 \rangle = \sqrt{\frac{N_{\mathbf{k}\alpha}}{2\omega}}. \tag{2.13}$$

The matrix elements of $P_{\mathbf{k}\alpha} = \dot{Q}_{\mathbf{k}\alpha}$ differ from those of $Q_{\mathbf{k}\alpha}$ only by a factor $\pm i\omega$.

For subsequent calculations it will, however, be more convenient to work not with $Q_{\mathbf{k}\alpha}$, $P_{\mathbf{k}\alpha}$ themselves but with their linear combinations $\omega Q_{\mathbf{k}\alpha} \pm iP_{\mathbf{k}\alpha}$, which have matrix elements only for the transitions $N_{\mathbf{k}\alpha} \to N_{\mathbf{k}\alpha} \pm 1$. Accordingly we introduce the operators

$$\hat{c}_{\mathbf{k}\alpha} = \frac{1}{\sqrt{2\omega}}(\omega \hat{Q}_{\mathbf{k}\alpha} + i\hat{P}_{\mathbf{k}\alpha}), \qquad \hat{c}^+_{\mathbf{k}\alpha} = \frac{1}{\sqrt{2\omega}}(\omega \hat{Q}_{\mathbf{k}\alpha} - i\hat{P}_{\mathbf{k}\alpha}) \tag{2.14}$$

(the classical quantities $c_{\mathbf{k}\alpha}$, $c^*_{\mathbf{k}\alpha}$ coincide, up to a factor $\sqrt{2\pi/\omega}$, with the coefficients $a_{\mathbf{k}\alpha}$, $a^*_{\mathbf{k}\alpha}$ in the expansion (2.4)).

The matrix elements of these operators are

$$\langle N_{\mathbf{k}\alpha} - 1 | \hat{c}_{\mathbf{k}\alpha} | N_{\mathbf{k}\alpha} \rangle = \langle N_{\mathbf{k}\alpha} | \hat{c}^+_{\mathbf{k}\alpha} | N_{\mathbf{k}\alpha} - 1 \rangle = \sqrt{N_{\mathbf{k}\alpha}}. \tag{2.15}$$

The commutation rule between $\hat{c}_{\mathbf{k}\alpha}$ and $\hat{c}^+_{\mathbf{k}\alpha}$ follows from the definition (2.14) and the rule (2.9):

$$\hat{c}_{\mathbf{k}\alpha}\hat{c}^+_{\mathbf{k}\alpha} - \hat{c}^+_{\mathbf{k}\alpha}\hat{c}_{\mathbf{k}\alpha} = 1. \tag{2.16}$$

Returning to the vector potential, we adopt an expansion of the same type as (2.4), but with the coefficients now promoted to operators:

$$\hat{\mathbf{A}} = \sum_{\mathbf{k}\alpha} (\hat{c}_{\mathbf{k}\alpha}\mathbf{A}_{\mathbf{k}\alpha} + \hat{c}^{+}_{\mathbf{k}\alpha}\mathbf{A}^{*}_{\mathbf{k}\alpha}), \tag{2.17}$$

where

$$\mathbf{A}_{\mathbf{k}\alpha} = \sqrt{4\pi}\,\frac{\mathbf{e}^{(\alpha)}}{\sqrt{2\omega}}\,e^{i\mathbf{kr}}. \tag{2.18}$$

Here $\mathbf{e}^{(\alpha)}$ denotes a unit vector specifying the polarization direction of the oscillator; the vectors $\mathbf{e}^{(\alpha)}$ are perpendicular to the wave vector $\mathbf{k}$, and for each $\mathbf{k}$ there are two independent polarizations.

Analogously, for the operators $\hat{\mathbf{E}}$ and $\hat{\mathbf{H}}$ we write

$$\hat{\mathbf{E}} = \sum_{\mathbf{k}\alpha} (\hat{c}_{\mathbf{k}\alpha}\mathbf{E}_{\mathbf{k}\alpha} + \hat{c}^{+}_{\mathbf{k}\alpha}\mathbf{E}^{*}_{\mathbf{k}\alpha}), \quad \hat{\mathbf{H}} = \sum_{\mathbf{k}\alpha} (\hat{c}_{\mathbf{k}\alpha}\mathbf{H}_{\mathbf{k}\alpha} + \hat{c}^{+}_{\mathbf{k}\alpha}\mathbf{H}^{*}_{\mathbf{k}\alpha}), \tag{2.19}$$

with

$$\mathbf{E}_{\mathbf{k}\alpha} = i\omega\mathbf{A}_{\mathbf{k}\alpha}, \qquad \mathbf{H}_{\mathbf{k}\alpha} = \mathbf{n}\times\mathbf{E}_{\mathbf{k}\alpha} \qquad (\mathbf{n} = \mathbf{k}/\omega). \tag{2.20}$$

The vectors $\mathbf{A}_{\mathbf{k}\alpha}$ are mutually orthogonal in the sense that

$$\int \mathbf{A}_{\mathbf{k}\alpha}\mathbf{A}^{*}_{\mathbf{k}'\alpha'}\,d^3x = \frac{2\pi}{\omega}\,\delta_{\alpha\alpha'}\,\delta_{\mathbf{kk}'}. \tag{2.21}$$

To see this, note that when $\mathbf{A}_{\mathbf{k}\alpha}$ and $\mathbf{A}^{*}_{\mathbf{k}'\alpha'}$ carry different wave vectors their product involves $e^{i(\mathbf{k}-\mathbf{k}')\mathbf{r}}$, which integrates to zero over the volume; when only the polarizations differ, the orthogonality $\mathbf{e}^{(\alpha)}\mathbf{e}^{(\alpha')*} = 0$ of the two independent polarization directions provides the required vanishing. Analogous relations hold for the vectors $\mathbf{E}_{\mathbf{k}\alpha}$, $\mathbf{H}_{\mathbf{k}\alpha}$. Their normalization is conveniently written as

$$\frac{1}{4\pi}\int (\mathbf{E}_{\mathbf{k}\alpha}\mathbf{E}^{*}_{\mathbf{k}'\alpha'} + \mathbf{H}_{\mathbf{k}\alpha}\mathbf{H}^{*}_{\mathbf{k}'\alpha'})\,d^3x = \omega\,\delta_{\mathbf{kk}'}\,\delta_{\alpha\alpha'}. \tag{2.22}$$

Substituting the operators (2.19) into (2.10) and integrating with the help of (2.22), we get the field Hamiltonian expressed through the operators $\hat{c}$, $\hat{c}^{+}$:

$$\hat{H} = \sum_{\mathbf{k}\alpha} \frac{1}{2}\omega(\hat{c}_{\mathbf{k}\alpha}\hat{c}^{+}_{\mathbf{k}\alpha} + \hat{c}^{+}_{\mathbf{k}\alpha}\hat{c}_{\mathbf{k}\alpha}). \tag{2.23}$$

In the representation under discussion (with the matrix elements of $\hat{c}$, $\hat{c}^{+}$ given by (2.15)) this operator is diagonal, and its eigenvalues coincide, naturally, with (2.12).

In classical theory the field momentum is defined by the integral

$$\mathbf{P} = \frac{1}{4\pi}\int \mathbf{E}\times\mathbf{H}\,d^3x.$$

On passing to the quantum theory by replacing $\mathbf{E}$ and $\mathbf{H}$ with the operators (2.19), one readily finds

$$\hat{\mathbf{P}} = \sum_{\mathbf{k}\alpha} \frac{1}{2}(\hat{P}^2_{\mathbf{k}\alpha} + \omega^2\hat{Q}^2_{\mathbf{k}\alpha})\mathbf{n} \tag{2.24}$$

— which mirrors the familiar classical link between energy and momentum of plane waves. This operator has eigenvalues

$$\mathbf{P} = \sum_{\mathbf{k}\alpha} \mathbf{k}\left(N_{\mathbf{k}\alpha} + \frac{1}{2}\right). \tag{2.25}$$

The operator representation realized by the matrix elements (2.15) is called the "occupation-number representation": it amounts to characterizing the state of the system (the field) through the quantum numbers $N_{\mathbf{k}\alpha}$ (*occupation numbers*). Within this representation the field operators (2.19) (and hence the Hamiltonian (2.11)) act on the system's wave function, written as a function of the numbers $N_{\mathbf{k}\alpha}$; we denote it $\Phi(N_{\mathbf{k}\alpha}, t)$. The field operators (2.19) do not depend on time explicitly. This corresponds to the Schrödinger representation of operators, familiar from non-relativistic quantum mechanics. What does depend on time is the state of the system, $\Phi(N_{\mathbf{k}\alpha}, t)$, its time dependence being governed by the Schrödinger equation

$$i\frac{\partial \Phi}{\partial t} = \hat{H}\Phi.$$

This description of the field is relativistically invariant in essence, since it rests on the invariant Maxwell equations. Yet the invariance is not exhibited explicitly — above all because the spatial coordinates and the time enter the description in a highly asymmetric way.

Within a relativistic framework it is desirable to give the description a more overtly invariant appearance. For this purpose one uses the so-called Heisenberg representation, wherein the explicit dependence on time is shifted onto the operators (see Vol. III, § 13). Time and spatial coordinates then appear symmetrically in the field-operator expressions, while the system state Φ depends only on the occupation numbers.

For the operator $\hat{\mathbf{A}}$ the passage to the Heisenberg representation amounts to replacing, in each term of the sum in (2.17), (2.18), the factor $e^{i\mathbf{kr}}$ by $e^{i(\mathbf{kr}-\omega t)}$; that is, one is to understand by $\mathbf{A}_{\mathbf{k}\alpha}$ the time-dependent functions

$$\mathbf{A}_{\mathbf{k}\alpha} = \sqrt{4\pi}\,\frac{\mathbf{e}^{(\alpha)}}{\sqrt{2\omega}}\,e^{-i(\omega t-\mathbf{kr})}. \tag{2.26}$$

This is easily verified by noting that the matrix element of a Heisenberg operator for the transition $i \to f$ must contain the factor $\exp\{-i(E_i - E_f)t\}$, where E_i and E_f are the energies of the initial and final states (see Vol. III, § 13). For a transition in which $N_{\mathbf{k}}$ decreases or increases by one, this factor reduces to $e^{-i\omega t}$ or $e^{i\omega t}$, respectively. The indicated replacement satisfies this requirement.

Henceforth (in treating both the electromagnetic field and the fields of particles) we shall always understand the Heisenberg representation of operators.

§ 3. Photons

Let us now turn to a discussion of the quantization formulae derived above.

To begin with, formula (2.12) for the energy of the field exposes the following difficulty. At the lowest energy level, all oscillator quantum numbers $N_{\mathbf{k}\alpha}$ vanish (this state is known as the *electromagnetic field vacuum*). Yet even in this state every oscillator retains a nonzero "zero-point energy" $\omega/2$. Summing over the entire infinite set of oscillators then produces an infinite result. One is thus confronted with one of the "divergences" stemming from the absence of full logical self-consistency in the existing theory.

As long as one is concerned only with the eigenvalues of the field energy, this difficulty can be eliminated by simply subtracting away the zero-point oscillation energy, i.e. by writing for the energy and momentum of the field[2]

$$E = \sum_{\mathbf{k}\alpha} N_{\mathbf{k}\alpha}\omega, \qquad \mathbf{P} = \sum_{\mathbf{k}\alpha} N_{\mathbf{k}\alpha}\mathbf{k}. \tag{3.1}$$

These formulae make it possible to introduce the concept fundamental to all of quantum electrodynamics: that of *light quanta*, or *photons*[3]. Namely, the free electromagnetic field can be regarded as a collection of particles, each carrying energy $\omega\ (=\hbar\omega)$ and momentum $\mathbf{k}\ (=\mathbf{n}\hbar\omega/c)$. The relationship between a photon's energy and momentum is precisely what relativistic mechanics requires for particles of zero rest mass that propagate at the speed of light. The occupation numbers $N_{\mathbf{k}\alpha}$ acquire the meaning of the number of photons with given momenta $\mathbf{k}$ and polarizations $\mathbf{e}^{(\alpha)}$. The polarization property of a photon is analogous to the notion of spin for other particles (the specific features of the photon in this regard will be examined below, in § 6).

It is easy to see that the mathematical formalism developed in the preceding section is fully consistent with the picture of the electromagnetic field as an assembly of photons; it is nothing other than the apparatus of so-called second quantization as applied to a system of photons[4]. In this method (see Vol. III, § 64) the occupation numbers serve as the independent variables, and the operators act on functions of those numbers. The central role is played by the "annihilation" and "creation" operators for particles, which respectively decrease or increase an occupation number by one. Exactly such operators are $\hat{c}_{\mathbf{k}\alpha}$ and $\hat{c}^+_{\mathbf{k}\alpha}$: the operator $\hat{c}_{\mathbf{k}\alpha}$ destroys a photon in the state $\mathbf{k}\alpha$, while $\hat{c}^+_{\mathbf{k}\alpha}$ creates one in that state.

The commutation rule (2.16) corresponds to particles obeying Bose statistics. Photons are therefore bosons, as one might have anticipated: the number of photons occupying any given state can be arbitrary (we shall return to the significance of this fact in § 5).

The plane waves $\mathbf{A}_{\mathbf{k}\alpha}$ (2.26), which appear in the operator $\hat{\mathbf{A}}$ (2.17) as coefficients multiplying the photon annihilation operators, can be interpreted as the wave functions of photons possessing definite momenta $\mathbf{k}$ and polarizations $\mathbf{e}^{(\alpha)}$. This interpretation corresponds to expanding the ψ-operator as a series in the stationary-state wave functions of the particle within the non-relativistic second-quantization framework (unlike the

[2]This subtraction can be carried out in a formally consistent manner by agreeing to interpret the operator products in (2.10) as "normal-ordered", i.e. arranged so that the operators $\hat{c}^+$ always stand to the left of the operators $\hat{c}$. Formula (2.23) then takes the form

$$\hat{H} = \sum_{\mathbf{k}\alpha}\left(\omega\, \hat{c}^+_{\mathbf{k}\alpha}\hat{c}_{\mathbf{k}\alpha}\right).$$

[3]The photon concept was first put forward by *Einstein* (A. Einstein, 1905).

[4]The method of second quantization as applied to radiation theory was developed by *Dirac* (P. A. M. Dirac, 1927).

non-relativistic case, however, the expansion (2.17) involves both annihilation and creation operators; the significance of this distinction will become clear later, see § 12).

The wave function (2.26) is normalized by the condition

$$\int \frac{1}{4\pi}\left(|\mathbf{E}_{\mathbf{k}\alpha}|^2 + |\mathbf{H}_{\mathbf{k}\alpha}|^2\right) d^3x = \omega \tag{3.2}$$

This amounts to normalizing to one photon in a volume $V = 1$. Indeed, the integral on the left-hand side of the equation represents the quantum-mechanical expectation value of the photon's energy in a state described by the given wave function[5]. On the right-hand side of (3.2) stands the energy of a single photon.

The role of the "Schrödinger equation" for the photon is played by Maxwell's equations. In the present case (for the potential $\mathbf{A}(\mathbf{r}, t)$ satisfying condition (2.1)) this is the wave equation

$$\frac{\partial^2 \mathbf{A}}{\partial t^2} - \Delta \mathbf{A} = 0.$$

In the general case of arbitrary stationary states, the photon "wave functions" are complex solutions of this equation whose time dependence enters through the factor $e^{-i\omega t}$.

Speaking of the photon wave function, let us emphasize once more that it cannot by any means be treated as a probability amplitude for the spatial localization of a photon — in contrast to the basic meaning of the wave function in non-relativistic quantum mechanics. This is linked to the fact that (as noted in § 1) the notion of photon coordinates has no physical meaning at all. We shall revisit the mathematical side of this issue at the end of the next section.

The Fourier-transform components of $\mathbf{A}(\mathbf{r}, t)$ with respect to the coordinates constitute the photon wave function in the momentum representation; we denote it $\mathbf{A}(\mathbf{k}, t) = \mathbf{A}(\mathbf{k})e^{-i\omega t}$. Thus, for a state with a definite momentum $\mathbf{k}$ and polarization $\mathbf{e}^{(\alpha)}$ the momentum-representation wave function is given simply by the coefficient of the exponential factor in (2.26):

$$\mathbf{A}_{\mathbf{k}\alpha}(\mathbf{k}', \alpha') = \sqrt{4\pi}\,\frac{\mathbf{e}^{(\alpha)}}{\sqrt{2\omega}}\,\delta_{\mathbf{k}'\mathbf{k}}\delta_{\alpha\alpha'}. \tag{3.3}$$

In keeping with the measurability of the momentum of a free particle, the momentum-representation wave function carries deeper physical content than its coordinate-space counterpart: it provides a way to compute the probabilities $w_{\mathbf{k}\alpha}$ of the various values of momentum and polarization for a photon in a given state. According to the general rules of quantum mechanics, $w_{\mathbf{k}\alpha}$ is given by the squared modulus of the expansion coefficients of $\mathbf{A}(\mathbf{k}')$ in the wave functions of states with definite $\mathbf{k}$ and $\mathbf{e}^{(\alpha)}$:

$$w_{\mathbf{k}\alpha} \propto \left|\sum_{\mathbf{k}'\alpha'} \mathbf{A}^*_{\mathbf{k}\alpha}(\mathbf{k}', \alpha')\mathbf{A}(\mathbf{k}')\right|^2$$

[5] Note that the coefficient $1/(4\pi)$ in the integral (3.2) is twice the usual coefficient $1/(8\pi)$ in (2.10). This difference is ultimately connected with the complex character of the vectors $\mathbf{E}_{\mathbf{k}\alpha}$, $\mathbf{H}_{\mathbf{k}\alpha}$, as opposed to the Hermitian field operators $\hat{\mathbf{E}}$, $\hat{\mathbf{H}}$.

(the proportionality coefficient depends on the choice of normalization for the functions). Substituting (3.3), we obtain

$$w_{\mathbf{k}\alpha} \propto |\mathbf{e}^{(\alpha)}\mathbf{A}(\mathbf{k})|^2. \tag{3.4}$$

Summing over the two polarizations, we find the probability that a photon has momentum $\mathbf{k}$:

$$w_{\mathbf{k}} \propto |\mathbf{A}(\mathbf{k})|^2. \tag{3.5}$$

§ 4. Gauge invariance

As is well known, the choice of field potentials in classical electrodynamics is not unique: the components of the 4-potential A_μ may be subjected to an arbitrary *gauge* (or *gradient*) transformation of the form

$$A_\mu \to A_\mu + \partial_\mu \chi, \tag{4.1}$$

where χ is an arbitrary function of the coordinates and time (see Vol. II, § 18). In the case of a plane wave, restricting oneself to transformations that preserve the functional form of the potential (its proportionality to $\exp(-ik_\mu x^\mu)$), the non-uniqueness amounts to the freedom of adding to the wave amplitude an arbitrary 4-vector proportional to k^μ.

The ambiguity of the potential persists, of course, in the quantum theory as well — as applied to the field operators or to the photon wave functions. Without prejudging the choice of potentials, one should write in place of (2.17) an analogous expansion for the operator 4-potential

$$\hat{A}^\mu = \sum_{\mathbf{k}\alpha} \left(\hat{c}_{\mathbf{k}\alpha} A^\mu_{\mathbf{k}\alpha} + \hat{c}^+_{\mathbf{k}\alpha} A^{\mu *}_{\mathbf{k}\alpha} \right), \tag{4.2}$$

where the wave functions $A^\mu_{\mathbf{k}\alpha}$ are 4-vectors of the form

$$A^\mu_{\mathbf{k}} = \sqrt{4\pi}\,\frac{e^\mu}{\sqrt{2\omega}}\,e^{-ik_\nu x^\nu}, \qquad e_\mu e^{\mu *} = -1,$$

or, in a compact notation that suppresses the four-dimensional vector indices:

$$A_{\mathbf{k}} = \sqrt{4\pi}\,\frac{e}{\sqrt{2\omega}}\,e^{-ikx}, \qquad ee^* = -1. \tag{4.3}$$

Here the 4-momentum is $k_\mu = (\omega, \mathbf{k})$ (so that $kx = \omega t - \mathbf{kr}$), and e is the unit polarization 4-vector[6].

If one restricts oneself to gauge transformations that do not alter how the function (4.3) depends on the coordinates and time, they amount to the replacement

$$e_\mu \to e_\mu + \chi k_\mu, \tag{4.4}$$

[6]Expression (4.3) does not have an entirely relativistically covariant (4-vector) form, which is related to the non-invariant character of our normalization to a finite volume $V = 1$. This is, however, of no fundamental significance and is fully compensated by the convenience of such a normalization scheme. We shall see later that it ensures simple and automatic derivation of real physical quantities in the proper invariant form.

where $\chi = \chi(k^\mu)$ is an arbitrary function. The transversality of the polarization means that one can always choose a gauge in which the 4-vector e takes the form

$$e^\mu = (0, \mathbf{e}), \qquad \mathbf{ek} = 0 \tag{4.5}$$

(we shall call such a gauge *three-dimensionally transverse*). In an invariant four-dimensional notation this requirement is written as the condition of *four-dimensional transversality*

$$ek = 0. \tag{4.6}$$

Note that this condition (as well as the normalization condition $ee^* = -1$) is not violated by the transformation (4.4), by virtue of $k^2 = 0$. On the other hand, the vanishing of the 4-momentum squared of a particle signifies that its mass is zero. A direct link between gauge invariance and the vanishing of the photon mass is thereby revealed (further aspects of this connection will be discussed in § 14).

No measurable physical quantity should change under a gauge transformation of the wave functions of the photons taking part in a given process. This requirement of *gauge invariance* plays an even greater role in quantum electrodynamics than in the classical theory. As we shall see from numerous examples, it serves here, alongside the requirement of relativistic invariance, as a powerful heuristic principle.

In turn, the gauge invariance of the theory is closely bound up with the law of conservation of electric charge; we shall dwell on this aspect in § 43.

We already remarked in the preceding section that the coordinate-space wave function of a photon cannot be interpreted as a probability amplitude for its spatial localization. From a mathematical standpoint this circumstance manifests itself in the impossibility of constructing, out of the wave function, a quantity whose formal properties alone would qualify it to serve as a probability density. Such a quantity would have to be an essentially positive bilinear combination of the wave function A_μ and its complex conjugate. Moreover, it would need to satisfy definite requirements of relativistic covariance — namely, to be the time component of a 4-vector (the point being that the continuity equation expressing conservation of particle number takes the four-dimensional form of a vanishing 4-divergence of the current 4-vector; the time component of this current is, in the present context, the probability density for the localization of the particle; see Vol. II, § 29). On the other hand, gauge invariance demands that A_μ enter the current only through the antisymmetric tensor $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu = -i(k_\mu A_\nu - k_\nu A_\mu)$. The current 4-vector would therefore have to be assembled bilinearly from $F_{\mu\nu}$ and $F^*_{\mu\nu}$ (together with components of the 4-vector k_μ). But no such 4-vector can be constructed at all: every expression satisfying the stated conditions (for instance, $k^\lambda F^*_{\mu\nu} F_\lambda{}^\nu$) vanishes by the transversality condition ($k^\lambda F_{\nu\lambda} = 0$), quite apart from the fact that it would not be essentially positive (since it contains odd powers of the components k_μ).

§ 5. The electromagnetic field in quantum theory

Treating the field as a collection of photons is the sole description that fully captures the physical meaning of the electromagnetic field within quantum theory. It replaces the classical characterization through field strengths. The field strengths themselves enter the mathematical apparatus of the photon picture as operators of second quantization.

It is well known that a quantum system behaves classically when the quantum numbers characterizing its stationary states become large. For the free electromagnetic field (within a given volume) this requires that the oscillator quantum numbers — that is, the photon occupation numbers $N_{\mathbf{k}\alpha}$ — be large. The fact that photons obey Bose statistics is of profound importance in this context. In the mathematical formalism, the link between Bose statistics and classical field behavior is encoded in the commutation rules for $\hat{c}_{\mathbf{k}\alpha}$ and $\hat{c}^{+}_{\mathbf{k}\alpha}$. When $N_{\mathbf{k}\alpha}$ is large, so that the matrix elements of these operators are also large, the unity appearing on the right-hand side of the commutation relation (2.16) may be dropped, giving

$$\hat{c}_{\mathbf{k}\alpha}\hat{c}^{+}_{\mathbf{k}\alpha} \simeq \hat{c}^{+}_{\mathbf{k}\alpha}\hat{c}_{\mathbf{k}\alpha},$$

i.e. the operators go over into mutually commuting classical quantities $c_{\mathbf{k}\alpha}$, $c^{*}_{\mathbf{k}\alpha}$ that determine the classical field strengths.

The quasi-classicality condition for the field requires, however, a further refinement. The point is that if all the numbers $N_{\mathbf{k}\alpha}$ are large, then upon summing over every state $\mathbf{k}\alpha$ the field energy is in any case infinite, rendering the condition meaningless.

A physically sensible formulation of the quasi-classicality criterion rests on examining values of the field averaged over some small time interval Δt. If one writes the classical electric field $\bar{\mathbf{E}}$ (or the magnetic field $\bar{\mathbf{H}}$) as a Fourier integral over time, then upon time-averaging over an interval Δt only those Fourier components whose frequencies satisfy $\omega\Delta t \lesssim 1$ contribute appreciably to the mean value of $\bar{\mathbf{E}}$; in the opposite limit the oscillating factor $e^{-i\omega t}$ nearly averages to zero. Consequently, in establishing the quasi-classicality condition for the time-averaged field, one need consider only those quantum oscillators whose frequencies satisfy $\omega \lesssim 1/\Delta t$. It suffices to require that the quantum numbers of these oscillators be large. The number of oscillators with frequencies between zero and $\omega \sim 1/\Delta t$ (referred to the volume $V = 1$) is of order[7]

$$\left(\frac{\omega}{c}\right)^{3} \sim \frac{1}{(c\Delta t)^{3}}. \tag{5.1}$$

The total energy of the field per unit volume is of order $\bar{\mathbf{E}}^2$. Dividing this quantity by the number of oscillators and by a characteristic energy of one photon ($\sim \hbar\omega$), one obtains an estimate for the photon occupation numbers

$$N_{\mathbf{k}} \sim \frac{\bar{\mathbf{E}}^2 c^3}{\hbar\omega^4}.$$

Demanding that this number be large, we arrive at the inequality

$$|\mathbf{E}| \gg \frac{\sqrt{\hbar c}}{(c\Delta t)^2}. \tag{5.2}$$

This is the sought-after condition permitting a classical treatment of the time-averaged (over intervals Δt) field. We see that the field must be sufficiently strong — all the stronger, the shorter the averaging interval Δt. For time-varying fields this interval must not, of course, exceed the characteristic periods over which the field changes appreciably.

[7] In this section we use ordinary units.

Consequently, sufficiently weak time-varying fields can under no circumstances be quasi-classical. Only for static (time-independent) fields may one set $\Delta t \to \infty$, so that the right-hand side of the inequality (5.2) vanishes. This means that a static field is always classical.

It has already been noted that the classical expressions for the electromagnetic field in the form of a superposition of plane waves must be treated as operator-valued in the quantum theory. The physical meaning of these operators is, however, rather limited. Indeed, a physically meaningful field operator ought to yield vanishing field values in the photon-vacuum state. Yet the expectation value of the squared-field operator $\hat{\mathbf{E}}^2$ in the ground state, which coincides up to a factor with the zero-point energy of the field, turns out to be infinite (by "expectation value" we mean the quantum-mechanical average, i.e. the corresponding diagonal matrix element of the operator). This divergence cannot be circumvented even by any formal subtraction procedure (as can be done for the field energy), since in the present case we would need to carry out the subtraction through some sensible modification of the operators $\hat{\mathbf{E}}$, $\hat{\mathbf{H}}$ themselves (rather than their squares), which proves to be impossible.

§ 6. Angular momentum and parity of the photon

As with any particle, the photon can possess a definite angular momentum. In order to elucidate the properties of this quantity for the photon, let us begin by recalling how the angular momentum of a particle is connected with its wave function in the formalism of quantum mechanics.

The total angular momentum $\mathbf{j}$ of a particle consists of the orbital part $\mathbf{l}$ and the intrinsic part — the spin $\mathbf{s}$. A particle of spin s is described by a symmetric spinor of rank $2s$, i.e. a set of $2s + 1$ components that transform into one another under coordinate-system rotations according to a specific law. The orbital angular momentum is instead linked to how the wave functions depend on coordinates: states with orbital angular momentum l have wave-function components that are expressed (linearly) in terms of spherical harmonics of order l.

The ability to distinguish consistently between spin and orbital angular momentum therefore demands that the "spin" and "coordinate" properties of the wave functions be independent: the coordinate dependence of each spinor component (at a given instant of time) should not be constrained by any supplementary conditions. Passing to the momentum representation, coordinate dependence becomes dependence on $\mathbf{k}$. In the three-dimensionally transverse gauge the photon wave function is the vector $\mathbf{A}(\mathbf{k})$. Because a vector is equivalent to a rank-2 spinor, one could formally ascribe spin 1 to the photon. This vector wave function, however, must satisfy the transversality requirement $\mathbf{k}\mathbf{A}(\mathbf{k}) = 0$, an extra constraint on the way $\mathbf{A}(\mathbf{k})$ depends on momentum. It follows that the dependence on $\mathbf{k}$ can no longer be chosen freely for every vector component at once, making it impossible to disentangle orbital angular momentum and spin.

We note also that for the photon one cannot define spin as the angular momentum of the particle at rest, since for a photon traveling at the speed of light no rest frame exists at all.

Thus, for a photon one can speak only of its total angular momentum. It is clear from the outset that the total angular momentum can assume only integer values. This is already evident from the fact that among the quantities characterizing the photon there are no spinors of odd rank.

As for any particle, the state of a photon is further characterized by its parity, associated with the behavior of the wave function under spatial inversion (see Vol. III, § 30). In the momentum representation, a reversal of the sign of the coordinates corresponds to reversing the sign of every component of $\mathbf{k}$. The action of the inversion operator $\hat{P}$ on a scalar function $\varphi(\mathbf{k})$ amounts simply to this sign change: $\hat{P}\varphi(\mathbf{k}) = \varphi(-\mathbf{k})$. When acting on the vector function $\mathbf{A}(\mathbf{k})$, one must additionally take into account that reversing the axes also reverses the sign of every component of the vector; hence[8]

$$\hat{P}\mathbf{A}(\mathbf{k}) = -\mathbf{A}(-\mathbf{k}). \tag{6.1}$$

Although the decomposition of the photon's angular momentum into orbital and spin parts lacks physical meaning, it is nevertheless convenient to introduce "spin" s and "orbital" angular momentum l in a formal sense as auxiliary concepts describing the transformation properties of the wave function with respect to rotations: the value $s = 1$ reflects the vectorial character of the wave function, while l is the order of the spherical harmonics entering it. Here we have in mind the wave functions of states with definite values of the photon's angular momentum, which for a free particle take the form of spherical waves. The number l determines, in particular, the parity of the photon state, equal to

$$P = (-1)^{l+1} \tag{6.2}$$

In the same formal sense one can write the angular-momentum operator $\hat{\mathbf{j}}$ as the sum $\hat{\mathbf{s}} + \hat{\mathbf{l}}$. As is well known, the angular-momentum operator is associated with infinitesimally small rotations of the coordinate system; here, with the action of such a rotation on a vector field. Within the decomposition $\hat{\mathbf{s}} + \hat{\mathbf{l}}$, the operator $\hat{\mathbf{s}}$ acts on the vector index and intermixes the various components of the vector. The operator $\hat{\mathbf{l}}$ acts on these same components viewed as functions of momentum (or of coordinates). Let us count the number of states (at a given energy) that are possible for a given value j of the photon's angular momentum (setting aside the trivial $(2j + 1)$-fold degeneracy with respect to the direction of the angular momentum).

For independent $\mathbf{l}$ and $\mathbf{s}$ this counting is carried out simply by enumerating the ways in which, according to the rules of the vector model, the angular momenta $\hat{\mathbf{l}}$ and $\mathbf{s}$ can be composed so as to yield the required value of j. For a particle with spin $s = 1$ we would find in this way (for a given nonzero value of j) three states with the following values of l and parity:

$$l = j,\ P = (-1)^{l+1} = (-1)^{j+1}; \quad l = j \pm 1,\ P = (-1)^{l+1} = (-1)^{j}.$$

[8] We adopt the convention of defining the parity of a state by the action of the inversion operator on the polar vector $\mathbf{A}$ (or, equivalently, the corresponding electric vector $\mathbf{E} = i\omega\mathbf{A}$). This differs in sign from the action on the axial vector $\mathbf{H} = i[\mathbf{kA}]$, since inversion does not reverse the direction of such a vector:

$$\hat{P}\mathbf{H}(\mathbf{k}) = \mathbf{H}(-\mathbf{k}).$$

If $j = 0$, there is only a single state (with $l = 1$) having parity $P = +1$.

This counting, however, has not accounted for the transversality condition on $\mathbf{A}$; all three components were regarded as independent. From the result obtained one must therefore subtract the number of states associated with a longitudinal vector. A longitudinal vector can be expressed as $\mathbf{k}\varphi(\mathbf{k})$, and it is apparent that, owing to its transformation behavior (with respect to rotations), its three components reduce to a single scalar φ [9]. The extra state that is excluded by transversality would therefore belong to a particle whose wave function is a scalar (a spinor of rank zero), that is, a particle with "spin 0" [10]. For this state the angular momentum j equals the order of the spherical harmonics appearing in φ. Its parity, viewed as a photon state, follows from how the inversion operator acts on the vector function $\mathbf{k}\varphi$:

$$\hat{P}(\mathbf{k}\varphi) = -(-\mathbf{k})\varphi(-\mathbf{k}) = (-1)^j \mathbf{k}\varphi(\mathbf{k}),$$

i.e. it equals $(-1)^j$. Thus, from the above count of states with parity $(-1)^j$ (two for $j \neq 0$ and one for $j = 0$) one must subtract one.

We arrive finally at the result that for nonzero photon angular momentum j there exist one even-parity and one odd-parity state. For $j = 0$ we obtain no states whatsoever. This means that the photon can never have zero angular momentum, so that j takes only the values $1, 2, 3, \ldots$ The impossibility of $j = 0$ is, moreover, obvious: the wave function of a state with vanishing angular momentum would have to be spherically symmetric, which is manifestly impossible for a transverse wave.

There is a standard nomenclature for the various photon states. When a photon carries angular momentum j with parity $(-1)^j$ it is called an *electric* 2^j*-pole* (or Ej) photon; when its parity is $(-1)^{j+1}$ it is called a *magnetic* 2^j*-pole* (or Mj) photon. In particular, the electric dipole photon has odd parity and $j = 1$; the electric quadrupole photon has even parity and $j = 2$; the magnetic dipole photon has even parity and $j = 1$ [11].

§ 7. Spherical waves of photons

Having established the admissible values of the photon's angular momentum, our task is now to find the wave functions that correspond to them [12].

Let us first consider a formal problem: to identify the vector functions that serve as eigenfunctions of the operators $\hat{\mathbf{j}}^2$ and $\hat{j}_z$; in doing so we do not stipulate in advance which particular functions enter the photon wave functions of interest, nor do we impose the transversality constraint.

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[9] Indeed, when one speaks of the transformation character of a quantity under rotation, one means a transformation at a given point, i.e. at a given $\mathbf{k}$. Under such a transformation $\mathbf{k}\varphi(\mathbf{k})$ does not change at all, i.e. it behaves as a scalar.

[10] Let us emphasize once more that what is meant here is not the state of any real particle. The counting carried out here is purely formal and amounts, mathematically, to sorting the complete collection of quantities that mix under rotations into irreducible representations of the rotation group.

[11] This nomenclature mirrors the usage in the classical theory of radiation: as will be shown later (§§ 46–47), the production of electric- and magnetic-type photons is controlled by the respective electric and magnetic multipole moments of the system of charges.

[12] This problem was first studied by Heitler (W. Heitler, 1936). The form of the solution presented here is due to V. B. Berestetskii (1947).

The functions will be sought in the momentum representation, where the coordinate operator takes the form $\hat{\mathbf{r}} = i\partial/\partial\mathbf{k}$ (see III, (15.12)). For the orbital angular momentum operator one then obtains

$$\hat{\mathbf{l}} = [\hat{\mathbf{r}}\mathbf{k}] = -i\left[\mathbf{k}\frac{\partial}{\partial\mathbf{k}}\right],$$

i.e. it differs from the angular momentum operator in coordinate space only by the replacement of $\mathbf{r}$ with $\mathbf{k}$. Accordingly, the formal solution of the problem is identical in both representations.

Let us denote the desired eigenfunctions by $\mathbf{Y}_{jm}$ and call them *spherical vectors*. They must satisfy the equations

$$\hat{\mathbf{j}}^2 = j(j+1)\mathbf{Y}_{jm}, \qquad \hat{j}_z\mathbf{Y}_{jm} = m\mathbf{Y}_{jm} \tag{7.1}$$

(the z axis being a fixed direction in space). We shall demonstrate that any function of the form $\mathbf{a}Y_{jm}$ possesses this property, where $\mathbf{a}$ is a vector constructed from the unit vector $\mathbf{n} = \mathbf{k}/\omega$ and Y_{jm} are the ordinary (scalar) spherical harmonics. The latter will everywhere be defined in accordance with III, § 28:

$$Y_{lm}(\mathbf{n}) = (-1)^{\frac{m+|m|}{2}} i^l \sqrt{\frac{(2l+1)(l-|m|)!}{4\pi(l+|m|)!}}\, P_l^{|m|}(\cos\theta)\, e^{im\varphi} \tag{7.2}$$

(θ, φ being the spherical angles specifying the direction of $\mathbf{n}$) [13].

To show this, recall the commutation rule III, (29.4):

$$\{\hat{l}_i, a_k\}_- = ie_{ikl}a_l.$$

The right-hand side can be written as $(-\hat{s}_i a_k)$, where $\hat{\mathbf{s}}$ is the spin-1 operator (its action on a vector function is precisely $\hat{s}_i a_k = -ie_{ikl}a_l$; see III, § 57, problem 2). Consequently,

$$\hat{l}_i a_k - a_k\hat{l}_i = -\hat{s}_i a_k.$$

Making use of this relation, we find

$$\hat{j}_i a_k = (\hat{l}_i + \hat{s}_i)a_k = a_k\hat{l}_i.$$

It follows that

$$\hat{\mathbf{j}}^2(\mathbf{a}Y_{jm}) = \mathbf{a}\hat{l}^2 Y_{jm}, \qquad \hat{j}_z(\mathbf{a}Y_{jm}) = \mathbf{a}\hat{l}_z Y_{jm}.$$

Since the spherical harmonic Y_{jm} is an eigenfunction of $\hat{l}^2$ and $\hat{l}_z$ with eigenvalues $j(j+1)$ and m respectively, we arrive at equations (7.1).

[13] For future reference, note the value of these functions at $\theta = 0$ ($\mathbf{n}$ along the z axis):

$$Y_{lm}(\mathbf{n}_z) = i^l\sqrt{\frac{2l+1}{4\pi}}\,\delta_{m0}. \tag{7.2a}$$

Three essentially distinct types of spherical vectors are obtained by choosing the vector **a** to be one of the following [14]:

$$\frac{\boldsymbol{\nabla}_{\mathbf{n}}}{\sqrt{j(j+1)}}, \qquad \frac{[\mathbf{n}\boldsymbol{\nabla}_{\mathbf{n}}]}{\sqrt{j(j+1)}}, \qquad \mathbf{n}. \tag{7.3}$$

The spherical vectors are thus defined as follows:

$$\begin{aligned}
\mathbf{Y}^{(\mathrm{e})}_{jm} &= \frac{1}{\sqrt{j(j+1)}}\,\boldsymbol{\nabla}_{\mathbf{n}} Y_{jm}, && P = (-1)^j;\\
\mathbf{Y}^{(\mathrm{m})}_{jm} &= [\mathbf{n}\mathbf{Y}^{(\mathrm{e})}_{jm}], && P = (-1)^{j+1};\\
\mathbf{Y}^{(\mathrm{l})}_{jm} &= \mathbf{n} Y_{jm}, && P = (-1)^j.
\end{aligned} \tag{7.4}$$

Next to each vector we have indicated its parity P. The spherical vectors of the three types are mutually orthogonal; $\mathbf{Y}^{(\mathrm{l})}_{jm}$ is longitudinal, while $\mathbf{Y}^{(\mathrm{e})}_{jm}$ and $\mathbf{Y}^{(\mathrm{m})}_{jm}$ are transverse with respect to **n**.

The spherical vectors can be expressed through scalar spherical harmonics. In this expansion, $\mathbf{Y}^{(\mathrm{m})}_{jm}$ involves harmonics of a single order $l = j$ only, whereas $\mathbf{Y}^{(\mathrm{e})}_{jm}$ and $\mathbf{Y}^{(\mathrm{l})}_{jm}$ involve harmonics of orders $l = j \pm 1$. This is self-evident: one need only compare the parities listed in (7.4) with the parity $(-1)^{l+1}$ of a vector field, expressed via the order of the spherical harmonics it contains.

The spherical vectors within each type are mutually orthogonal and normalized according to

$$\int \mathbf{Y}_{jm}\mathbf{Y}^*_{j'm'}\,do = \delta_{jj'}\delta_{mm'}. \tag{7.5}$$

For the vectors $\mathbf{Y}^{(\mathrm{l})}_{jm}$ this follows immediately from the normalization of the spherical harmonics Y_{jm}. For $\mathbf{Y}^{(\mathrm{e})}_{jm}$ the normalization integral becomes

$$\frac{1}{j(j+1)}\int \boldsymbol{\nabla}_{\mathbf{n}} Y_{jm}\,\boldsymbol{\nabla}_{\mathbf{n}} Y^*_{j'm'}\,do = -\frac{1}{j(j+1)}\int Y^*_{j'm'}\,\Delta_{\mathbf{n}} Y_{jm}\,do,$$

and since $\Delta_{\mathbf{n}} Y_{jm} = -j(j+1)Y_{jm}$, we recover (7.5). The normalization of the vectors $\mathbf{Y}^{(\mathrm{m})}_{jm}$ reduces to an identical integral.

We remark that the spherical vectors (7.4) could also have been obtained without the direct verification of equations (7.1) carried out above — simply on the basis of general transformation-property arguments. Such reasoning led us in the preceding section to conclude that a vector function of the form $\mathbf{n}\varphi$ corresponds to a value of j equal to

[14] The operator $\boldsymbol{\nabla}_{\mathbf{n}} \equiv |\mathbf{k}|\boldsymbol{\nabla}_{\mathbf{k}}$ acts on functions that depend only on the direction of **n**. In spherical coordinates it has just two components:

$$\boldsymbol{\nabla}_{\mathbf{n}} = \left(\frac{\partial}{\partial\theta}, \ \frac{1}{\sin\theta}\frac{\partial}{\partial\varphi}\right).$$

The operator denoted below by $\Delta_{\mathbf{n}}$ is the angular part of the Laplacian:

$$\Delta_{\mathbf{n}} = \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\sin\theta\frac{\partial}{\partial\theta} + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}.$$

the order of the spherical harmonics appearing in φ; setting $\varphi = Y_{jm}$ further fixes the projection m. In this way one immediately arrives at the spherical vectors $\mathbf{Y}^{(1)}_{jm}$. The transformation-property arguments of § 6, however, remain valid if the factor $\mathbf{n}$ in the product $\mathbf{n}\varphi$ is replaced by $\boldsymbol{\nabla}_{\mathbf{n}}$ or by $[\mathbf{n}\boldsymbol{\nabla}_{\mathbf{n}}]$, yielding in this manner the spherical vectors of the other two types.

Let us return to the photon wave functions. For an electric-type photon (Ej) the vector $\mathbf{A}(\mathbf{k})$ must have parity $(-1)^j$. Among the spherical vectors, $\mathbf{Y}^{(e)}_{jm}$ and $\mathbf{Y}^{(1)}_{jm}$ possess this parity; of the two, however, only the former satisfies the transversality condition. For a magnetic-type photon (Mj) the vector $\mathbf{A}(\mathbf{k})$ must have parity $(-1)^{j+1}$; only the spherical vector $\mathbf{Y}^{(m)}_{jm}$ has this parity. The wave functions of a photon with definite angular momentum j, projection m, and energy ω are therefore

$$\mathbf{A}_{\omega jm}(\mathbf{k}) = \frac{4\pi^2}{\omega^{3/2}}\,\delta(|\mathbf{k}| - \omega)\mathbf{Y}_{jm}(\mathbf{n}), \tag{7.6}$$

where one should write $\mathbf{Y}^{(e)}_{jm}$ or $\mathbf{Y}^{(m)}_{jm}$ for $\mathbf{Y}_{jm}$ according as the photon is of electric or magnetic type; the specified energy is incorporated through the factor $\delta(|\mathbf{k}| - \omega)$. The functions (7.6) are normalized by the condition

$$\frac{1}{(2\pi)^4}\int \omega\omega'\mathbf{A}^*_{\omega' j' m'}(\mathbf{k})\mathbf{A}_{\omega jm}(\mathbf{k})\,d^3k = \omega\delta(\omega' - \omega)\delta_{jj'}\delta_{mm'}. \tag{7.7}$$

For the coordinate-representation wave functions, condition (7.7) is equivalent to [15]

$$\frac{1}{4\pi}\int \left\{\mathbf{E}^*_{\omega' j' m'}(\mathbf{r})\mathbf{E}_{\omega jm}(\mathbf{r}) + \mathbf{H}^*_{\omega' j' m'}(\mathbf{r})\mathbf{H}_{\omega jm}(\mathbf{r})\right\} d^3x = \omega\delta(\omega' - \omega)\delta_{jj'}\delta_{mm'}. \tag{7.8}$$

Indeed, when written in terms of potentials, the integral on the left-hand side takes the form

$$\frac{1}{2\pi}\int \mathbf{A}^*_{\omega' j' m'}(\mathbf{r})\mathbf{A}_{\omega jm}(\mathbf{r})\,\omega'\omega\,d^3x.$$

One must substitute here the Fourier expansions

$$\begin{aligned} \mathbf{A}_{\omega jm}(\mathbf{r}) &= \int \mathbf{A}_{\omega jm}(\mathbf{k})\,e^{i\mathbf{kr}}\,\frac{d^3k}{(2\pi)^3}, \\ \mathbf{A}^*_{\omega' j' m'}(\mathbf{r}) &= \int \mathbf{A}^*_{\omega' j' m'}(\mathbf{k}')\,e^{-i\mathbf{k}'\mathbf{r}}\,\frac{d^3k'}{(2\pi)^3}. \end{aligned} \tag{7.9}$$

After that, integration over d^3x produces the delta function $(2\pi)^3\delta(\mathbf{k}' - \mathbf{k})$, which is removed by the integration over d^3k', bringing the integral to the form (7.7).

Up to this point we have assumed the transverse gauge for the potentials, in which the scalar potential $\Phi = 0$. In various applications, however, other gauge choices for a spherical wave may prove more convenient. An admissible gauge transformation of the potentials in the momentum representation consists of the substitution

$$\mathbf{A} \to \mathbf{A} + \mathbf{n}f(\mathbf{k}), \qquad \Phi \to \Phi + f(\mathbf{k}),$$

[15]This condition is of the same type as (2.22). The appearance of the factor $\delta(\omega' - \omega)$ on the right-hand side is due to the fact that the field (a spherical wave) is being considered in all of infinite space rather than in a finite volume $V = 1$.

with $f(\mathbf{k})$ an arbitrary function. We choose it here so that the new potentials are still expressed through the same spherical harmonics and continue to have definite parity. For an electric-type photon these requirements restrict the potentials to the following form:

$$\begin{aligned} \mathbf{A}^{(e)}_{\omega jm}(\mathbf{k}) &= \frac{4\pi^2}{\omega^{3/2}}\,\delta(|\mathbf{k}|-\omega)\big(\mathbf{Y}^{(e)}_{jm}+C\mathbf{n}Y_{jm}\big),\\ \Phi^{(e)}_{\omega jm}(\mathbf{k}) &= \frac{4\pi^2}{\omega^{3/2}}\,\delta(|\mathbf{k}|-\omega)\,CY_{jm}, \end{aligned} \tag{7.10}$$

where C is an arbitrary constant. For a magnetic-type photon, on the other hand, such an addition to $\mathbf{A}^{(m)}(\mathbf{k})$ would destroy its definite parity, so under the same conditions the choice (7.6) is uniquely fixed.

The probability that a photon of given angular momentum and parity is detected traveling in the direction $\mathbf{n}$ within the solid-angle element do equals, by (3.5) and (7.6),

$$w(\mathbf{n})do = \left|\mathbf{Y}^{(e)}_{jm}\right|^2 do. \tag{7.11}$$

We have written the expression for an E-type photon. Since, however, $\left|\mathbf{Y}^{(m)}_{jm}\right|^2 = \left|\mathbf{Y}^{(e)}_{jm}\right|^2$, the probability distributions $w(\mathbf{n})$ for photons of both types are identical.

The squared modulus $\left|\mathbf{Y}^{(e)}_{jm}\right|^2$ is independent of the azimuthal angle φ (the factors $e^{\pm im\varphi}$ in the spherical harmonics cancel). The probability distribution $w(\mathbf{n})$ is therefore axially symmetric about the z axis. Furthermore, since each spherical vector has definite parity, the squared moduli are even under inversion, i.e. under the replacement $\theta \to \pi-\theta$; this means that the function $w(\theta)$, when expanded in Legendre polynomials, contains only polynomials of even order. The determination of the expansion coefficients reduces to computing integrals of products of three spherical harmonics followed by summation over components. Both steps are carried out with the aid of formulas derived in III, §§ 107–108, and lead to the following result:

$$\begin{aligned} w(\theta) = (-1)^{m+1}\frac{(2j+1)^2}{4\pi}\sum_{n=0}^{\infty}(4n+1)\begin{pmatrix} j & j & 2n\\ 0 & 0 & 0\end{pmatrix}\begin{pmatrix} j & j & 2n\\ m & -m & 0\end{pmatrix}\times\\ \times\begin{Bmatrix} j & j & 1\\ j & j & 2n\end{Bmatrix}P_{2n}(\cos\theta). \end{aligned} \tag{7.12}$$

Finally, we present expressions for the components of the spherical vectors as expansions in spherical harmonics. We employ the "spherical components" of a vector $\mathbf{f}$, defined in accordance with III, § 107; the components f_λ of the vector $\mathbf{f}$ are

$$f_0 = if_z,\quad f_{+1} = -\frac{i}{\sqrt{2}}(f_x+if_y),\quad f_{-1} = \frac{i}{\sqrt{2}}(f_x-if_y). \tag{7.13}$$

Introducing "circular basis vectors":

$$\mathbf{e}^{(0)} = i\mathbf{e}^{(z)},\quad \mathbf{e}^{(+1)} = -\frac{i}{\sqrt{2}}\big(\mathbf{e}^{(x)}+i\mathbf{e}^{(y)}\big),\quad \mathbf{e}^{(-1)} = \frac{i}{\sqrt{2}}\big(\mathbf{e}^{(x)}-i\mathbf{e}^{(y)}\big) \tag{7.14}$$

($\mathbf{e}^{(x,y,z)}$ being the unit vectors along the x, y, z axes), we have

$$\mathbf{f} = \sum_{\lambda} (-1)^{1-\lambda} f_{-\lambda} \mathbf{e}^{(\lambda)}, \qquad f_\lambda = (-1)^{1-\lambda} \mathbf{f}\mathbf{e}^{(-\lambda)*} = \mathbf{f}\mathbf{e}^{(\lambda)}. \tag{7.15}$$

The spherical components of the spherical vectors are expressed through $3j$-symbols and spherical harmonics by the following formulas:

$$\begin{aligned}
(-1)^{j+m+\lambda+1} \big(\mathbf{Y}^{(\mathrm{e})}_{jm}\big)_\lambda &= -\sqrt{j} \begin{pmatrix} j+1 & 1 & j \\ m+\lambda & -\lambda & -m \end{pmatrix} Y_{j+1,m+\lambda} \\
&\quad + \sqrt{j+1} \begin{pmatrix} j-1 & 1 & j \\ m+\lambda & -\lambda & -m \end{pmatrix} Y_{j-1,m+\lambda}, \\
(-1)^{j+m+\lambda+1} \big(\mathbf{Y}^{(\mathrm{m})}_{jm}\big)_\lambda &= -\sqrt{2j+1} \begin{pmatrix} j & 1 & j \\ m+\lambda & -\lambda & -m \end{pmatrix} Y_{j,m+\lambda}, \\
(-1)^{j+m+\lambda+1} \big(\mathbf{Y}^{(\mathrm{l})}_{jm}\big)_\lambda &= \sqrt{j+1} \begin{pmatrix} j+1 & 1 & j \\ m+\lambda & -\lambda & -m \end{pmatrix} Y_{j+1,m+\lambda} \\
&\quad + \sqrt{j} \begin{pmatrix} j-1 & 1 & j \\ m+\lambda & -\lambda & -m \end{pmatrix} Y_{j-1,m+\lambda}.
\end{aligned} \tag{7.16}$$

These formulas are derived as follows. Each of the three spherical vectors has the form $\mathbf{Y}_{jm} = \mathbf{a} Y_{jm}$, where $\mathbf{a}$ is one of the three vectors (7.3). Therefore

$$\mathbf{Y}_{jm} = \sum_{lm'} \langle lm' | \mathbf{a} | jm \rangle Y_{lm'},$$

so that one must determine the matrix elements of the vectors $\mathbf{a}$ between eigenstates of the orbital angular momentum. From formula (107.6) of III it follows that

$$\langle lm' | a_\lambda | jm \rangle = i(-1)^{j_{\max} - m'} \begin{pmatrix} l & 1 & j \\ -m' & \lambda & m \end{pmatrix} \langle l \| a \| j \rangle,$$

where $j_{\max}$ is the larger of the numbers l and j. It therefore suffices to know the nonvanishing reduced matrix elements $\langle l \| a \| j \rangle$. The relevant formulas are:

$$\begin{aligned}
\langle l-1 \| n \| l \rangle &= \langle l \| n \| l-1 \rangle^* = i\sqrt{l}, \\
\langle l \| \boldsymbol{\nabla}_{\mathbf{n}} \| l-1 \rangle &= i(l-1)\sqrt{l}, \\
\langle l-1 \| \boldsymbol{\nabla}_{\mathbf{n}} \| l \rangle &= i(l+1)\sqrt{l}, \\
\langle l \| [\mathbf{n} \boldsymbol{\nabla}_{\mathbf{n}}] \| l \rangle &= i\sqrt{l(l+1)(2l+1)}.
\end{aligned} \tag{7.17}$$

§ 8. Polarization of the photon

The polarization vector $\mathbf{e}$ plays for the photon the role of the "spin part" of the wave function (with the reservations expressed in § 6 regarding the notion of photon spin).

The various cases arising for the polarization of a photon are in no way distinct from the possible types of polarization of a classical electromagnetic wave (see II, § 48).

An arbitrary polarization $\mathbf{e}$ can be represented as a superposition of two mutually orthogonal polarizations $\mathbf{e}^{(1)}$ and $\mathbf{e}^{(2)}$ ($\mathbf{e}^{(1)}\mathbf{e}^{(2)*} = 0$), chosen in some definite manner. In the decomposition

$$\mathbf{e} = e_1\mathbf{e}^{(1)} + e_2\mathbf{e}^{(2)} \tag{8.1}$$

the squared moduli of the coefficients e_1 and e_2 give the probability that the photon has polarization $\mathbf{e}^{(1)}$ or $\mathbf{e}^{(2)}$.

As these two basis polarizations one may choose two mutually perpendicular linear polarizations. Alternatively, any polarization can be decomposed into two circular ones with opposite senses of rotation. We shall denote the right and left circular polarization vectors by $\mathbf{e}^{(+1)}$ and $\mathbf{e}^{(-1)}$ respectively; in a coordinate system $\xi\eta\zeta$ with the ζ axis along the photon's direction $\mathbf{n} = \mathbf{k}/\omega$,

$$\mathbf{e}^{(+1)} = -\frac{i}{\sqrt{2}}\left(\mathbf{e}^{(\xi)} + i\mathbf{e}^{(\eta)}\right), \qquad \mathbf{e}^{(-1)} = \frac{i}{\sqrt{2}}\left(\mathbf{e}^{(\xi)} - i\mathbf{e}^{(\eta)}\right). \tag{8.2}$$

The existence of two distinct polarizations of the photon (at a given momentum) means, in other words, that every eigenvalue of the momentum is doubly degenerate. This circumstance is intimately connected with the fact that the photon mass equals zero.

For a freely moving particle of nonvanishing mass a rest frame always exists. It is clear that the intrinsic symmetry properties of the particle as such manifest themselves precisely in this frame. The symmetry to be considered is with respect to all possible rotations about the center, that is, with respect to the full spherical symmetry group. The quantity characterizing the symmetry properties of the particle under this group is its spin s, which determines the multiplicity of the degeneracy (the number $2s + 1$ of distinct wave functions that transform into one another). In particular, a particle whose wave function is a vector (three components) has spin 1.

For a particle of zero mass, however, no rest frame exists — in every reference frame it travels at the speed of light. For such a particle there is always a preferred spatial direction — the direction of the momentum vector $\mathbf{k}$ (the ζ axis). In this situation the full three-dimensional rotation symmetry is evidently absent, and one may speak only of axial symmetry about the distinguished axis.

Under axial symmetry only the *helicity* of the particle is conserved — the projection of angular momentum onto the ζ axis; we denote it by λ [16]. If one further demands symmetry under reflections in planes passing through the ζ axis, then states differing in the sign of λ are mutually degenerate; for $\lambda \neq 0$ there is therefore a twofold degeneracy [17]. The state of a photon with a definite momentum belongs to precisely this type of twofold-degenerate state. It is described by a "spin" wave function that is a vector $\mathbf{e}$ in the $\xi\eta$ plane; the two components of this vector transform into each other under all rotations about the ζ axis and under reflections in planes containing that axis.

The various polarization cases of a photon are in a definite correspondence with the possible values of its helicity. This correspondence can be established using formulas III,

[16] In contrast to the projection m of the angular momentum onto a fixed direction (the z axis) in space, discussed in the preceding section.

[17] We note that the electronic terms of a diatomic molecule are classified in exactly the same way (see III, § 78).

(57.9), which relate the components of a vector wave function to the components of the equivalent rank-2 spinor [18]. The projections $\lambda = +1$ and -1 correspond to vectors **e** that have only the component $e_\xi - ie_\eta$ or $e_\xi + ie_\eta$ nonzero, i.e. $\mathbf{e} = \mathbf{e}^{(+1)}$ or $\mathbf{e} = \mathbf{e}^{(-1)}$ respectively. In other words, $\lambda = +1$ and -1 correspond to the right and left circular polarizations of the photon (in § 16 the same result will be derived by directly computing the eigenfunctions of the spin-projection operator). Thus the projection of the photon's angular momentum onto the direction of its motion can take only the two values (± 1); the value 0 is not possible.

A photon state with definite momentum and polarization is a pure state (in the sense explained in III, § 14); it is described by a wave function and constitutes a complete quantum-mechanical specification of the particle (photon) state. There also exist "mixed" photon states corresponding to a less complete description, carried out not by a wave function but only by a density matrix.

Let us examine a photon state that is mixed in its polarization but possesses a definite momentum **k**. Such a state (called a state of *partial polarization*) admits a "coordinate" wave function.

The polarization density matrix of a photon is a second-rank tensor $\rho_{\alpha\beta}$ defined in the plane orthogonal to **n** (the $\xi\eta$ plane; the indices α, β assume only two values). This tensor is Hermitian:

$$\rho_{\alpha\beta} = \rho^*_{\beta\alpha}, \tag{8.3}$$

and normalized by the condition

$$\rho_{\alpha\alpha} = \rho_{11} + \rho_{22} = 1. \tag{8.4}$$

By virtue of (8.3) the diagonal components ρ_{11} and ρ_{22} are real, and one determines the other through condition (8.4). The component ρ_{12} is complex, while $\rho_{21} = \rho^*_{12}$. Altogether, then, the density matrix is characterized by three real parameters.

Once the polarization density matrix is known, one can find the probability that the photon has any given definite polarization **e**. This probability is obtained by "projecting" the tensor $\rho_{\alpha\beta}$ onto the direction of the vector **e**, i.e. it equals the quantity

$$\rho_{\alpha\beta} e^*_\alpha e_\beta. \tag{8.5}$$

Thus the components ρ_{11} and ρ_{22} are the probabilities of linear polarizations along the ξ and η axes. Projection onto the vectors (8.2) yields the probabilities of the two circular polarizations:

$$\tfrac{1}{2}\left[1 \pm i(\rho_{12} - \rho_{21})\right]. \tag{8.6}$$

In both form and substance the properties of the tensor $\rho_{\alpha\beta}$ coincide with those of the tensor $J_{\alpha\beta}$ that describes partially polarized light in the classical theory (see II, § 50). Let us recall some of these properties here.

For a pure state of definite polarization **e**, the tensor $\rho_{\alpha\beta}$ reduces to products of the components of the vector **e**:

$$\rho_{\alpha\beta} = e_\alpha e^*_\beta. \tag{8.7}$$

[18]Recall that the wave-function components, viewed as probability amplitudes for the various values of the angular-momentum projection of the particle (which is what is meant here), correspond to the contravariant components of the spinor.

In this case the determinant $|\rho_{\alpha\beta}| = 0$. In the opposite case of an unpolarized photon, all polarization directions are equally probable, i.e.

$$\rho_{\alpha\beta} = \delta_{\alpha\beta}/2, \tag{8.8}$$

and then $|\rho_{\alpha\beta}| = 1/4$.

For the general case of partial polarization it is convenient to introduce three real Stokes parameters ξ_1, ξ_2, ξ_3 [19]; the density matrix is then expressible as

$$\rho_{\alpha\beta} = \frac{1}{2}\begin{pmatrix} 1+\xi_3 & \xi_1 - i\xi_2 \\ \xi_1 + i\xi_2 & 1-\xi_3 \end{pmatrix}. \tag{8.9}$$

All three parameters range between -1 and $+1$. In the unpolarized state $\xi_1 = \xi_2 = \xi_3 = 0$; for a completely polarized photon $\xi_1^2 + \xi_2^2 + \xi_3^2 = 1$.

The parameter ξ_3 characterizes linear polarization along the ξ or η axis; the probability of linear polarization along these axes is $(1+\xi_3)/2$ and $(1-\xi_3)/2$ respectively. The values $\xi_3 = +1$ or -1 therefore correspond to complete polarization in these directions.

The parameter ξ_1 characterizes linear polarization along directions making an angle $\varphi = \pi/4$ or $\varphi = -\pi/4$ with the ξ axis. The probability of linear polarization in these directions is $(1+\xi_1)/2$ or $(1-\xi_1)/2$; this is easily verified by projecting the tensor $\rho_{\alpha\beta}$ onto the directions $\mathbf{e} = (1, \pm 1)/\sqrt{2}$.

Lastly, the parameter ξ_2 represents the degree of circular polarization; by (8.6), the probability of right or left circular polarization is $(1+\xi_2)/2$ or $(1-\xi_2)/2$ respectively. Because these two polarizations answer to helicities $\lambda = \pm 1$, it is evident that in general ξ_2 coincides with the mean helicity of the photon. Note further that for a pure state with polarization $\mathbf{e}$,

$$\xi_2 = i[\mathbf{e}\mathbf{e}^*]\mathbf{n}. \tag{8.10}$$

Recall (see II, § 50) that the quantities ξ_2 and $\sqrt{\xi_1^2 + \xi_3^2}$ are invariant under Lorentz transformations.

In what follows we shall also need to know how the Stokes parameters transform under time reversal. One readily verifies that they are invariant under this operation. Since the result clearly does not depend on whether the polarization state is pure or mixed, it is enough to check invariance for a pure state. In quantum mechanics, time reversal replaces the wave function with its complex conjugate (see III, § 18). For a plane-polarized wave the corresponding substitution is [20]

$$\mathbf{k} \to -\mathbf{k}, \qquad \mathbf{e} \to -\mathbf{e}^*. \tag{8.11}$$

Under this transformation the symmetric part of the density matrix, $(1/2)(e_\alpha e_\beta^* + e_\beta e_\alpha^*)$, and hence the parameters ξ_1 and ξ_3 remain unchanged. The invariance of ξ_2 under the

[19] The notation for the parameters should not be confused with that for the ξ axis!

[20] The extra sign reversal of $\mathbf{e}$ is due to the fact that time reversal changes the sign of the electromagnetic vector potential. The scalar potential is unaffected; consequently, for the 4-vector e time reversal takes the form

$$(e_0, \mathbf{e}) \to (e_0^*, -\mathbf{e}^*). \tag{8.11a}$$

same transformation is apparent from (8.10); it is also obvious already from the physical meaning of ξ_2 as the mean helicity. Indeed, helicity is the projection of the angular momentum **j** onto the direction **n**, that is, the product **jn**; time reversal reverses the sign of both these vectors.

For the calculations that follow, the photon density matrix must be cast in four-dimensional form, that is, expressed as a certain 4-tensor $\rho_{\mu\nu}$. When the photon is polarized and described by a 4-vector e_μ, a natural definition of this tensor is

$$\rho_{\mu\nu} = e_\mu e_\nu^*. \tag{8.12}$$

In the three-dimensionally transverse gauge, $e = (0, \mathbf{e})$, if one of the spatial coordinate axes is chosen along **n**, the nonvanishing components of this 4-tensor coincide with (8.7).

For an unpolarized photon the three-dimensionally transverse gauge yields a tensor $\rho_{\mu\nu}$ with components

$$\rho_{ik} = \tfrac{1}{2}(\delta_{ik} - n_i n_k), \qquad \rho_{0i} = \rho_{i0} = \rho_{00} = 0 \tag{8.13}$$

(if one of the axes coincides with the direction of **n**, we return to (8.8)). Using the tensor $\rho_{\mu\nu}$ directly in this three-dimensional form is, however, inconvenient. We may instead perform a gauge transformation; for the density matrix this is a transformation of the type

$$\rho_{\mu\nu} \to \rho_{\mu\nu} + \chi_\mu k_\nu + \chi_\nu k_\mu, \tag{8.14}$$

where χ_μ are arbitrary functions. Setting

$$\chi_0 = -1/(4\omega), \qquad \chi_i = k_i/(4\omega^2)$$

we obtain in place of (8.13) the simple four-dimensional expression

$$\rho_{\mu\nu} = -g_{\mu\nu}/2. \tag{8.15}$$

The four-dimensional representation of the density matrix of a partially polarized photon is readily obtained by first rewriting the two-dimensional tensor (8.9) in three-dimensional form. In the three-dimensionally transverse gauge: $e^{(1)} = (0, \mathbf{e}^{(1)})$, $e^{(2)} = (0, \mathbf{e}^{(2)})$. The four-dimensional photon density matrix is therefore

$$\begin{aligned} \rho_{\mu\nu} = (1/2)(e_\mu^{(1)} e_\nu^{(1)} + e_\mu^{(2)} e_\nu^{(2)}) + (\xi_1/2)(e_\mu^{(1)} e_\nu^{(2)} + e_\mu^{(2)} e_\nu^{(1)}) - \\ - (i\xi_2/2)(e_\mu^{(1)} e_\nu^{(2)} - e_\mu^{(2)} e_\nu^{(1)}) + (\xi_3/2)(e_\mu^{(1)} e_\nu^{(1)} - e_\mu^{(2)} e_\nu^{(2)}). \end{aligned} \tag{8.17}$$

The convenience of any particular choice of the 4-vectors $e^{(1)}$, $e^{(2)}$ depends on the specific conditions of the problem at hand.

It should be borne in mind that the conditions (8.16) do not fix the choice of $e^{(1)}$ and $e^{(2)}$ uniquely. If any 4-vector e_μ satisfies these conditions, so does any 4-vector of the form $e_\mu + \chi k_\mu$ (since $k^2 = 0$). This ambiguity is tied to the gauge freedom of the density matrix.

The first term in (8.17) corresponds to the unpolarized state. It could therefore be replaced, by (8.15), with $-g_{\mu\nu}/2$. Such a replacement is again equivalent to a gauge transformation. When working with 4-tensors of the form (8.17), decomposed in two independent 4-vectors, it is useful to adopt the following formal device. Writing (8.17) as

$$\rho_{\mu\nu} = \sum_{a,b=1}^{2} \rho^{(ab)} e_\mu^{(a)} e_\nu^{(b)},$$

we represent the coefficients $\rho^{(ab)}$ as a 2×2 matrix

$$\rho = \begin{pmatrix} \rho^{(11)} & \rho^{(12)} \\ \rho^{(21)} & \rho^{(22)} \end{pmatrix}.$$

Any Hermitian 2×2 matrix can be expanded over four linearly independent 2×2 matrices — the Pauli matrices σ_x, σ_y, σ_z and the identity matrix 1:

$$\rho = (1/2)(1 + \boldsymbol{\xi}\boldsymbol{\sigma}), \qquad \boldsymbol{\xi} = (\xi_1, \xi_2, \xi_3), \tag{8.18}$$

as is easily verified by direct comparison with (8.17) using the well-known explicit forms of the Pauli matrices (18.5) (grouping the three quantities ξ_1, ξ_2, ξ_3 into a "vector" $\boldsymbol{\xi}$ is of course purely formal and serves only to shorten the notation).

The requisite generalization is achieved by replacing these 3-vectors by spacelike real 4-vectors $e^{(1)}$, $e^{(2)}$, orthogonal to each other and to the photon 4-momentum k:

$$e^{(1)2} = e^{(2)2} = -1, \qquad e^{(1)}e^{(2)} = 0, \qquad e^{(1)}k = e^{(2)}k = 0. \tag{8.16}$$

Problem

Write the photon density matrix in the representation whose "coordinate axes" are the circular basis vectors (8.2).

Solution. The components of the tensor $\rho'_{\alpha\beta}$ in the new axes ($\alpha, \beta = \pm 1$) are obtained by projecting the tensor (8.9) onto the basis vectors (8.2):

$$\rho'_{11} = \rho_{\alpha\beta} e_\alpha^{(+1)*} e_\beta^{(+1)}, \qquad \rho'_{1,-1} = \rho_{\alpha\beta} e_\alpha^{(+1)*} e_\beta^{(-1)}, \qquad \dots,$$

$$\rho' = \frac{1}{2}\begin{pmatrix} 1 + \xi_2 & -\xi_3 + i\xi_1 \\ -\xi_3 - i\xi_1 & 1 - \xi_2 \end{pmatrix}.$$

§ 9. The two-photon system

Arguments analogous to those developed in § 6 make it possible to count the number of allowed states in the more involved case of a system of two photons as well (*L. Landau*, 1948).

Let us work in the center-of-inertia frame of the two photons, where $\mathbf{k}_1 = -\mathbf{k}_2 \equiv \mathbf{k}$[21]. In the momentum representation the wave function of the pair can be cast as a three-dimensional rank-2 tensor $A_{ik}(\mathbf{n})$ built bilinearly from the components of the individual

[21] A center-of-inertia frame can be found in every case except when the two photons propagate in the same direction. For such a pair the total momentum $\mathbf{k}_1 + \mathbf{k}_2$ and total energy $\omega_1 + \omega_2$ are connected by exactly the same relation as for one photon, so no frame with $\mathbf{k}_1 + \mathbf{k}_2 = 0$ exists.

photon vector wave functions; each of its two indices refers to one photon ($\mathbf{n}$ denotes the unit vector in the direction of $\mathbf{k}$). The transversality requirement for each photon is encoded in the orthogonality of A_{ik} to $\mathbf{n}$:

$$A_{il}n_l = 0, \qquad A_{lk}n_l = 0. \tag{9.1}$$

Interchanging the two photons amounts to transposing the indices of the tensor A_{ik} together with a simultaneous reversal of the sign of $\mathbf{n}$. Because photons obey Bose statistics,

$$A_{ik}(-\mathbf{n}) = A_{ki}(\mathbf{n}). \tag{9.2}$$

The tensor A_{ik} is, in general, not symmetric in its indices. We decompose it into its symmetric part (s_{ik}) and antisymmetric part (a_{ik}): $A_{ik} = s_{ik} + a_{ik}$. Each of these parts must individually satisfy the relation (9.2) (as well as the orthogonality conditions (9.1)). This gives

$$s_{ik}(-\mathbf{n}) = s_{ik}(\mathbf{n}), \tag{9.3}$$

$$a_{ik}(-\mathbf{n}) = -a_{ik}(\mathbf{n}). \tag{9.4}$$

Spatial inversion does not by itself reverse the sign of the components of a second-rank tensor, but it does reverse the sign of $\mathbf{n}$. From (9.3) it is therefore apparent that the wave function s_{ik} is symmetric under inversion, i.e. it corresponds to even-parity states of the photon system; the wave function a_{ik}, on the other hand, corresponds to odd-parity states.

An antisymmetric second-rank tensor is equivalent (dual) to an axial vector $\mathbf{a}$ whose components are expressed via the tensor components as $a_i = \frac{1}{2}e_{ikl}a_{kl}$, where e_{ikl} is the antisymmetric unit tensor (see II, § 6). The orthogonality of the tensor a_{kl} to the vector $\mathbf{n}$ implies that the vectors $\mathbf{a}$ and $\mathbf{n}$ are parallel[22]. One may therefore write $\mathbf{a} = \mathbf{n}\varphi(\mathbf{n})$, where φ is a scalar; by (9.4) we must have $\mathbf{a}(-\mathbf{n}) = -\mathbf{a}(\mathbf{n})$, and hence

$$\varphi(-\mathbf{n}) = \varphi(\mathbf{n}).$$

This relation means that the scalar φ can be built linearly only from spherical harmonics of even order L (including zero).

We see that the antisymmetric tensor a_{ik}, by its transformation properties (under rotations), is equivalent to a single scalar (cf. the footnote on p. 34). Assigning to the latter a "spin" of 0, we find that the angular momentum of the state is $J = L$. Thus the tensor a_{ik} corresponds to odd-parity states of the photon system with even total angular momentum J.

Let us turn to the symmetric tensor s_{ik}. Since it is even under reversal of the sign of $\mathbf{n}$, it describes even-parity states of the photon system. It follows likewise that all components of s_{ik} are expressed through spherical harmonics of even order L (including $L = 0$). An arbitrary symmetric second-rank tensor can be decomposed, as is well known, into a scalar (s_{ii}) and a traceless symmetric tensor (s'_{ik}) with $s'_{ii} = 0$.

[22]We have: $a_{ik} = e_{ikl}a_l$, and the orthogonality condition yields

$$a_{ik}n_k = e_{ikl}a_l n_k = [\mathbf{na}]_i = 0.$$

The scalar s_{ii} is assigned "spin" 0, so the angular momentum of the corresponding states is $J = L$, which is even. The tensor s'_{ik} corresponds to "spin" 2 (see III, § 57). Composing this "spin" with the even "orbital angular momentum" L according to the addition rule for angular momenta, we find that for a given even $J \neq 0$ there are three states (with $L = J \pm 2, J$), while for odd $J \neq 1$ there are two states (with $L = J \pm 1$). The exceptions are $J = 0$ with a single state ($L = 2$) and $J = 1$ with a single state ($L = 2$).

In this counting, however, we have not yet accounted for the orthogonality of the tensor s_{ik} to the vector $\mathbf{n}$. One must therefore subtract from the number of states obtained those corresponding to a symmetric second-rank tensor that is "parallel" to the vector $\mathbf{n}$. Such a tensor (denote it s''_{ik}) can be written as

$$s''_{ik} = n_i b_k + n_k b_i,$$

where $\mathbf{b}$ is some vector. By (9.3) this vector must satisfy $\mathbf{b}(-\mathbf{n}) = -\mathbf{b}(\mathbf{n})$. The tensor s''_{ik} responsible for the "extra" states is thus equivalent to an odd-parity vector. This vector must consequently be expressed through spherical harmonics of odd orders L only. Noting further that a vector corresponds to "spin" 1, we find that for each even $J \neq 0$ there are two states (with $L = J \pm 1$), and for each odd J a single state (with $L = J$); the special case $J = 0$ has one state (with $L = 1$).

Collecting all the results obtained, we arrive at the following table giving the number of possible even- and odd-parity states of a system of two photons (with vanishing total momentum) for the various values of the total angular momentum J:

J	Even	Odd
0	1	1
1	–	–
$2k$	2	1
$2k+1$	1	–

(9.5)

(k being a positive integer other than zero). We observe that for odd J no odd-parity states exist, while the value $J = 1$ is altogether excluded[23].

The wave function of the two-photon system A_{ik} determines the correlation between their polarizations. The probability that both photons simultaneously possess definite polarizations $\mathbf{e}_1$ and $\mathbf{e}_2$ is proportional to

$$A_{ik} e^*_{1i} e^*_{2k}.$$

In other words, if the polarization $\mathbf{e}_1$ of one photon is specified, then the polarization of the second photon, $\mathbf{e}_2$, is proportional to

$$e_{2k} \propto A_{ik} e^*_{1i}. \tag{9.6}$$

In odd-parity states of the system, A_{ik} coincides with the antisymmetric tensor a_{ik}. In that case

$$\mathbf{e}_2 \mathbf{e}^*_1 \propto a_{ik} e^*_{1i} e^*_{1k} = 0,$$

[23] For an alternative derivation of these results, see problem 1 to § 69.

i.e. the polarizations of the two photons are mutually orthogonal. For linear polarization this signifies that their directions are perpendicular, while for circular polarization it means opposite senses of rotation.

The even-parity state with $J = 0$ is described by a symmetric tensor reducing to a scalar,

$$s_{ik} = \text{const} \cdot (\delta_{ik} - n_i n_k).$$

From (9.6) we then obtain $\mathbf{e}_1 = \mathbf{e}_2^*$. For linear polarization this means that their directions are parallel, whereas for circular polarization it means opposite senses of rotation. The latter fact is self-evident: when $J = 0$ the sum of the angular-momentum projections of the two photons onto any common direction $\mathbf{k}$ must in all cases vanish (the projections onto the opposite directions $\mathbf{k}_1$ and $\mathbf{k}_2$, i.e. the helicities, are accordingly equal).

CHAPTER II

Bosons

§ 10. Wave equation for particles with spin 0

Chapter I demonstrated the construction of a quantum description for the free electromagnetic field, proceeding from the known classical-limit properties of the field and relying on the apparatus of ordinary quantum mechanics. The scheme of field description as a photon system obtained in this way retains many features that carry over to the relativistic quantum-theoretical treatment of particles as well.

The electromagnetic field is a system with an infinite number of degrees of freedom. For it there is no conservation law for the number of particles (photons), and among its possible states are those with any number of particles [1]. However, the same property must, generally speaking, also hold in relativistic theory for systems of any particles. The conservation of particle number in non-relativistic theory is tied to the conservation of mass: the sum of the masses (rest masses) of the particles does not change during their interaction; conservation of the total mass in a system of electrons, say, implies the constancy of their number as well. In relativistic mechanics, however, no law of mass conservation exists; only the total energy of the system (which includes the rest energies of the particles) must be conserved. For this reason the particle number need no longer be conserved, and consequently every relativistic particle theory must be a theory of systems possessing an infinite number of degrees of freedom. In other words, such a particle theory acquires the character of a field theory.

The adequate mathematical apparatus for treating systems whose particle number varies is that of second quantization (see III, §§ 64, 65). In the quantum treatment of the electromagnetic field the 4-potential $\hat{A}$ serves as the second-quantization operator. It is built from the (coordinate-space) wave functions of the individual particles (photons) together with their creation and annihilation operators. A completely analogous role in the description of a system of particles belongs to the operator of the quantized wave function. Its construction requires, first and foremost, knowledge of the wave-function form for a single free particle and of the equation governing this function.

It should be stressed that the notion of free-particle fields is auxiliary in character. Real particles interact, and the task of the theory lies in studying these interactions. But every interaction reduces to a collision, before and after which the system can be regarded as a collection of free particles. In § 1 it was noted that these are the only directly measurable objects. We therefore use the fields of free particles as a means for describing initial and final states.

We begin the relativistic description of free particles with the case of spin-0 particles. The mathematical simplicity of this case makes it possible to bring out most clearly the basic ideas and characteristic features of such a description.

[1] In practice, of course, the photon number changes only through various interaction processes.

The state of a free particle (without spin) can be completely specified by giving its momentum $\mathbf{p}$ alone. The energy ε of the particle[2] satisfies $\varepsilon^2 = \mathbf{p}^2 + m^2$ (where m is the particle mass), or in four-dimensional form:

$$p^2 = m^2. \tag{10.1}$$

It is well known that the conservation of momentum and energy is tied to the homogeneity of space and time, that is, to symmetry with respect to any parallel translation of the 4-coordinate system. Within the quantum description, this symmetry requirement implies that under such a coordinate transformation the wave function of a particle possessing a definite 4-momentum may only acquire a phase factor (whose modulus equals unity). This requirement is met solely by an exponential with an exponent that depends linearly on the 4-coordinates. Put differently, the wave function of a free particle in a state of definite 4-momentum $p^\mu = (\varepsilon, \mathbf{p})$ must have the form of a plane wave:

$$\text{const} \cdot e^{-ipx}, \qquad px = \varepsilon t - \mathbf{pr} \tag{10.2}$$

(the choice of sign in the exponent is in itself a convention within relativistic theory; it is made so as to agree with the non-relativistic case).

The wave equation must possess the functions (10.2) as particular solutions for every 4-vector p obeying the condition (10.1). It must be linear—this being a manifestation of the superposition principle: every linear combination of functions (10.2) likewise describes a possible particle state and hence must also satisfy the equation. Lastly, the equation should have the lowest attainable order; raising the order would bring in extraneous solutions.

Spin is the angular momentum that a particle possesses in the frame where it is at rest. When the particle's spin equals s, the wave function in the rest frame takes the form of a three-dimensional spinor of rank $2s$. For describing the particle in an arbitrary reference frame one must cast the wave function in terms of four-dimensional objects.

A particle of spin 0 is described in its rest frame by a three-dimensional scalar. This scalar may, however, have a different four-dimensional "ancestry": it can be a four-scalar ψ, but it can equally well be the temporal component of a timelike 4-vector ψ_μ whose only component that does not vanish in the rest frame is ψ_0[3].

For a free particle the only operator that can enter the wave equation is the 4-momentum operator $\hat{p}$. Its components are the differentiation operators with respect to the coordinates and time:

$$\hat{p}^\mu = i\partial^\mu = \left(i\frac{\partial}{\partial t}, -i\boldsymbol{\nabla} \right). \tag{10.3}$$

The wave equation must represent a differential relation between the quantities ψ and ψ_μ, effected by means of the operator $\hat{p}$. This relation must, of course, be expressed through relativistically invariant equations. Such equations are

$$m\psi_\mu = \hat{p}_\mu \psi, \qquad \hat{p}^\mu \psi_\mu = m\psi, \tag{10.4}$$

[2] We denote the energy of an individual particle by ε, as distinct from the energy E of the system of particles.

[3] Or, by the same token, the temporal component of a higher-rank 4-tensor; that possibility, however, would give rise to equations of higher order.

where m is a dimensional constant characterizing the particle[4].

Substituting ψ_μ from the first equation into the second, we obtain

$$(\hat{p}^2 - m^2)\psi = 0 \tag{10.5}$$

(*O. Klein*, *V. A. Fock*, 1926; *W. Gordon*, 1927). Written out explicitly, this equation reads

$$-\partial_\mu \partial^\mu \psi \equiv \left(-\frac{\partial^2}{\partial^2 t} + \Delta\right)\psi = m^2\psi. \tag{10.6}$$

If one inserts a plane wave (10.2) for ψ, the result is $p^2 = m^2$, showing that m is indeed the particle mass. It is worth noting that the structure of equation (10.5) could, of course, have been anticipated, since $\hat{p}^2$ is the sole scalar operator one can form from $\hat{p}$ (and for this reason every component of the wave function of a particle of arbitrary spin obeys the same equation—something we shall encounter many times below).

Thus a spin-0 particle is described essentially by a single (four-dimensional) scalar ψ obeying the second-order equation (10.5). In the first-order equations (10.4) the role of the wave function is played by the aggregate of the quantities ψ and ψ_μ, the 4-vector ψ_μ reducing to the 4-gradient of the scalar ψ. In the rest frame the particle's wave function does not depend on the (spatial) coordinates, so that the spatial components of the 4-vector ψ_μ vanish, as they should.

For carrying out second quantization it proves useful to write the particle's energy and momentum as spatial integrals of certain bilinear (in ψ and ψ^*) combinations playing the role of spatial densities of these quantities. This amounts to finding the energy–momentum tensor $T_{\mu\nu}$ that corresponds to equation (10.5). With the aid of this tensor, the energy–momentum conservation law takes the form

$$\partial_\mu T^\mu_\nu = 0. \tag{10.7}$$

Following the general rules of field theory (see II, § 32), let us write down the variational principle whose consequence would be equation (10.5). Such a principle must consist in requiring the minimality of the "action integral"

$$S = \int L\, d^4x \tag{10.8}$$

with respect to a certain real 4-scalar L—the Lagrangian density of the field[5]. From the scalar ψ (and the operator ∂^μ) one can construct a real scalar bilinear expression of the form

$$L = \partial_\mu \psi^* \cdot \partial^\mu \psi - m^2 \psi^* \psi, \tag{10.9}$$

where m is a dimensional constant. Treating ψ and ψ^* as independent variables describing the field ("generalized coordinates" of the field q), it is easy to verify that the Lagrange

[4] The constants m are introduced in (10.4) in such a way that ψ_μ and ψ have the same dimensionality. It would be pointless to introduce different constants m_1 and m_2 in these two equations, since they could always be made equal by redefining ψ or ψ_μ.

[5] The corresponding second-quantized operator $\hat{L}$ is called the field *Lagrangian*. For brevity of terminology we shall use this term both for the "quantized" and the "unquantized" Lagrangian density.

equations

$$\frac{\partial}{\partial x^\mu}\frac{\partial L}{\partial q_{,\mu}} = \frac{\partial L}{\partial q} \tag{10.10}$$

($q_{,\mu} \equiv \partial_\mu q$) do indeed coincide with the equations (10.5) for ψ and ψ^*, with m being the particle mass. Note also that the expression (10.9) is written with an overall sign such that the square of the time derivative, $|\partial\psi/\partial t|^2$, enters L with a plus sign; in the opposite case the action could not have a minimum (cf. II, § 27). The choice of the overall numerical coefficient of L is a matter of convention (and is reflected only in the normalization coefficient in ψ).

The energy–momentum tensor is now computed by the formula

$$T_\mu^\nu = \sum q_{,\mu}\frac{\partial L}{\partial q_{,\nu}} - L\delta_\mu^\nu \tag{10.11}$$

(summation over all q). Substituting (10.9), we obtain

$$T_{\mu\nu} = \partial_\mu\psi^* \cdot \partial_\nu\psi + \partial_\nu\psi^* \cdot \partial_\mu\psi - Lg_{\mu\nu} \tag{10.12}$$

(these quantities are, as they should be, real, which is ensured by the reality of L). In particular,

$$T_{00} = 2\frac{\partial\psi^*}{\partial t}\frac{\partial\psi}{\partial t} - L = \frac{\partial\psi^*}{\partial t}\frac{\partial\psi}{\partial t} + \nabla\psi^* \cdot \nabla\psi + m^2\psi^*\psi, \tag{10.13}$$

$$T_{i0} = \frac{\partial\psi^*}{\partial t}\frac{\partial\psi}{\partial x^i} + \frac{\partial\psi^*}{\partial x^i}\frac{\partial\psi}{\partial t}. \tag{10.14}$$

The 4-momentum of the field is given by the integral

$$P_\mu = \int T_{\mu 0}\, d^3x, \tag{10.15}$$

i.e. T_{00} and T_{i0} play the role of the energy and momentum densities. We note that the quantity T_{00} is essentially positive.

Formula (10.13) can be used to normalize the wave function. A plane wave normalized to one particle in a volume $V = 1$ takes the form

$$\psi_p = \frac{1}{\sqrt{2\varepsilon}}e^{-ipx}. \tag{10.16}$$

For this function one has $T_{00} = \varepsilon$, whence the total energy within the volume $V = 1$ coincides with the energy of one particle.

The angular momentum, whose conservation follows from the isotropy of space, likewise admits a representation as a spatial integral; we shall, however, have no occasion to use such a representation later on.

Finally, besides the conservation laws linked directly with space–time symmetry, equations (10.4) admit one more conservation law. It is easy to verify that, by virtue of (10.4) (and the analogous equations for ψ^*), the following equation holds:

$$\partial_\mu j^\mu = 0, \tag{10.17}$$

where

$$j_\mu = m(\psi^*\psi_\mu + \psi^*_\mu\psi) = i[\psi^*\partial_\mu\psi - (\partial_\mu\psi^*)\psi]. \tag{10.18}$$

It is apparent from this that j^μ plays the role of the 4-vector current density. Equation (10.17) is then the continuity equation expressing the conservation of the quantity

$$Q = \int j_0\, d^3x, \tag{10.19}$$

where

$$j_0 = j^0 = i\left(\psi^*\frac{\partial\psi}{\partial t} - \frac{\partial\psi^*}{\partial t}\psi\right). \tag{10.20}$$

Let us call attention to the fact that j_0 is not a positive-definite quantity. This circumstance alone shows that in the general case it certainly cannot be interpreted as the probability density of the spatial localization of the particle. The meaning of the conservation law expressed by equation (10.17) will be clarified in the next section.

§ 11. Particles and antiparticles

In keeping with the general procedure of second quantization, one must expand an arbitrary wave function over the eigenfunctions spanning the full set of possible free-particle states, for example over the plane waves ψ_p:

$$\psi = \sum_{\mathbf{p}} a_{\mathbf{p}}\psi_p, \qquad \psi^* = \sum_{\mathbf{p}} a^*_{\mathbf{p}}\psi^*_p.$$

After that, the coefficients $a_{\mathbf{p}}$, $a^*_{\mathbf{p}}$ would have to be promoted to operators $\hat{a}_{\mathbf{p}}$, $\hat{a}^+{}_{\mathbf{p}}$ for annihilating and creating particles in the corresponding states [6].

At this point, however, we immediately encounter the following fundamentally new circumstance (compared with the non-relativistic theory). In a plane wave that solves equation (10.5), the energy ε must satisfy (for a given momentum $\mathbf{p}$) only the condition $\varepsilon^2 = \mathbf{p}^2 + m^2$, so that it can take two values: $\pm\sqrt{\mathbf{p}^2 + m^2}$. Physically meaningful values of the free-particle energy can, however, be only positive ones. Yet simply discarding the negative values is inadmissible: the general solution of the wave equation is formed only by a superposition of all its independent particular solutions. This points to the need for a certain re- interpretation of the expansion coefficients of ψ and ψ^* upon second quantization.

Let us write this expansion in the form

$$\psi = \sum_{\mathbf{p}} \frac{1}{\sqrt{2\varepsilon}} a^{(+)}_{\mathbf{p}}\, e^{i(\mathbf{pr}-\varepsilon t)} + \sum_{\mathbf{p}} \frac{1}{\sqrt{2\varepsilon}} a^{(-)}_{\mathbf{p}}\, e^{i(\mathbf{pr}+\varepsilon t)}, \tag{11.1}$$

where the first sum contains plane waves normalized according to (10.16) with positive "frequencies," and the second with negative ones; everywhere ε stands for the positive

[6] We label the ψ-functions by the 4-momentum index p; functions with "negative frequency" will henceforth be denoted ψ_{-p}. The operators $\hat{a}$, $\hat{a}^+$, on the other hand, carry the three-momentum index $\mathbf{p}$, which completely specifies the state of a real particle.

quantity: $\varepsilon = +\sqrt{\mathbf{p}^2 + m^2}$. In performing second quantization, the coefficients $a_{\mathbf{p}}^{(+)}$ in the first sum are replaced in the usual way by particle annihilation operators $\hat{a}_{\mathbf{p}}$. In the second sum we observe that upon subsequent evaluation of matrix elements the time dependence of its terms will correspond not to annihilation but to creation of particles; the factor $e^{i\varepsilon t} = (e^{-i\varepsilon t})^*$ corresponds to one extra particle with energy ε in the final state (cf. the end of § 2). Accordingly, the coefficients $a_{\mathbf{p}}^{(-)}$ are replaced by creation operators $\hat{b}^+_{-\mathbf{p}}$ for certain other particles. Changing the summation variable in the second sum of (11.1) from $\mathbf{p}$ to $-\mathbf{p}$ (so that the exponential factor takes the form $e^{-i(\mathbf{pr}-\varepsilon t)}$), we arrive at the ψ-operators

$$\hat{\psi} = \sum_{\mathbf{p}} \frac{1}{\sqrt{2\varepsilon}} (\hat{a}_{\mathbf{p}} e^{-ipx} + \hat{b}^+_{\mathbf{p}} e^{ipx}), \quad \hat{\psi}^+ = \sum_{\mathbf{p}} \frac{1}{\sqrt{2\varepsilon}} (\hat{a}^+_{\mathbf{p}} e^{ipx} + \hat{b}_{\mathbf{p}} e^{-ipx}). \tag{11.2}$$

As a result, every operator $\hat{a}_{\mathbf{p}}$ or $\hat{b}_{\mathbf{p}}$ appears multiplied by a function carrying the "correct" time factor ($\sim e^{-i\varepsilon t}$), whereas each $\hat{a}^+_{\mathbf{p}}$ or $\hat{b}^+_{\mathbf{p}}$ is accompanied by the complex conjugate of that function. This permits the identification, following the general rules, of $\hat{a}_{\mathbf{p}}$ and $\hat{b}_{\mathbf{p}}$ as annihilation operators and of $\hat{a}^+_{\mathbf{p}}$ and $\hat{b}^+_{\mathbf{p}}$ as creation operators for particles possessing momenta $\mathbf{p}$ and energies ε.

We thus arrive at the picture of particles of two kinds, appearing together and on an equal footing. They are referred to as *particles* and *antiparticles* (the reason for this name will become clear below). One kind corresponds, within the second-quantization formalism, to the operators $\hat{a}_{\mathbf{p}}$, $\hat{a}^+_{\mathbf{p}}$, and the other to $\hat{b}_{\mathbf{p}}$, $\hat{b}^+_{\mathbf{p}}$. Both species of particles, whose operators enter the same ψ-operator, thereby possess identical masses.

These conclusions can also be reached by starting directly from the demands of relativistic invariance.

From a mathematical viewpoint, Lorentz transformations are rotations of the four-dimensional coordinate system that change the orientation of the time axis (combined with purely spatial rotations, which leave the time axis unchanged, they constitute the transformation group known as the *Lorentz group*[7]). All these transformations share the property that they do not carry the t axis out of the corresponding half of the light cone, which expresses the physical principle of the existence of a limiting signal-propagation speed.

From a purely mathematical standpoint, however, a simultaneous reversal of the sign of all four coordinates is likewise a rotation (*four-dimensional inversion*): the determinant of this transformation equals $+1$, just as for every other rotational transformation. Under it the time axis passes from one half of the light cone to the other. Although this circumstance signifies the physical unrealizability of such a transformation (as a change of reference frame), mathematically the distinction reduces merely to the fact that (because of the pseudo-Euclidean character of the metric) such a rotation cannot be carried out without admitting a complex transformation of the coordinates in the process.

It is natural to expect that this distinction should be immaterial when one speaks of four-dimensional invariance. Then every expression that is invariant under Lorentz

[7]Note that the totality of all three-dimensional (spatial) rotations constitutes by itself a group, entering the Lorentz group as a subgroup. The totality of Lorentz transformations does not by itself form a group: the result of successive Lorentz transformations may reduce to a purely spatial rotation.

transformations should also be invariant under 4-inversion. The precise formulation of this requirement as applied to the scalar ψ-operator will be given in § 13. But let us immediately note that in any case it entails the simultaneous presence in the ψ-operators of terms with both signs in front of ε in the exponents, since the replacement $t \to -t$ reverses precisely this sign.

Returning to expressions (11.2), let us now determine the commutation relations among the operators $\hat{a}_{\mathbf{p}}$, $\hat{a}^+_{\mathbf{p}}$ (and $\hat{b}_{\mathbf{p}}$, $\hat{b}^+_{\mathbf{p}}$). In the photon case this was done (for $\hat{c}_{\mathbf{p}}$, $\hat{c}^+_{\mathbf{p}}$) by drawing on the oscillator analogy, that is, ultimately from the classical-limit behavior of the electromagnetic field. Here no comparable analogy exists. In order to decide whether the operators obey Bose or Fermi commutation rules, one can appeal only to the structure of the Hamiltonian built from these operators.

The Hamiltonian is obtained (see III, § 64) by substituting $\hat{\psi}$ and $\hat{\psi}^+$ for ψ and ψ^* in the integral $\int T_{00}\, d^3x$ [8]. In this way we find

$$\hat{H} = \sum_{\mathbf{p}} \varepsilon(\hat{a}^+_{\mathbf{p}}\hat{a}_{\mathbf{p}} + \hat{b}_{\mathbf{p}}\hat{b}^+_{\mathbf{p}}). \tag{11.3}$$

It is easy to see that a sensible result for the eigenvalues of this Hamiltonian is obtained only if the operators satisfy Bose commutation rules:

$$\{\hat{a}_{\mathbf{p}},\quad \hat{a}^+_{\mathbf{p}}\}_- = \{\hat{b}_{\mathbf{p}},\quad \hat{b}^+_{\mathbf{p}}\}_- \tag{11.4}$$

(all other pairs of operators commute; in particular, all particle operators $\hat{a}_{\mathbf{p}}$, $\hat{a}^+_{\mathbf{p}}$ commute with all antiparticle operators $\hat{b}_{\mathbf{p}}$, $\hat{b}^+_{\mathbf{p}}$). Indeed, in that case

$$\hat{H} = \sum_{\mathbf{p}} \varepsilon(\hat{a}^+_{\mathbf{p}}\hat{a}_{\mathbf{p}} + \hat{b}^+_{\mathbf{p}}\hat{b}_{\mathbf{p}} + 1).$$

The eigenvalues of the products $\hat{a}^+_{\mathbf{p}}\hat{a}_{\mathbf{p}}$ and $\hat{b}^+_{\mathbf{p}}\hat{b}_{\mathbf{p}}$ are non-negative integers $N_{\mathbf{p}}$ and $\bar{N}_{\mathbf{p}}$—the numbers of particles and antiparticles. The infinite additive constant $\sum\varepsilon$ ("vacuum energy") can once again simply be dropped:

$$E = \sum_{\mathbf{p}} \varepsilon(N_{\mathbf{p}} + \bar{N}_{\mathbf{p}}) \tag{11.5}$$

(cf. formula (3.1) and the remark following it). This expression is manifestly positive and accords with the picture of two species of actually existing particles. In an analogous fashion, for the total momentum of the particle system we obtain

$$\mathbf{P} = \sum_{\mathbf{p}} \mathbf{p}(N_{\mathbf{p}} + \bar{N}_{\mathbf{p}}). \tag{11.6}$$

Had we adopted, in place of (11.4), the Fermi commutation relations (anticommutators instead of commutators), we would have obtained

$$\hat{H} = \sum_{\mathbf{p}} \varepsilon(\hat{a}^+_{\mathbf{p}}\hat{a}_{\mathbf{p}} - \hat{b}^+_{\mathbf{p}}\hat{b}_{\mathbf{p}} + 1)$$

[8] In non-relativistic theory one customarily places the conjugate operator $\hat{\psi}^+$ to the left of ψ. Here, however, the ordering is immaterial, since interchanging $\hat{\psi}^+$ and $\hat{\psi}$ would merely interchange the equivalent operators $\hat{a}_{\mathbf{p}}$ and $\hat{b}_{\mathbf{p}}$. It is, however, necessary, once a particular ordering has been chosen, always to adhere to the same convention.

and, instead of formula (11.5), the physically meaningless expression $\sum \varepsilon(N_{\mathbf{p}} - \bar{N}_{\mathbf{p}})$. This expression is not positive- definite and therefore cannot represent the energy of a system of free particles.

Thus spin-0 particles are bosons.

Next, consider the integral Q (10.19). Replacing the functions ψ and ψ^* in j^0 by the operators $\hat{\psi}$ and $\hat{\psi}^+$ and carrying out the integration, we obtain

$$\hat{Q} = \sum_{\mathbf{p}} (\hat{a}^+_{\mathbf{p}} \hat{a}_{\mathbf{p}} - \hat{b}_{\mathbf{p}} \hat{b}^+_{\mathbf{p}}) = \sum_{\mathbf{p}} (\hat{a}^+_{\mathbf{p}} \hat{a}_{\mathbf{p}} - \hat{b}^+_{\mathbf{p}} \hat{b}_{\mathbf{p}} - 1). \tag{11.7}$$

The eigenvalues of this operator (apart from the inessential additive constant $\sum 1$) are

$$Q = \sum_{\mathbf{p}} (N_{\mathbf{p}} - \bar{N}_{\mathbf{p}}), \tag{11.8}$$

i.e. they equal the difference between the total numbers of particles and antiparticles.

As long as we deal with free particles, abstracting from all interaction among them, the content of the conservation law for Q (as well as the conservation of total energy and momentum (11.5, 11.6)) remains, of course, largely formal: what is actually conserved is not merely this sum, but each of the individual numbers $N_{\mathbf{p}}$, $\bar{N}_{\mathbf{p}}$ separately. Whether the quantity Q will remain conserved as a result of the interaction depends on the nature of that interaction. If Q is conserved (i.e. if the operator $\hat{Q}$ commutes with the interaction Hamiltonian), then expression (11.8) shows what restriction this imposes on the possible changes in particle number: only pairs "particle + antiparticle" can appear or disappear.

If a particle carries an electric charge, its antiparticle must bear a charge of the opposite sign: were both to carry identical charges, the creation or annihilation of a pair would violate the strict law of nature—conservation of the total electric charge. We shall see below (§ 32) how this opposition of charges arises automatically within the theory (in the interaction of particles with the electromagnetic field).

The quantity Q is sometimes called the *charge* of the field of the given particles. For electrically charged particles, Q determines, in particular, the total electric charge of the system (in units of the elementary charge e). Let us stress, however, that particles and antiparticles may well be electrically neutral.

We see in this way how the character of the relativistic energy–momentum relation (the two-valuedness of the root of the equation $\varepsilon^2 = \mathbf{p}^2 + m^2$), combined with the requirements of relativistic invariance, leads in quantum theory to the emergence of a new classification principle for particles—the possibility of existence of pairs of distinct particles ("particle + antiparticle"), standing in the correspondence described above. This remarkable prediction was first made (for spin-1/2 particles) by *Dirac* in 1930, even before the actual discovery of the first antiparticle—the positron[9].

§ 12. Strictly neutral particles

When carrying out the second quantization of the ψ-function (11.1), the coefficients $a^{(+)}_{\mathbf{p}}$ and $a^{(-)}_{\mathbf{p}}$ were treated as operators pertaining to different particles. This is, however,

[9] The notion of antiparticles was extended to bosons by *Weisskopf* and *Pauli* (*V. Weisskopf, W. Pauli*, 1934).

not obligatory: as a special case, the annihilation and creation operators appearing in $\hat{\psi}$ may refer to one and the same type of particle (as was the case for photons—cf. (2.17)). Denoting these operators $\hat{c}_{\mathbf{p}}$ and $\hat{c}^+_{\mathbf{p}}$ in this case, we write the ψ-operator as

$$\hat{\psi} = \sum_{\mathbf{p}} \frac{1}{\sqrt{2\varepsilon}} (\hat{c}_{\mathbf{p}} e^{-ipx} + \hat{c}^+_{\mathbf{p}} e^{ipx}). \tag{12.1}$$

The field described by such an operator corresponds to a system of identical particles that can be said to "coincide with their own antiparticles."

The operator (12.1) is Hermitian ($\hat{\psi}^+ = \hat{\psi}$); in this sense such a field has half as many "degrees of freedom" as a complex field, for which the operators $\hat{\psi}$ and $\hat{\psi}^+$ do not coincide.

In connection with this, the Lagrangian of the field, when expressed through the Hermitian operator $\hat{\psi}$, must contain an extra factor of $1/2$ (compared with (10.9)) [10]

$$\hat{L} = (1/2)(\partial_\mu \hat{\psi} \cdot \partial^\mu \hat{\psi} - m^2 \hat{\psi}^2). \tag{12.2}$$

The corresponding energy–momentum tensor is

$$\hat{T}_{\mu\nu} = \partial_\mu \hat{\psi} \cdot \partial_\nu \hat{\psi} - \hat{L} g_{\mu\nu}, \tag{12.3}$$

so that the energy-density operator reads

$$\hat{T}_{00} = \left(\frac{\partial \hat{\psi}}{\partial t}\right)^2 - \hat{L} = \frac{1}{2}\left[\left(\frac{\partial \hat{\psi}}{\partial t}\right)^2 + (\boldsymbol{\nabla}\hat{\psi})^2 + m^2 \hat{\psi}^2\right]. \tag{12.4}$$

Inserting (12.1) into the integral $\int \hat{T}_{00}\, d^3x$, we obtain the field Hamiltonian

$$\hat{H} = \frac{1}{2} \sum_{\mathbf{p}} \varepsilon (\hat{c}^+_{\mathbf{p}} \hat{c}_{\mathbf{p}} + \hat{c}_{\mathbf{p}} \hat{c}^+_{\mathbf{p}}). \tag{12.5}$$

This again reveals the necessity of Bose quantization:

$$\{\hat{c}_{\mathbf{p}},\ \hat{c}^+_{\mathbf{p}}\}_- = 1, \tag{12.6}$$

and the energy eigenvalues (once more dropping the additive constant) are

$$E = \sum_{\mathbf{p}} \varepsilon_{\mathbf{p}} N_{\mathbf{p}}. \tag{12.7}$$

Fermi quantization, on the other hand, would yield a meaningless result—a value of E independent of $N_{\mathbf{p}}$.

The "charge" Q of the field under consideration vanishes. This is already clear from the fact that the charge Q must change sign when particles are replaced by antiparticles,

[10] This is analogous to the extra factor $1/2$ in the operator (2.10) for the electromagnetic field energy density (expressed via the Hermitian operators $\hat{\mathbf{E}}$ and $\hat{\mathbf{H}}$), as compared to the photon energy density (3.2) expressed through the photon's complex wave function; cf. the remark on p. 26.

and here the two coincide. Correspondingly, no 4-vector current density exists either. Indeed, the expression

$$\hat{j}_\mu = i[\hat{\psi}^+\partial_\mu\hat{\psi} - (\partial_\mu\hat{\psi}^+)\hat{\psi}\,] \tag{12.8}$$

for the operator of the conserved 4-current $\hat{j}$ vanishes when $\hat{\psi} = \hat{\psi}^+$ (and the vector $\hat{\psi}\partial_\mu\hat{\psi}$ is not by itself conserved). This in turn signifies the absence of any special conservation law that would restrict the possible changes of the particle number. It is evident that such particles are, in any case, electrically neutral.

Particles of this kind are called *strictly neutral*, as distinct from electrically neutral particles that possess an antiparticle. While the latter can annihilate (converting into photons) only in pairs, strictly neutral particles can annihilate singly.

The structure of the ψ-operator (12.1) is the same as that of the operators (2.17)–(2.20) of the electromagnetic field. In this sense one can say that photons themselves are strictly neutral particles. In the case of the electromagnetic field, the Hermiticity of the operators was linked to the reality of the field strengths as measurable (in the classical limit) physical quantities. For the ψ-operators of particles, however, no such link exists, since they do not correspond to any directly measurable quantities at all.

The absence of a conserved 4-vector current is a general property of strictly neutral particles and is not connected with zero spin (it holds for photons as well). Physically it expresses the absence of the corresponding prohibitions on changes in particle number. From a formal standpoint there is a direct connection between the absence of a conserved current and the reality of the field—the Hermiticity of the operator $\hat{\psi}$.

The Lagrangian of a complex field

$$\hat{L} = \partial_\mu\hat{\psi}^+ \cdot \partial^\mu\hat{\psi} - m^2\hat{\psi}^+\hat{\psi} \tag{12.9}$$

is invariant under multiplication of the ψ-operator by an arbitrary phase factor, i.e. under transformations of the form

$$\hat{\psi} \to e^{i\alpha}\hat{\psi}, \qquad \hat{\psi}^+ \to e^{-i\alpha}\hat{\psi}^+ \tag{12.10}$$

(these are termed *gauge* transformations). In particular, the Lagrangian remains unchanged under an infinitesimally small gauge transformation

$$\hat{\psi} \to \hat{\psi} + i\delta\alpha \cdot \hat{\psi}, \qquad \hat{\psi}^+ \to \hat{\psi}^+ - i\delta\alpha \cdot \hat{\psi}^+. \tag{12.11}$$

Under an infinitesimally small variation of the "generalized coordinates" q, the Lagrangian undergoes the change

$$\delta\hat{L} = \sum\left(\frac{\partial\hat{L}}{\partial q}\delta q + \frac{\partial\hat{L}}{\partial q_{,\mu}}\delta q_{,\mu}\right) =$$

$$= \sum\left(\frac{\partial\hat{L}}{\partial q} - \frac{\partial}{\partial x^\mu}\frac{\partial\hat{L}}{\partial q_{,\mu}}\right)\delta q + \sum\frac{\partial}{\partial x^\mu}\left(\frac{\partial\hat{L}}{\partial q_{,\mu}}\delta q\right)$$

(summation over all q). The first term vanishes by virtue of the "equations of motion" (Lagrange's equations). Taking $\hat{\psi}$ and $\hat{\psi}^+$ as the "coordinates" q and setting

$$\delta\hat{\psi} = i\delta\alpha \cdot \hat{\psi}, \quad \delta\hat{\psi}^+ = -i\delta\alpha \cdot \hat{\psi}^+,$$

we get

$$\delta\hat{L} = i\delta\alpha \frac{\partial}{\partial x^\mu}\left(\hat{\psi}\frac{\partial \hat{L}}{\partial \hat{\psi}_{,\mu}} - \hat{\psi}^+ \frac{\partial \hat{L}}{\partial \hat{\psi}^+_{,\mu}}\right).$$

It is apparent that the condition of invariance of the Lagrangian ($\delta\hat{L} = 0$) is equivalent to the continuity equation ($\partial_\mu \hat{j}^\mu$=0) for the 4-vector

$$\hat{j}^\mu = i\left(\hat{\psi}^+ \frac{\partial \hat{L}}{\partial \hat{\psi}^+_{,\mu}} - \hat{\psi}\frac{\partial \hat{L}}{\partial \hat{\psi}_{,\mu}}\right). \tag{12.12}$$

It is readily verified that for the Lagrangian (12.9) this formula yields the current (12.8).

We see, therefore, that in the mathematical framework of the theory the existence of a conserved current is tied to gauge invariance of the Lagrangian (*W. Pauli*, 1941). The Lagrangian (12.2) of a strictly neutral field, however, lacks this symmetry.

§ 13. The transformations C, P, T

Unlike four-dimensional inversion, a three-dimensional (spatial) inversion cannot be reduced to any rotation of the 4-coordinate system: the determinant of this transformation equals not $+1$ but -1. The symmetry properties of particles with respect to inversion (P-transformation) are therefore not dictated by the requirements of relativistic invariance [11].

Applied to a scalar wave function, the inversion operation amounts to the transformation

$$\hat{P}\psi(t, \mathbf{r}) = \pm\psi(t, -\mathbf{r}), \tag{13.1}$$

where the sign "+" or "−" on the right-hand side corresponds, respectively, to a true scalar or a pseudoscalar.

It is apparent from this that one must distinguish two aspects of the wave function's behavior under inversion. One of them is related to the dependence of the wave function on the coordinates. In nonrelativistic quantum mechanics only this aspect was considered—it leads to the notion of the parity of a state (which we shall henceforth call the *orbital parity*), characterizing the symmetry properties of the particle's motion. If a state possesses a definite orbital parity ($+1$ or -1), this means that

$$\psi(t, -\mathbf{r}) = \pm\psi(t, \mathbf{r}).$$

The second aspect is the behavior of the wave function at a given point (which one may conveniently take to be the coordinate origin) when the coordinate axes are inverted. This behavior gives rise to the concept of the *intrinsic parity of the particle*. For a spin-0 particle the two signs in definition (13.1) correspond to intrinsic-parity values of $+1$ and -1. The overall parity of a particle system equals the product of their individual intrinsic parities and the orbital parity associated with the relative motion.

The "intrinsic" symmetry properties of different particles manifest themselves, of course, only in the processes of their mutual transformations. The analogue of intrinsic

[11] The Lorentz group supplemented by spatial inversion is called the *extended Lorentz group* (as opposed to the original group, which does not contain P and which in this context is termed *proper*). The extended group includes all transformations that do not carry the t axis out of the corresponding half of the light cone.

parity in nonrelativistic quantum mechanics is the parity of a bound state of a composite system (for instance, a nucleus). From the standpoint of relativistic theory, which draws no fundamental distinction between composite and elementary particles, this kind of intrinsic parity is no different from the intrinsic parity of the particles that figure as elementary ones in nonrelativistic theory. In the nonrelativistic domain, where the latter behave as immutable objects, their intrinsic symmetry properties are unobservable, and consequently any consideration of them would be devoid of physical content.

Within the second-quantization formalism, the intrinsic parity is expressed through the behavior of the ψ-operators under inversion. For scalar and pseudoscalar fields the transformation laws read

$$P : \hat{\psi}(t, \mathbf{r}) \to \pm\hat{\psi}(t, -\mathbf{r}). \tag{13.2}$$

The very meaning of the action of inversion on the ψ-operator must, however, be formulated as a definite transformation of the particle annihilation and creation operators—one such that the change (13.2) results from it. It is easy to see that the required transformation is

$$P : \hat{a}_{\mathbf{p}} \to \pm\hat{a}_{-\mathbf{p}}, \quad \hat{b}_{\mathbf{p}} \to \pm\hat{b}_{-\mathbf{p}} \tag{13.3}$$

(and likewise for the conjugate operators). Indeed, carrying out this substitution in the operator

$$\hat{\psi}(t, \mathbf{r}) = \sum_{\mathbf{p}} \frac{1}{\sqrt{2\varepsilon}} \left(\hat{a}_{\mathbf{p}} e^{-i\omega t + i\mathbf{pr}} + \hat{b}_{\mathbf{p}}^{+} e^{i\omega t - i\mathbf{pr}} \right) \tag{13.4}$$

and then relabeling the summation variable ($\mathbf{p} \to -\mathbf{p}$), we bring it to the form $\pm\hat{\psi}(t, -\mathbf{r})$. Thus, denoting by $\hat{\psi}^P(t, \mathbf{r})$ the operator in which transformation (13.3) has been performed, one may write

$$\hat{\psi}^P(t, \mathbf{r}) = \pm\hat{\psi}(t, -\mathbf{r}). \tag{13.5}$$

Note that transformation (13.3) has a perfectly natural form: inversion reverses the sign of the polar vector $\mathbf{p}$, so that particles with momentum $\mathbf{p}$ are replaced by particles with momentum $-\mathbf{p}$.

In (13.3) the operators $\hat{a}_{\mathbf{p}}$ and $\hat{b}_{\mathbf{p}}$ both transform with either the upper or the lower signs. In the second-quantization formalism this expresses the equality of the intrinsic parities of particle and antiparticle (with spin 0). This equality is in itself already obvious from the fact that particles and antiparticles (of spin 0) are described by the very same (scalar or pseudoscalar) wave functions.

In relativistic theory there also arises a symmetry with respect to a transformation that has no counterpart in nonrelativistic theory; it is called *charge conjugation* (C-transformation). If all operators $\hat{a}_{\mathbf{p}}$ and $\hat{b}_{\mathbf{p}}$ are mutually interchanged:

$$C : \hat{a}_{\mathbf{p}} \to \hat{b}_{\mathbf{p}}, \quad \hat{b}_{\mathbf{p}} \to \hat{a}_{\mathbf{p}} \tag{13.6}$$

(i.e. particles and antiparticles are interchanged), then $\hat{\psi}$ goes over into the "charge-conjugate" operator $\hat{\psi}^C$, with

$$\hat{\psi}^C(t, \mathbf{r}) = \hat{\psi}^+(t, \mathbf{r}) \tag{13.7}$$

This equality expresses the symmetry with which the notions of particles and antiparticles enter the theory.

Let us note that the definition of the charge-conjugation transformation contains a certain inessential formal arbitrariness. The content of the transformation is unaltered if one introduces into definition (13.6) an arbitrary phase factor:

$$\hat{a}_{\mathbf{p}} \to e^{i\alpha}\hat{b}_{\mathbf{p}}, \quad \hat{b}_{\mathbf{p}} \to e^{-i\alpha}\hat{a}_{\mathbf{p}}.$$

Then one would have

$$\hat{\psi} \to e^{i\alpha}\hat{\psi}^{+}, \quad \hat{\psi}^{+} \to e^{-i\alpha}\hat{\psi},$$

and a twofold application of this transformation would still yield the identity ($\hat{\psi} \to \hat{\psi}$). All such definitions are, however, equivalent to one another. Since the properties of the ψ-operators are unaffected by multiplication by a phase factor (cf. the end of the preceding section), one may simply relabel $\hat{\psi}$ as $\hat{\psi}e^{i\alpha/2}$ and then return to the definition of charge conjugation in the form (13.6),(13.7).

Because charge conjugation replaces a particle by a non-identical antiparticle, it does not in the general case give rise to any new characteristic of a particle or a system of particles as such.

Systems containing equal numbers of particles and antiparticles constitute an exception. The operator $\hat{C}$ carries such a system into itself, so that eigenstates exist with eigenvalues $C = \pm 1$ (which follow from $\hat{C}^2 = 1$). When describing charge symmetry one may treat particle and antiparticle as two distinct "charge states" of a single particle, distinguished by the charge quantum number $Q = \pm 1$. The system's wave function then factorizes into an orbital part and a "charge" part, and must be symmetric with respect to the simultaneous interchange of all variables (spatial and charge) belonging to any pair of particles. The symmetry character of the "charge" factor fixes the charge parity of the system (see the Problem) [12].

The concept of charge parity, which arises naturally for "strictly neutral" systems, should apply also to strictly neutral "elementary" particles. In the second-quantization formalism this concept is described by the relation

$$\hat{\psi}^{C} = \pm\hat{\psi}; \tag{13.8}$$

the signs "+" and "−" correspond to charge-even and charge-odd particles.

In § 11 it was pointed out that relativistic invariance must also entail invariance under 4-inversion. For the operator of a scalar (in the sense of 4-rotations) field this means that under such a transformation one must have

$$\hat{\psi}(t, \mathbf{r}) = \hat{\psi}(-t, -\mathbf{r})$$

always with the same sign "+" on the right-hand side. In terms of the operators $\hat{a}_{\mathbf{p}}$, $\hat{b}_{\mathbf{p}}$, the conversion of $\hat{\psi}(t, \mathbf{r})$ into $\hat{\psi}(-t, -\mathbf{r})$ is accomplished by interchanging the coefficients of e^{-ipx} and e^{ipx} in (13.4), i.e. by the replacement

$$\hat{a}_{\mathbf{p}} \to \hat{b}^{+}_{\mathbf{p}}, \quad \hat{b}_{\mathbf{p}} \to \hat{a}^{+}_{\mathbf{p}} \tag{13.9}$$

[12] In these arguments we have in mind particles of spin 0. The method of analysis described here generalizes directly to other cases—see, for example, the Problem in § 27.

Since it replaces the a-operators by b-operators, this transformation incorporates the mutual interchange of particles and antiparticles. We see that in relativistic theory there naturally arises a requirement of invariance under a transformation in which spatial inversion (P) and time reversal (T) are accompanied by charge conjugation (C); this statement is known as the *CPT theorem* [13].

In this connection it is, however, appropriate to stress that although the arguments presented here and in §§ 11, 12 appear to be a natural development of the concepts of ordinary quantum mechanics and classical relativity, the results obtained in this way go beyond that framework both in form (ψ-operators that simultaneously contain creation and annihilation operators for particles) and in substance (particles and antiparticles). These results cannot therefore be regarded as a matter of pure logical necessity. They embody new physical principles whose correctness can be judged only by experiment.

If we denote by $\hat{\psi}^{CPT}(t, \mathbf{r})$ the operator (13.4) in which the transformation (13.9) has been carried out, one may write

$$\hat{\psi}^{CPT}(t, \mathbf{r}) = \hat{\psi}(-t, -\mathbf{r}). \tag{13.10}$$

Having thus formulated 4-inversion as the transformation (13.9), we thereby establish for the ψ-operator also the formulation of the time-reversal transformation: together with the CP transformation (called the *combined inversion*) it must yield (13.9). Taking into account the definitions (13.3) and (13.6), we therefore find

$$T : \hat{a}_{\mathbf{p}} \to \pm\hat{a}^{+}_{-\mathbf{p}}, \quad \hat{b}_{\mathbf{p}} \to \pm\hat{b}^{+}_{-\mathbf{p}} \tag{13.11}$$

(the signs "±" match those in (13.3)). The meaning of this transformation is perfectly natural: time reversal not only converts motion with momentum $\mathbf{p}$ into motion with momentum $-\mathbf{p}$, but also interchanges the initial and final states in matrix elements; the annihilation operators for particles with momenta $\mathbf{p}$ are therefore replaced by creation operators for particles with momenta $-\mathbf{p}$. Performing the substitution (13.11) in (13.4) and relabeling the summation variable ($\mathbf{p} \to -\mathbf{p}$), we find that [14]

$$\hat{\psi}^{T}(t, \mathbf{r}) = \pm\hat{\psi}^{+}(-t, \mathbf{r}). \tag{13.12}$$

This result mirrors the standard time-reversal prescription of quantum mechanics: when a state has wave function $\psi(t, \mathbf{r})$, the "temporally reversed" state is represented by $\psi^{*}(-t, \mathbf{r})$; complex conjugation is needed in order to recover the "proper" character of the time dependence, which the sign flip of t would otherwise spoil (*E. P. Wigner*, 1932).

Since the transformation T (and with it CPT) interchange initial and final states, the concepts of eigenstates and eigenvalues have no meaning for them. They therefore do not lead to new characteristics of particles as such. The consequences to which they lead when applied to scattering processes will be discussed in §§ 69, 71.

Let us examine how the operator 4-vector of the current $\hat{j}^{\mu}$ (12.8) transforms under C, P, and T. Transformation (13.2) together with the replacement $(\partial_0, \partial_i) \to (\partial_0, -\partial_i)$

[13] It was formulated by *Lüders* (G. Lüders, 1954) and *Pauli* (W. Pauli, 1955).

[14] If one were to define the operation T independently of the other transformations, the same arbitrariness in the choice of a phase factor would arise as for the operation C. The requirement of CPT symmetry, however, leaves the phase-factor choice free in only one of the two transformations, C or T.

gives

$$P : (\hat{j}^0, \hat{\mathbf{j}})_{t,\mathbf{r}} \to (\hat{j}^0, -\hat{\mathbf{j}})_{t,-\mathbf{r}}, \tag{13.13}$$

as it must for a true 4-vector. Transformation (13.7) would simply give

$$C : (\hat{j}^0, \hat{\mathbf{j}})_{t,\mathbf{r}} \to (-\hat{j}^0, -\hat{\mathbf{j}})_{t,\mathbf{r}}, \tag{13.14}$$

were the operators $\hat{\psi}$ and $\hat{\psi}^+$ commutative. The noncommutativity of these operators arises, however, solely from the noncommutativity of $\hat{a}_\mathbf{p}$ and $\hat{a}^+_\mathbf{p}$ (or of $\hat{b}_\mathbf{p}$ and $\hat{b}^+_\mathbf{p}$) at the same $\mathbf{p}$; but by virtue of the commutation rules (11.4) the interchange of these operators merely introduces terms independent of the occupation numbers, i.e. of the field state. Discarding these terms as inessential (just as in (11.5),(11.6)), we recover rule (13.14), which has a natural meaning: by replacing particles with antiparticles, charge conjugation reverses the sign of every component of the 4-current.

Since the time-reversal operation is linked with the transposition of initial and final states, its application to a product of operators reverses the order of the factors. Thus,

$$\left(\hat{\psi}^+ \partial_\mu \hat{\psi}\right)^T = \left(\partial_\mu \hat{\psi}\right)^T \left(\hat{\psi}^+\right)^T .$$

In the present case, however, this is immaterial: by virtue of the commutativity of the ψ-operators (in the sense indicated above), reverting to the original order of the factors does not affect the result. Noting also that under time reversal $(\partial_0, \partial_i) \to (-\partial_0, \partial_i)$, we find the transformation rule for the current:

$$T : (\hat{j}^0, \hat{\mathbf{j}})_{t,\mathbf{r}} \to (\hat{j}^0, -\hat{\mathbf{j}})_{-t,\mathbf{r}}. \tag{13.15}$$

The three-dimensional vector $\mathbf{j}$ changes sign in accordance with the classical meaning of this quantity.

Finally, under the CPT transformation we have

$$CPT : (\hat{j}^0, \hat{\mathbf{j}})_{t,\mathbf{r}} \to (-\hat{j}^0, -\hat{\mathbf{j}})_{-t,-\mathbf{r}}, \tag{13.16}$$

in accord with the meaning of this operation as a 4-inversion. Let us stress in this connection that, since 4-inversion reduces to a rotation of the 4-coordinate system, with respect to it there is no distinction at all between two types (true and pseudo) of 4-tensors of any rank.

Up to now we have been considering free particles. But the quantum numbers of parity acquire real significance only when one examines interacting particles, where definite selection rules are associated with them, permitting or forbidding particular processes. Such significance, however, can attach only to conserved quantities—eigenvalues of operators that commute with the Hamiltonian of the interacting particles.

Relativistic invariance guarantees that the CPT-transformation operator commutes with the Hamiltonian in every case. As for the individual transformations C and P (together with T), experiment demonstrates that electromagnetic and strong interactions respect these symmetries, so that the associated parity quantum numbers are conserved within those interactions. In weak processes, however, these conservation laws break down [15].

[15]The suggestion that parity conservation might fail in weak interactions was first put forward by *Lee* and *Yang* (*T. D. Lee*, *C. N. Yang*, 1956). Earlier still, the broader idea that P- and T-invariance of physical laws is not mandatory was advanced by *Dirac* (1949).

Anticipating later developments, we point out that the operator governing the coupling of charged particles to the electromagnetic field is formed as the product of the operator 4-vectors $\hat{A}$ and $\hat{j}$. Because charge conjugation flips the sign of $\hat{j}$, electromagnetic-interaction invariance under this transformation requires that $\hat{A}$ likewise change sign. Put differently, photons are charge-odd particles.

The stated behavior of the operators $\hat{A}$ is in agreement with the properties of the 4-potential in classical theory. Indeed, from the transformations

$$C : (\hat{A}_0, \hat{\mathbf{A}}) \to (-\hat{A}_0, -\hat{\mathbf{A}})_{t,\mathbf{r}},$$

$$P : (\hat{A}_0, \hat{\mathbf{A}}) \to (\hat{A}_0, -\hat{\mathbf{A}})_{t,-\mathbf{r}},$$

$$CPT : (\hat{A}_0, \hat{\mathbf{A}}) \to (-\hat{A}_0, -\hat{\mathbf{A}})_{-t,-\mathbf{r}},$$

it follows that

$$T : (\hat{A}_0, \hat{\mathbf{A}}) \to (\hat{A}_0, -\hat{\mathbf{A}})_{-t,\mathbf{r}},$$

which corresponds to the classical rule for transforming the electromagnetic-field potentials under time reversal.

The requirement of CPT invariance does not impose any restrictions on the properties of particles in themselves. It does, however, lead to a definite connection between the properties of particles and antiparticles. This includes, first and foremost, the equality of their masses—a fact already clear from the connection, set out in § 11, between 4-inversion and the very origin of the particle–antiparticle concept.

Furthermore, CPT invariance implies that the proportionality coefficients relating the electric and magnetic moment vectors to the spin vector differ for particle and antiparticle only in sign. Indeed, the magnetic moment changes sign under C and T transformations and (being an axial vector) is P-invariant. Consequently, the CPT transformation, while converting a particle into its antiparticle, does not at the same time change the sign of the magnetic moment; the spin vector, however, does change sign. The same applies to the electric moment, which remains unchanged under time reversal and changes sign under C-transformation and (by the properties of a polar vector) under spatial inversion.

The requirements of P- or T-invariance (if they are obeyed), on the other hand, constrain the properties of each individual particle: they forbid the particle from possessing an electric dipole moment. Indeed, the only vector that can be constructed for an elementary particle at rest from its ψ-operators is the vector of its spin operator. This vector is P-even and T-odd; it can therefore determine only the magnetic, but not the electric moment. Let us emphasize that for this prohibition it suffices to require merely one of the two—P- or T-invariance.

Problem

Determine the charge parity and spatial parity of a system consisting of a spin-0 particle and its antiparticle, with orbital angular momentum l of the relative motion.

Solution. Permutation of the particle coordinates is equivalent to inversion (about the center of inertia) and therefore multiplies the orbital function by $(-1)^l$; permutation of the charge variables is equivalent to charge conjugation and multiplies the "charge" factor in the wave function by the sought quantity C. From the condition $C(-1)^l = 1$ we have

$$C = (-1)^l.$$

The spatial parity P of the system is the product of the orbital parity with the intrinsic parities of the two particles. Because particle and antiparticle share the same intrinsic parity, here P simply equals the orbital parity: $P = (-1)^l$.

§ 14. Wave equation for spin-1 particles

In its rest frame a particle of spin 1 is characterized by a wave function with three components—a three-dimensional vector (one therefore often speaks of a *vector* particle). Viewed four-dimensionally, these three components can be either the spatial part of a (spacelike) 4-vector ψ^μ, or the mixed components of an antisymmetric 4-tensor of rank two, $\psi^{\mu\nu}$, for which the time (ψ^0) and purely spatial (ψ^{ik}) components vanish in the rest frame [16].

The wave equation—a differential relation between the quantities ψ^μ and $\psi^{\mu\nu}$—is established by the relations that we write as

$$i\psi_{\mu\nu} = \hat{p}_\mu\psi_\nu - \hat{p}_\nu\psi_\mu, \tag{14.1}$$

$$im^2\psi_\mu = \hat{p}^\nu\psi_{\mu\nu}, \tag{14.2}$$

where $\hat{p} = i\partial$ (*A. Proca*, 1936). Applying the operator $\hat{p}^\mu$ to both sides of equation (14.2), we obtain (on account of the antisymmetry of $\psi_{\mu\nu}$)

$$\hat{p}^\mu\psi_\mu = 0. \tag{14.3}$$

From (14.1, 14.2) one can eliminate $\psi_{\mu\nu}$ by inserting the first equation into the second. With the aid of (14.3) this yields

$$(\hat{p}^2 - m^2)\psi_\mu = 0, \tag{14.4}$$

from which it is again apparent (cf. § 10) that m is the particle mass. A free spin-1 particle can therefore be described entirely by a single 4-vector ψ^μ whose components satisfy the second-order equation (14.4) together with the supplementary condition (14.3), which removes from ψ^μ the part belonging to spin 0.

In the rest frame, where ψ_μ is independent of the spatial coordinates, we find that $\hat{p}^0\psi_0 = 0$. Since at the same time $\hat{p}^0\psi_0 = m\psi_0$, it follows that $\psi_0 = 0$ in the rest frame, as required. Together with ψ_0 the quantities ψ_{ik} also vanish.

A spin-1 particle can possess different intrinsic parities, depending on whether ψ^μ is a true vector or a pseudovector. In the first case

$$\hat{P}\psi^\mu = \left(\psi^0, -\psi^i\right),$$

[16] Looking ahead, let us mention that the set consisting of the 4-vector ψ_μ and the 4-tensor $\psi^{\nu\mu}$ corresponds to the set of four-dimensional second-rank spinors $\xi^{\alpha\beta}$, $\eta_{\dot{\alpha}\dot{\beta}}$, $\zeta^{\alpha\dot{\beta}}$, where $\xi^{\alpha\beta}$ and $\eta_{\dot{\alpha}\dot{\beta}}$ are symmetric spinors that transform into each other under inversion (see § 19).

and in the second

$$\hat{P}\psi^{\mu} = \left(-\psi^{0}, \psi^{i}\right).$$

Equations (14.1),(14.2) can be derived from a variational principle with the Lagrangian

$$L = (1/2)\,\psi_{\mu\nu}\psi^{\mu\nu *} - (1/2)\psi^{\mu\nu *}(\partial_{\mu}\psi_{\nu} - \partial_{\nu}\psi_{\mu}) - \\ - (1/2)\psi^{\mu\nu}(\partial_{\mu}\psi^{*}_{\nu} - \partial_{\nu}\psi^{*}_{\mu}) + m^{2}\psi_{\mu}\psi^{\mu *}. \quad (14.5)$$

The independent generalized coordinates here are ψ_{μ}, ψ^{*}_{μ}, $\psi_{\mu\nu}$, $\psi^{*}_{\mu\nu}$ [17].

For obtaining the energy–momentum tensor, formula (10.11) is not entirely convenient in the present case, since it would lead to a non-symmetric tensor requiring additional symmetrization. One may instead use the formula

$$\frac{1}{2}\,T_{\mu\nu}\sqrt{-g} = -\frac{\partial}{\partial x^{\lambda}}\frac{\partial\sqrt{-g}\,L}{\partial g^{\mu\nu}_{,\lambda}} + \frac{\partial\sqrt{-g}\,L}{\partial g^{\mu\nu}}, \quad (14.6)$$

in which it is assumed that L is written in a form valid for arbitrary curvilinear coordinates (see II, § 94). If L involves only the components of the metric tensor $g_{\mu\nu}$ itself (but not their coordinate derivatives), the formula simplifies to

$$T_{\mu\nu} = \frac{2}{\sqrt{-g}}\frac{\partial\sqrt{-g}\,L}{\partial g^{\mu\nu}} = 2\frac{\partial L}{\partial g^{\mu\nu}} - g_{\mu\nu}L$$

(recall that $d\ln g = -g_{\mu\nu}dg^{\mu\nu}$).

Since the differentiation in formula (14.6) is not performed with respect to ψ_{μ} or $\psi_{\mu\nu}$, when applying it there is no need to treat these quantities as independent; one may directly use relation (14.1) and rewrite the Lagrangian (14.5) as

$$L = (-1/2)\,\psi_{\mu\nu}\psi^{*}_{\lambda\rho}g^{\mu\lambda}g^{\nu\rho} + m^{2}\psi_{\mu}\psi^{*}_{\nu}g^{\mu\nu}. \quad (14.7)$$

Then

$$T_{\mu\nu} = -\psi_{\mu\lambda}\psi_{\nu}{}^{\lambda *} - \psi^{*}_{\mu\lambda}\psi_{\nu}{}^{\lambda} + m^{2}\left(\psi^{*}_{\mu}\psi_{\nu} + \psi^{*}_{\nu}\psi_{\mu}\right) + \\ + g_{\mu\nu}\left((1/2)\,\psi_{\lambda\rho}\psi^{\lambda\rho *} - m^{2}\psi^{*}_{\lambda}\psi^{\lambda}\right). \quad (14.8)$$

In particular, the energy density is given by a manifestly positive expression

$$T_{00} = (1/2)\,\psi_{ik}\psi^{*}_{ik} + \psi_{0i}\psi^{*}_{0i} + m^{2}\left(\psi_{0}\psi^{*}_{0} + \psi_{i}\psi^{*}_{i}\right). \quad (14.9)$$

The conserved 4-vector of the current density reads

$$j^{\mu} = i\left(\psi^{\mu\nu *}\psi_{\nu} - \psi^{\mu\nu}\psi^{*}_{\nu}\right). \quad (14.10)$$

[17] Had we carried out the variation with respect to ψ_{μ} alone (assuming from the outset that $\psi_{\mu\nu}$ is expressed through ψ_{μ} via (14.1)), equation (14.3) would have to be imposed as an auxiliary condition unrelated to the variational principle.

It can be obtained from formula (12.12) by differentiating the Lagrangian (14.5) with respect to the derivatives $\partial_\mu \psi_\nu$. In particular,

$$j^0 = i\left(\psi^{0k*}\psi_k - \psi^{0k}\psi_k^*\right) \tag{14.11}$$

and is not a positive-definite quantity.

A plane wave normalized to one particle in volume $V = 1$:

$$\psi_\mu = \frac{1}{\sqrt{2\varepsilon}} u_\mu e^{-ipx}, \quad u_\mu u^{\mu*} = -1, \tag{14.12}$$

where u_μ is a unit polarization 4-vector obeying (by virtue of (14.3)) the four-dimensional transversality condition

$$u_\mu p^\mu = 0. \tag{14.13}$$

Indeed, substituting the function (14.12) into (14.9) and (14.11) gives

$$T_{00} = -2\varepsilon^2 \psi_\mu \psi^{\mu*} = \varepsilon, \qquad j^0 = 1.$$

In contrast to the photon, a vector particle of nonzero mass possesses three independent polarization directions. For the corresponding amplitudes see (16.21).

For partially polarized vector particles the polarization density matrix is introduced in such a way that for a pure state it becomes simply the product

$$\rho_{\mu\nu} = u_\mu u_\nu^*$$

(analogously to expression (8.7) for photons). According to (14.12), (14.13) it satisfies the conditions

$$p^\mu \rho_{\mu\nu} = 0, \qquad \rho_\mu^\mu = -1. \tag{14.14}$$

For unpolarized particles the matrix $\rho_{\mu\nu}$ must have the form $a g_{\mu\nu} + b p_\mu p_\nu$. Fixing the coefficients a and b from (14.14), one finds

$$\rho_{\mu\nu} = -\frac{1}{3}\left(g_{\mu\nu} - \frac{p_\mu p_\nu}{m^2}\right). \tag{14.15}$$

Quantization of the vector-particle field proceeds in the same way as for the scalar case, and there is no need to repeat the whole argument anew. The ψ-operators of the vector field take the form

$$\begin{aligned} \hat\psi_\mu &= \sum_{\mathbf{p}\alpha} \frac{1}{\sqrt{2\varepsilon}}\left(\hat a_{\mathbf{p}\alpha} u_\mu^{(\alpha)} e^{-ipx} + \hat b^+_{\mathbf{p}\alpha} u_\mu^{(\alpha)*} e^{ipx}\right), \\ \hat\psi^+_\mu &= \sum_{\mathbf{p}\alpha} \frac{1}{\sqrt{2\varepsilon}}\left(\hat a^+_{\mathbf{p}\alpha} u_\mu^{(\alpha)*} e^{ipx} + \hat b_{\mathbf{p}\alpha} u_\mu^{(\alpha)} e^{-ipx}\right), \end{aligned} \tag{14.16}$$

where the index α labels the three independent polarizations.

The positive definiteness of expression (14.9) for T_{00} and the indefiniteness of j^0 (14.11) lead, just as in the scalar case, to the requirement of Bose quantization.

There exists a close connection between the properties of the strictly neutral vector field and the electromagnetic field. A neutral vector field is described by a Hermitian ψ-operator:

$$\hat{\psi}_\mu = \sum_{\mathbf{p}\alpha} \frac{1}{\sqrt{2\varepsilon}} \left(\hat{c}_{\mathbf{p}\alpha} u_\mu^{(\alpha)} e^{-ipx} + \hat{c}^+_{\mathbf{p}\alpha} u_\mu^{(\alpha)*} e^{ipx} \right). \tag{14.17}$$

The Lagrangian of this field is

$$\hat{L} = \frac{1}{4} \hat{\psi}_{\mu\nu} \hat{\psi}^{\mu\nu} - \frac{1}{2} \hat{\psi}^{\mu\nu} (\partial_\mu \hat{\psi}_\nu - \partial_\nu \hat{\psi}_\mu) + \frac{1}{2} m^2 \hat{\psi}_\mu \hat{\psi}^\mu. \tag{14.18}$$

The electromagnetic field corresponds to the case $m = 0$. The 4-vector ψ^μ then becomes the 4-potential A^μ, and the 4-tensor $\psi^{\mu\nu}$ becomes the field tensor $F^{\mu\nu}$, related to the potential through definition (14.1). Equation (14.2) turns into $\partial^\nu \psi_{\mu\nu} = 0$, which corresponds to the second pair of Maxwell's equations. Condition (14.3) no longer follows from it and thus ceases to be obligatory. In the absence of the supplementary condition there is no need to treat $\hat{\psi}_\mu$ and $\hat{\psi}_{\mu\nu}$ as independent "coordinates" in the Lagrangian, and (14.18) reduces to

$$\hat{L} = -(1/4) \hat{\psi}_{\mu\nu} \hat{\psi}^{\mu\nu} \tag{14.19}$$

in agreement with the familiar classical expression for the electromagnetic-field Lagrangian. This Lagrangian, together with the tensor $\hat{\psi}_{\mu\nu}$, is invariant under an arbitrary gauge transformation of the "potentials" $\hat{\psi}_\mu$. The connection between this circumstance and the vanishing mass is clearly visible: the Lagrangian (14.18) lacks this property on account of the term $m^2 \hat{\psi}_\mu \hat{\psi}^\mu$.

§ 15. Wave equation for particles with higher integer spins

Since the wave equations (14.3, 14.4) follow directly from specifying the mass and spin of the particle, the practical use of the Lagrangian comes down not so much to deriving these equations as to constructing the expressions for the energy, momentum, and charge of the field.

For this purpose, as already noted, one may use expression (14.7) in place of (14.5), and the former can be further transformed. Using (14.1) and rewriting the Lagrangian (14.7) as

$$L = -\left(\partial_\mu \psi^*_\nu\right) \left(\partial^\mu \psi^\nu\right) + \left(\partial_\nu \psi^*_\mu\right) \left(\partial^\mu \psi^\nu\right) + m^2 \psi_\mu \psi^{\mu*} = \tag{15.1}$$

$$= -\left(\partial_\mu \psi^*_\nu\right) \left(\partial^\mu \psi^\nu\right) + m^2 \psi^*_\mu \psi^\mu + \partial_\nu \left(\psi^*_\mu \partial^\mu \psi^\nu\right) - \psi^*_\mu \partial^\mu \partial_\nu \psi^\nu.$$

By virtue of (14.3) the last term vanishes, while the penultimate one is a total derivative. Dropping it, we arrive at the Lagrangian

$$L' = -\left(\partial_\mu \psi^*_\nu\right) \left(\partial^\mu \psi^\nu\right) + m^2 \psi^*_\mu \psi^\mu.$$

It has the same structure as the Lagrangian (10.9) for a spin-0 particle, differing only in that the scalar ψ is replaced by the 4-vector ψ_μ and the overall sign is reversed. The sign change is due to the spacelike character of ψ_μ, so that $\psi_\mu \psi^{\mu *} < 0$, whereas for a scalar particle $\psi\psi^* > 0$.

If one constructs the energy–momentum 4-tensor and the current 4-vector with the help of Lagrangian (15.1), one obtains expressions of the same form as expressions (10.12) and (10.18) for the scalar field:

$$T_{\mu\nu} = -\partial_\mu \psi^{\lambda *} \cdot \partial_\nu \psi_\lambda - \partial_\nu \psi^{\lambda *} \cdot \partial_\mu \psi_\lambda - L' g_{\mu\nu}, \tag{15.2}$$

$$j_\mu = -i \left[\psi^*_\lambda \partial_\mu \psi^\lambda - \left(\partial_\mu \psi^*_\lambda \right) \psi^\lambda \right]. \tag{15.3}$$

Their difference from (14.8) and (14.10) likewise reduces to total derivatives. But the local values of these quantities lack (as was already stressed) any deep physical significance. Only the volume integrals P_μ (10.15) and Q (10.19) matter, and they coincide for either choice of $T_{\mu\nu}$ and j_μ.

This method of description extends straightforwardly to particles of arbitrary (integer) spin. The wave function of a particle with spin s is an irreducible 4-tensor of rank s, that is, a tensor symmetric in all of its indices and vanishing upon contraction over any pair of them:

$$\psi_{\ldots\mu\ldots\nu\ldots} = \psi_{\ldots\nu\ldots\mu\ldots}, \quad \psi_{\ldots\mu\ldots}{}^{\mu}{}_{\ldots} = 0. \tag{15.4}$$

This tensor must obey the supplementary four-dimensional transversality condition

$$\hat{p}^\mu \psi_{\ldots\mu\ldots} = 0, \tag{15.5}$$

and each of its components must satisfy the second-order equation

$$(\hat{p}^2 - m^2)\psi_{\ldots} = 0. \tag{15.6}$$

In the rest frame, condition (15.5) causes every component of the 4-tensor that carries a 0 among its indices to vanish. In other words, the wave function in the rest frame (i.e. in the nonrelativistic limit) reduces, as expected, to an irreducible 3-tensor of rank s with $2s + 1$ independent components.

The Lagrangian, the energy–momentum tensor, and the current vector for particles of spin s differ from (15.1)–(15.3) solely by the replacement of ψ_λ with $\psi_{\lambda\mu\ldots}$.

The normalized plane wave is

$$\psi^{\mu\nu\cdots} = \frac{1}{\sqrt{2\varepsilon}} u^{\mu\nu\cdots} e^{-ipx}, \quad u^*_{\mu\nu\ldots} u^{\mu\nu\cdots} = -1, \tag{15.7}$$

with the wave amplitude satisfying the conditions

$$u^{\cdots\mu\cdots} p_\mu = 0. \tag{15.8}$$

There are $2s + 1$ independent polarization states.

Field quantization proceeds as an obvious generalization of the spin-0 or spin-1 cases.

The scheme presented above is entirely sufficient for the stated purpose—describing the field of free particles. The situation changes, however, if one poses the problem of describing the interaction of particles with the electromagnetic field. Such an interaction would need to be incorporated into a Lagrangian from which all equations could be obtained without having to impose supplementary conditions separately. In practice, however, it turns out that this approach to describing the interaction works only for electrons—particles of spin $1/2$ (see § 32). For other spins the problem would therefore be of purely methodological interest.

Let us note that for all (integer and half-integer) spins $s > 1$ it proves impossible to formulate a variational principle using only a single function (tensorial or spinorial) whose rank matches the given spin. For this purpose it becomes necessary to introduce, as auxiliary quantities, tensorial or spinorial objects of lower rank as well. The Lagrangian is then chosen so that these auxiliary quantities automatically vanish by virtue of the free-particle field equations that follow from the variational principle [18].

§ 16. Helicity states of a particle[19]

Within relativistic theory, the orbital angular momentum $\mathbf{l}$ and the spin $\mathbf{s}$ of a particle in motion lack individual conservation laws—only their sum, the total angular momentum $\mathbf{j} = \mathbf{l} + \mathbf{s}$, is conserved. Consequently, the spin projection onto a given direction (the z axis) is also nonconserved, rendering it unsuitable for labeling the polarization (spin) states of a particle in motion.

What is conserved, however, is the projection of the spin on the direction of the momentum: since $\mathbf{l} = [\mathbf{rp}]$, the product $\mathbf{sn}$ coincides with the conserved product $\mathbf{jn}$ ($\mathbf{n} = \mathbf{p}/|\mathbf{p}|$). This quantity is called the *helicity* [20] (we have already considered it for the photon in § 8). Its eigenvalues will be denoted by λ ($\lambda = -s, \dots, +s$), and states of the particle with definite values of λ will be called helicity states.

Let $\psi_{\mathbf{p}\lambda}$ be the wave function (a plane wave) describing a state of the particle with definite $\mathbf{p}$ and λ, and let $u^{(\lambda)}(\mathbf{p})$ be its amplitude; for brevity we suppress the component indices of this function (for integer spin these are 4-tensor indices).

We saw in the preceding sections that a relativistic description of particles with nonzero (integer) spin requires introducing a wave function whose number of components exceeds $2s+1$. The number of independent components, however, remains equal to $2s+1$; the "extra" components are eliminated by imposing supplementary conditions, by virtue of which these components vanish in the rest frame (in the next chapter we shall see the same for half-integer s).

According to the transformation rules for angular momentum (see II, § 14), the helicity is invariant under Lorentz transformations that leave unchanged the direction of $\mathbf{p}$ onto which the angular momentum is projected. Therefore the number λ retains its meaning as a quantum number under such transformations, and to study the symmetry properties of helicity states one may use a reference frame in which $|\mathbf{p}| \ll m$ (in the limit, the rest frame). Then $\psi_{\mathbf{p}\lambda}$ reduces to a nonrelativistic $(2s + 1)$-component wave function. Let us denote its amplitude by $w^{(\lambda)}(\mathbf{n})$, indicating as the argument the direction $\mathbf{n} = \mathbf{p}/|\mathbf{p}|$ along which the angular momentum is quantized. The amplitude $w^{(\lambda)}$ is an eigenfunction

[18] See *Fierz M., Pauli W.*//Proc. Roy. Soc. — 1939. — V. A 173. — P. 211. In that work the program indicated here was carried out for particles of spin $3/2$ and 2.

[19] The content of this section applies to particles of any (integer or half-integer) spin.

[20] In the Russian literature the term is *spiralnost'*.

of the operator $\mathbf{n}\hat{\mathbf{s}}$:

$$\mathbf{n}\hat{\mathbf{s}}\, w^{(\lambda)}(\mathbf{n}) = \lambda w^{(\lambda)}(\mathbf{n}). \tag{16.1}$$

In the spinor representation $w^{(\lambda)}$ is a contravariant symmetric spinor of rank $2s$; by the correspondence formulas (57.2) (see III) its components can also be labeled by the values of the spin projection σ on a fixed axis z [21].

In the momentum representation the wave functions of the states under consideration essentially coincide with the amplitudes $u^{(\lambda)}(\mathbf{p})$. Specifically:

$$\psi_{\mathbf{p}\lambda}(\mathbf{k}) = u^{(\lambda)}(\mathbf{k})\delta^{(2)}(\boldsymbol{\nu} - \mathbf{n}) = u^{(\lambda)}(\mathbf{p})\delta^{(2)}(\boldsymbol{\nu} - \mathbf{n}), \tag{16.2}$$

where the momentum as an independent variable is denoted $\mathbf{k}$, in contrast to its eigenvalue $\mathbf{p}$, and $\boldsymbol{\nu} = \mathbf{k}/|\mathbf{k}|$, in contrast to $\mathbf{n} = \mathbf{p}/|\mathbf{p}|$ [22]. In the nonrelativistic limit

$$\psi_{\mathbf{n}\lambda}(\boldsymbol{\nu}) = w_\sigma^{(\lambda)}(\boldsymbol{\nu})\delta^{(2)}(\boldsymbol{\nu} - \mathbf{n}) = w^{(\lambda)}(\mathbf{n})\delta^{(2)}(\boldsymbol{\nu} - \mathbf{n}). \tag{16.3}$$

In more detail this expression should be written as

$$\psi_{\mathbf{n}\lambda}(\boldsymbol{\nu}, \sigma) = w_\sigma^{(\lambda)}(\boldsymbol{\nu})\delta^{(2)}(\boldsymbol{\nu} - \mathbf{n}),$$

where the discrete independent variable σ is also displayed explicitly.

The helicity operator $\hat{\mathbf{s}}\mathbf{n}$ is commutative with $\hat{j}_z$ and $\hat{\mathbf{j}}^2$. This follows from the fact that the angular-momentum operator corresponds to an infinitesimal coordinate rotation, while the dot product of any two vectors remains unchanged under arbitrary rotations. Consequently, stationary states exist in which the particle has simultaneously well-defined angular momentum j, its projection $j_z = m$, and helicity λ. We shall refer to such states as *spherical helicity states*.

We now find the wave functions of these states in the momentum representation. The construction proceeds by direct analogy with the expressions derived in III, § 103 for the wave functions of a symmetric top. Those results were obtained from the transformation formulas for wave functions under finite rotations (see III, § 58). Since that derivation relies exclusively on rotational-symmetry properties, it applies to momentum-representation functions just as it does to coordinate-space functions.

Along with the space-fixed coordinate system xyz (with respect to which the functions $\psi_{jm\lambda}$ are written), let us also introduce a "body-fixed" frame $\xi\eta\zeta$ with the ζ axis along the direction of $\boldsymbol{\nu}$. Without repeating the corresponding reasoning (cf. the derivation of formula (103.8) (see III)), we write

$$\psi_{jm\lambda}(\mathbf{k}) = \psi_{j\lambda}^{(0)} D_{\lambda m}^{(j)}(\boldsymbol{\nu}),$$

[21]The reasoning given here (together with the listing of allowed values of λ) pertains to particles whose mass differs from zero. For a particle of zero mass no rest frame exists, and the helicity admits only the two values $\lambda = \pm s$. This restriction traces back to the circumstance noted already in § 8: states of such a particle fall into representations of the axial-symmetry group, which permits at most a twofold degeneracy of levels (from the viewpoint of the wave equation this signifies that upon passing to the limit $m \to 0$ the system of equations for a spin-s particle separates into independent equations describing massless particles of spins s, $s-1, \dots$). In particular, for the photon $\lambda = \pm 1$, and the three-dimensional vectors $\mathbf{e}^{(\pm 1)}$ (8.2) assume the role of the corresponding amplitudes $w^{(\lambda)}$.

[22]Here the delta function $\delta^{(2)}$ is defined so that $\int \delta^{(2)}(\boldsymbol{\nu} - \mathbf{n})\, do_\nu = 1$. In (16.2) (and in the analogous case below, see (16.4)) the delta function ensuring the given value of the energy has been omitted.

where $\psi^{(0)}_{j\lambda}$ is the wave function in the "body-fixed" frame describing a state with a definite value of the ζ-projection of the angular momentum: $j_\zeta = \lambda$; in the momentum representation this function evidently coincides with the amplitude $u^{(\lambda)}$. The normalized (see below) wave function is

$$\psi_{jm\lambda}(\mathbf{k}) = \sqrt{\frac{2j+1}{4\pi}}\, D^{(j)}_{\lambda m}(\boldsymbol{\nu}) u^{(\lambda)}(\mathbf{k}). \tag{16.4}$$

Here, however, a question arises regarding the choice of phases, connected with the following ambiguity. The rotation of the system xyz relative to $\xi\eta\zeta$ is specified by three Euler angles α, β, γ; the direction of $\boldsymbol{\nu}$, on which alone the wave function of the particle can depend, is determined only by two spherical angles $\alpha \equiv \varphi$, $\beta \equiv \theta$. One must therefore adopt some convention for the angle γ. We shall set $\gamma = 0$, i.e. define $D^{(j)}_{\lambda m}(\boldsymbol{\nu})$ as

$$D^{(j)}_{\lambda m}(\boldsymbol{\nu}) = D^{(j)}_{\lambda m}(\varphi, \theta, 0) = e^{im\varphi} d^{(j)}_{\lambda m}(\theta). \tag{16.5}$$

By virtue of (58.21) (see III) the functions (16.5) satisfy the orthogonality and normalization conditions:

$$\int D^{(j_1)*}_{\lambda_1 m_1}(\boldsymbol{\nu}) D^{(j_2)}_{\lambda_2 m_2}(\boldsymbol{\nu}) \frac{do_\nu}{4\pi} = \frac{1}{2j+1}\,\delta_{j_1 j_2}\delta_{m_1 m_2} \tag{16.6}$$

($do_\nu = \sin\theta\, d\theta\, d\varphi$). Orthogonality in the index λ is guaranteed by the amplitude factor $u^{(\lambda)}$. The functions $\psi_{jm\lambda}$ are therefore mutually orthogonal, as required, across all three labels $jm\lambda$, and the prefactor adopted in (16.4) ensures their normalization according to

$$\int |\psi_{jm\lambda}|^2\, do_\nu = 1. \tag{16.7}$$

Here it is assumed that the amplitudes $u^{(\lambda)}$ are normalized to unity: $u^{(\lambda)}u^{(\lambda)*} = 1$.

Let us consider the behavior of helicity-state wave functions under coordinate inversion. The product of the polar vector $\boldsymbol{\nu}$ and the axial vector $\mathbf{j}$ is a pseudoscalar. It is therefore clear that upon inversion a state of helicity λ goes over into a state with $-\lambda$; one must, however, determine the phase factors in these transformations.

Under inversion $\boldsymbol{\nu} \to -\boldsymbol{\nu}$. The vector $\boldsymbol{\nu}$ is specified by two angles φ, θ, and the transformation $\boldsymbol{\nu} \to -\boldsymbol{\nu}$ is effected by the replacement $\varphi \to \varphi + \pi$, $\theta \to \pi - \theta$. This determines the axis ζ, but the position of the axes ξ and η, which also depends on the third Euler angle γ, remains unspecified; the transformation of θ and φ alone does not allow one to distinguish between an inversion and a rotation of the axis ζ in this sense. In terms of all three Euler angles the inversion is the transformation

$$\alpha \equiv \varphi \to \varphi + \pi, \quad \beta \equiv \theta \to \pi - \theta, \quad \gamma \to \pi - \gamma. \tag{16.8}$$

Therefore, if $D^{(j)}_{\lambda m}(\boldsymbol{\nu})$ is defined according to (16.5) (i.e. with $\gamma = 0$), and the substitution $\boldsymbol{\nu} \to -\boldsymbol{\nu}$ is understood as the result of an inversion, then

$$D^{(j)}_{\lambda m}(-\boldsymbol{\nu}) = D^{(j)}_{\lambda m}(\varphi + \pi, \pi - \theta, \pi). \tag{16.9}$$

Using formulas (58.9), (58.16), (58.18) (see III) we therefore find

$$D^{(j)}_{\lambda m}(-\boldsymbol{\nu}) = e^{i\lambda\pi} d^{(j)}_{\lambda m}(\pi - \theta) e^{im(\varphi+\pi)} =$$

$$= (-1)^{j-\lambda} e^{im\varphi} d^{(j)}_{-\lambda m}(\theta) = (-1)^{j-\lambda} D^{(j)}_{-\lambda m}(\boldsymbol{\nu}),$$

or

$$D^{(j)}_{\lambda m}(-\boldsymbol{\nu}) = (-1)^{-\lambda} D^{(j)}_{-\lambda m}(\boldsymbol{\nu}) \tag{16.10}$$

($j - \lambda$ is an integer).

An analogous formula for the spinor $w^{(\lambda)}$ can be obtained by noting that its components $w^{(\lambda)}_\sigma$ coincide, up to a factor, with the functions

$$w^{(\lambda)}_\sigma(\boldsymbol{\nu}) \propto D^{(s)}_{\lambda\sigma}(\boldsymbol{\nu})^*. \tag{16.11}$$

Indeed, applying the transformation formula (58.7) (see III) to the spin eigenfunctions and setting the ζ-projection of the spin to the definite value λ (i.e. replacing $\psi_{jm'}$ on the right-hand side of (58.7) (see III) by $\delta_{m'\lambda}$), we find that $D^{(s)}_{\lambda\sigma}(\boldsymbol{\nu})$ are spin wave functions corresponding to definite values of the z- and ζ-projections (σ and λ). The totality of these functions ($\sigma = -s, \ldots, +s$) forms (by the correspondence formulas (57.6) (see III)) a covariant spinor of rank $2s$. The components of the contravariant spinor (57.2) (see III), however—corresponding to the components $w^{(\lambda)}_\sigma$—transform as the complex conjugates of the components of a covariant spinor of the same rank. From (16.10),(16.11) we have

$$w^{(\lambda)}(-\boldsymbol{\nu}) = (-1)^{s-\lambda} w^{(-\lambda)}(\boldsymbol{\nu}) \tag{16.12}$$

($s - \lambda$ is an integer). The inversion operation applied to $w^{(\lambda)}$, however, consists not only in the replacement $\boldsymbol{\nu} \to -\boldsymbol{\nu}$, but also in multiplication by an overall phase factor (the "intrinsic parity" of the particle), which we denote by η:

$$\hat{P} w^{(\lambda)}(\boldsymbol{\nu}) = \eta w^{(\lambda)}(-\boldsymbol{\nu}) = \eta(-1)^{s-\lambda} w^{(-\lambda)}(\boldsymbol{\nu}). \tag{16.13}$$

For the relativistic amplitude $u^{(\lambda)}(\mathbf{k})$ this transformation takes the form

$$\hat{P} u^{(\lambda)}(\mathbf{k}) = \eta\beta u^{(\lambda)}(-\mathbf{k}) = \eta(-1)^{s-\lambda} u^{(-\lambda)}(\mathbf{k}), \tag{16.14}$$

where β is a certain matrix that acts as the identity on the components of $u^{(\lambda)}$ surviving in the limit $|\mathbf{p}| \to 0$. It is important that this matrix is independent of the quantum numbers of the state, so that the difference between (16.13) and (16.14) is in this sense inessential [24].

Applying (16.14) to (16.2), we obtain the transformation law for the wave functions of the states $|\mathbf{n}\lambda\rangle$:

$$\hat{P}\psi_{\mathbf{n}\lambda}(\boldsymbol{\nu}) = \eta(-1)^{s-\lambda}\psi_{-\mathbf{n}\,-\lambda}(\boldsymbol{\nu}). \tag{16.15}$$

For the spherical helicity states, making use of (16.10) and (16.12), we obtain the transformation law:

$$\hat{P}\psi_{jm\lambda}(\boldsymbol{\nu}) = \eta(-1)^{j-s}\psi_{jm\,-\lambda}(\boldsymbol{\nu}). \tag{16.16}$$

[24] Thus, for $s = 1$ the amplitudes $u^{(\lambda)}$ are 4-vectors (16.22); in this case β is the full identity matrix with respect to 4-vector indices: $\beta_{\mu\nu} = \delta_{\mu\nu}$. For $s = 1/2$, $u^{(\lambda)}$ is a bispinor; the phase factor is then $\eta = i$, and β is the Dirac matrix γ^0 (see (21.10)).

The states ψ_{jm0} transform, according to (16.16), into themselves, i.e. they possess a definite parity. If, however, $\lambda \neq 0$, then only superpositions of states with opposite helicities have a definite parity:

$$\psi^{(\pm)}_{jm|\lambda|} = \frac{1}{\sqrt{2}}\left(\psi_{jm\lambda} \pm \psi_{jm-\lambda}\right). \tag{16.17}$$

Under inversion they transform into themselves according to

$$\hat{P}\psi^{(\pm)}_{jm|\lambda|}(\boldsymbol{\nu}) = \pm\eta(-1)^{j-s}\psi^{(\pm)}_{jm|\lambda|}(\boldsymbol{\nu}). \tag{16.18}$$

Let us draw attention to the fact that in this section we have classified the states of a free particle with given angular momentum by operating solely with conserved quantities, without appealing to the concept of orbital angular momentum (which was used, for instance, in §§ 6, 7 for the classification of photon states).

As an illustration consider the case of spin 1. In the rest frame the amplitudes $u^{(\lambda)}$ (4-vectors) reduce to three-dimensional vectors $\mathbf{e}^{(\lambda)}$, which play here the role of the amplitudes $w^{(\lambda)}$. The action of the spin-1 operator on a vector function $\mathbf{e}$ is given by

$$(\hat{s}_i\mathbf{e})_k = -ie_{ikl}e_l \tag{16.19}$$

(see III, § 57, Problem 2). Equation (16.1) therefore assumes the form

$$i[\mathbf{n}\mathbf{e}^{(\lambda)}] = \lambda\mathbf{e}^{(\lambda)}. \tag{16.20}$$

Its solutions (in a coordinate frame $\xi\eta\zeta$ with the ζ axis along $\mathbf{n}$) coincide with the circular unit vectors (7.14) [26]:

$$\mathbf{e}^{(0)} = i(0,0,1), \qquad \mathbf{e}^{(\pm 1)} = \mp\frac{i}{\sqrt{2}}(1, \pm i, 0). \tag{16.21}$$

In a frame where the particle has momentum $\mathbf{p}$, the helicity-state amplitudes are the 4-vectors

$$u^{(0)\mu} = \left(\frac{|\mathbf{p}|}{m}, \frac{\varepsilon}{m}\mathbf{e}^{(0)}\right), \quad u^{(\pm 1)\mu} = (0, \mathbf{e}^{(\pm 1)}). \tag{16.22}$$

If $\mathbf{e}$ is a polar vector, then $\eta = -1$. In that case the functions (16.17) (which for $s = 1$ are three-dimensional vectors) have the following parities:

$$\psi^{(+)}_{jm|\lambda|}: \quad P = (-1)^j,$$

$$\psi^{(-)}_{jm|\lambda|}: \quad P = (-1)^{j+1},$$

$$\psi_{jm0}: \quad P = (-1)^j.$$

Comparing with the definition of vector spherical harmonics (7.4), we see that these functions are identical (up to phase factors) with $\mathbf{Y}^{(\mathrm{e})}_{jm}$, $\mathbf{Y}^{(\mathrm{m})}_{jm}$, $\mathbf{Y}^{(\mathrm{l})}_{jm}$ respectively.

[26] The choice of phase factors is fixed by the requirement that the matrix elements of the spin operators computed with the eigenfunctions (16.21) conform to the general definitions in III, §§ 27, 107.

Determining the phase factors (say, by comparing the values at $\theta = 0$), we arrive at the equalities

$$\mathbf{Y}^{(\mathrm{e})}_{jm} = i^{j-1}\sqrt{\frac{2j+1}{8\pi}}\left(\mathbf{e}^{(1)}D^{(j)}_{1m}(\boldsymbol{\nu}) + \mathbf{e}^{(-1)}D^{(j)}_{-1m}(\boldsymbol{\nu})\right),$$

$$\mathbf{Y}^{(\mathrm{m})}_{jm} = i^{j-1}\sqrt{\frac{2j+1}{8\pi}}\left(\mathbf{e}^{(1)\prime}D^{(j)}_{1m}(\boldsymbol{\nu}) + \mathbf{e}^{(-1)\prime}D^{(j)}_{-1m}(\boldsymbol{\nu})\right), \tag{16.23}$$

$$\mathbf{Y}^{(\mathrm{l})}_{jm} = i^{j-1}\sqrt{\frac{2j+1}{4\pi}}\,\mathbf{e}^{(0)}D^{(j)}_{0m}$$

(j is an integer!); $\mathbf{e}^{(\lambda)\prime} = [\mathbf{n}\mathbf{e}^{(\lambda)}]$ are circular unit vectors in axes $\xi'\eta'\zeta$ that are rotated relative to $\xi\eta\zeta$ by 90° about the ζ axis.

The last of formulas (16.23) is equivalent to the expression (58.23) (see III) for $d^{(j)}_{0m}(\theta)$. From the first (or the second) formula one can obtain a simple expression for the functions $d^{(j)}_{\pm 1m}$. We have

$$i^{j-1}\sqrt{\frac{2j+1}{8\pi}}\,D^{(j)}_{\pm 1m} = \mathbf{Y}^{(\mathrm{e})}_{jm}\mathbf{e}^{(\pm 1)*} = \frac{1}{\sqrt{j(j+1)}}\,\mathbf{e}^{(\pm 1)*}\boldsymbol{\nabla}Y_{jm}.$$

We expand the scalar product on the right-hand side in the frame $\xi\eta\zeta$, where

$$\left(\frac{\partial}{\partial\xi}, \frac{\partial}{\partial\eta}\right) \to \left(\frac{\partial}{\partial\theta}, \frac{1}{\sin\theta}\frac{\partial}{\partial\varphi}\right).$$

Recalling definition (7.2) of the function Y_{jm} and definition (16.5), we obtain as the final result

$$d^{(j)}_{\pm 1m}(\theta) = (-1)^{m+1}\sqrt{\frac{(j-m)!}{(j+m)!\,j(j+1)}}\left(\pm\frac{\partial}{\partial\theta} + \frac{m}{\sin\theta}\right)P^m_j(\cos\theta), \quad m \geqslant 0. \tag{16.24}$$

CHAPTER III

Fermions

§ 17. Four-dimensional spinors

In nonrelativistic theory a particle of arbitrary spin s is described by a $(2s+1)$-component quantity—a symmetric spinor of rank $2s$. From the mathematical standpoint these are quantities that furnish the irreducible representations of the spatial rotation group.

In relativistic theory this group appears only as a subgroup of the broader group of four-dimensional rotations—the Lorentz group. In connection with this there arises the need to construct a theory of four-dimensional spinors (4-spinors)—quantities that carry the irreducible representations of the Lorentz group; the exposition of this theory occupies §§ 17–19. In §§ 17, 18 only the proper Lorentz group is considered, which does not include spatial inversion; the latter will be treated in § 19.

The construction of the 4-spinor theory proceeds along the same lines as the three-dimensional spinor theory (*B. L. van der Waerden*, 1929; *G. E. Uhlenbeck*, *O. Laporte*, 1931).

A spinor ξ^α is a two-component quantity ($\alpha = 1, 2$); regarded as components of the wave function of a spin-$1/2$ particle, ξ^1 and ξ^2 correspond to eigenvalues of the z-projection of the spin equal to $+1/2$ and $-1/2$ respectively. Under any (proper) Lorentz transformation the two quantities ξ^1, ξ^2 transform through each other:

$$\xi^{1\prime} = \alpha\xi^1 + \beta\xi^2, \qquad \xi^{2\prime} = \gamma\xi^1 + \delta\xi^2. \tag{17.1}$$

The coefficients α, β, γ, δ are definite functions of the rotation angles of the 4-coordinate system, subject to the condition

$$\alpha\delta - \beta\gamma = 1, \tag{17.2}$$

i.e. the determinant of the binary transformation (17.1) equals 1, just as the determinants of coordinate transformations in the Lorentz group do.

By virtue of conditions (17.2) the bilinear form $\xi^1\Xi^2 - \xi^2\Xi^1$ (where ξ^α and Ξ^α are two spinors) is invariant under the transformation (17.1) (it corresponds to a spin-0 particle "composed" of two spin-$1/2$ particles). For a natural way of writing such invariant expressions, alongside the "contravariant" components of the spinor ξ^α one also introduces "covariant" components ξ_α. The passage from one set to the other is carried out with the aid of the "spinor metric" $g_{\alpha\beta}$ [1]:

$$\xi_\alpha = g_{\alpha\beta}\xi^\beta, \tag{17.3}$$

where

$$g = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \tag{17.4}$$

[1] Spinor indices will be denoted by the first letters of the Greek alphabet: $\alpha, \beta, \gamma, \ldots$.

so that

$$\xi_1 = \xi^2, \qquad \xi_2 = -\xi^1. \tag{17.5}$$

The invariant $\xi^1\Xi^2 - \xi^2\Xi^1$ is then written as a scalar product $\xi^\alpha\Xi_\alpha$. Note that $\xi^\alpha\Xi_\alpha = -\xi_\alpha\Xi^\alpha$.

Up to this point the properties listed above have formally coincided with those of three-dimensional spinors. A difference arises, however, when one turns to complex-conjugate spinors.

In nonrelativistic theory the sum

$$\psi^1\psi^{1*} + \psi^2\psi^{2*}, \tag{17.6}$$

which governs the probability density of finding particles at a given point in space, had to be a scalar, and for this purpose the components $\psi^{\alpha*}$ had to transform as covariant spinor components; put differently, the transformation (17.1) had to be unitary ($\alpha = \delta^*$, $\beta = -\gamma^*$). In the relativistic case, however, the particle density does not behave as a scalar; it forms the time component of a 4-vector. Owing to this, the requirement just stated ceases to hold, and no further restrictions on the transformation coefficients arise (apart from (17.2)). Four complex quantities $\alpha, \beta, \gamma, \delta$ constrained only by (17.2) amount to $8 - 2 = 6$ real parameters—in agreement with the number of angles specifying a 4-coordinate rotation (rotations in six coordinate planes).

Consequently, complex-conjugate binary transformations prove to be essentially different from one another, and in relativistic theory two varieties of spinors arise. To distinguish between them one adopts special notation: the indices of spinors whose transformation law is the complex conjugate of (17.1) are written as digits with dots placed above them (*dotted indices*). Thus, by definition,

$$\eta^{\dot\alpha} \sim \xi^{\alpha*}, \tag{17.7}$$

where the sign $\sim$ stands for "transforms as." In other words, the transformation formulas for a "dotted" spinor are:

$$\eta^{\dot 1'} = \alpha^*\eta^{\dot 1} + \beta^*\eta^{\dot 2}, \qquad \eta^{\dot 2'} = \gamma^*\eta^{\dot 1} + \delta^*\eta^{\dot 2}. \tag{17.8}$$

The operations of lowering and raising dotted indices are performed in the same way as for undotted ones:

$$\eta_{\dot 1} = \eta^{\dot 2}, \qquad \eta_{\dot 2} = -\eta^{\dot 1}. \tag{17.9}$$

With respect to spatial rotations the behavior of 4-spinors coincides with that of 3-spinors. For the latter, as we know, $\psi^*_\alpha \sim \psi^\alpha$. By virtue of definition (17.7) the 4-spinor $\eta_{\dot\alpha}$ therefore behaves under rotations as a contravariant 3-spinor ψ^α. The eigenvalues of the spin projection $1/2$ and $-1/2$ accordingly correspond to the covariant components $\eta_{\dot 1}$ and $\eta_{\dot 2}$.

Spinors of higher rank are defined as sets of quantities that transform like products of components of several first-rank spinors. The indices of a higher-rank spinor may include both dotted and undotted ones. For instance, three types of second-rank spinors exist:

$$\xi^{\alpha\beta} \sim \xi^\alpha\Xi^\beta, \qquad \zeta^{\alpha\dot\beta} \sim \xi^\alpha\eta^{\dot\beta}, \qquad \eta^{\dot\alpha\dot\beta} \sim \eta^{\dot\alpha}H^{\dot\beta}.$$

Merely stating the total rank of a spinor is therefore insufficient for an unambiguous definition of the concept; when necessary we shall specify the rank as a pair of numbers (k, l)—the number of undotted and the number of dotted indices.

Since the transformations (17.1) and (17.8) are algebraically independent, there is no need to fix the ordering of dotted and undotted indices (in this sense, for example, the spinors $\zeta^{\alpha\dot{\beta}}$ and $\zeta^{\dot{\beta}\alpha}$ are one and the same).

For an expression involving spinors to have an invariant character, it must carry the same number of undotted and dotted indices on both sides; otherwise the expression will certainly be violated upon passing from one reference frame to another. One should keep in mind here that complex conjugation entails the interchange of dotted and undotted indices. Therefore the relation $\eta^{\dot{\alpha}\dot{\beta}} = \left(\xi^{\alpha\beta}\right)^*$ between two spinors does have an invariant character.

Contraction of spinors or their products can be performed only over pairs of indices of the same kind—two dotted or two undotted. Summation over a pair of indices of different kinds is not an invariant operation. For this reason, from the spinor

$$\zeta^{\alpha_1\alpha_2\ldots\alpha_k\dot{\beta}_1\dot{\beta}_2\ldots\dot{\beta}_l}, \tag{17.10}$$

which is symmetric in all k undotted and in all l dotted indices, it is impossible to form a spinor of lower rank (recall that contracting over a pair of indices in which the spinor is symmetric gives zero). This means that from the quantities (17.10) one cannot assemble any smaller set of their linear combinations that would transform among themselves under every group transformation. Stated differently, symmetric 4-spinors realize the irreducible representations of the proper Lorentz group. Every such representation is labeled by a pair of numbers (k, l).

Since each spinorial index runs over two values, there are $k + 1$ essentially distinct sets of numbers $\alpha_1\alpha_2\ldots\alpha_k$ in (17.10) (containing $0, 1, 2, \ldots, k$ ones and $k, k-1, \ldots, 0$ twos) and $l + 1$ sets of numbers $\dot{\beta}_1\dot{\beta}_2\ldots\dot{\beta}_l$. Altogether, therefore, a symmetric spinor of rank (k, l) has $(k + 1)(l + 1)$ independent components; this is the dimensionality of the irreducible representation it carries.

§ 18. Relation of spinors to 4-vectors

A spinor $\zeta^{\alpha\dot{\beta}}$ carrying one dotted and one undotted index possesses $2 \cdot 2 = 4$ independent components—precisely the number of components belonging to a 4-vector. It is therefore evident that both objects furnish one and the same irreducible representation of the proper Lorentz group, and a definite correspondence must hold between their respective components.

To establish this correspondence we turn first of all to the analogous correspondence in the three-dimensional case, bearing in mind that with respect to purely spatial rotations the behavior of 3- and 4-spinors must be identical.

For a three-dimensional spinor $\psi^{\alpha\beta}$ the correspondence formulas hold (see III, § 57), which we write here in the form

$$
\begin{aligned}
a_x &= \tfrac{1}{2}(\psi^{22} - \psi^{11}) = \tfrac{1}{2}(\psi^2_1 + \psi^1_2),\\
a_y &= -\tfrac{i}{2}(\psi^{22} + \psi^{11}) = \tfrac{i}{2}(\psi^1_2 - \psi^2_1),\\
a_z &= \tfrac{1}{2}(\psi^{12} + \psi^{21}) = \tfrac{1}{2}(\psi^1_1 - \psi^2_2),
\end{aligned}
\tag{III.1}
$$

where a_x, a_y, a_z are the components of a certain three-dimensional vector **a**. In passing to the four-dimensional case one must replace the components ψ^α_β by $\zeta^{\alpha\dot\beta}$, and interpret a_x, a_y, a_z as the contravariant components a^1, a^2, a^3 of a 4-vector. As for the expression for the fourth component, a^0, its form is already clear from the circumstance noted in § 17: the quantity (17.6) must transform as a^0. Hence $a^0 \sim \zeta^{1\dot1} + \zeta^{2\dot2}$; the proportionality coefficient is fixed by requiring that the scalar $\zeta_{\alpha\dot\beta}\zeta^{\alpha\dot\beta}$ coincide with $2a_\mu a^\mu \equiv 2a^2$.

We thus arrive at the following correspondence formulas:

$$
\begin{aligned}
a^1 &= \tfrac{1}{2}(\zeta^{1\dot2} + \zeta^{2\dot1}), & a^2 &= \tfrac{i}{2}(\zeta^{1\dot2} - \zeta^{2\dot1}),\\
a^3 &= \tfrac{1}{2}(\zeta^{1\dot1} - \zeta^{2\dot2}), & a^0 &= \tfrac{1}{2}(\zeta^{1\dot1} + \zeta^{2\dot2}).
\end{aligned}
\tag{18.1}
$$

The inverse formulas are:

$$
\begin{aligned}
\zeta^{1\dot1} &= \zeta_{2\dot2} = a^3 + a^0, & \zeta^{2\dot2} &= \zeta_{1\dot1} = a^0 - a^3,\\
\zeta^{1\dot2} &= -\zeta_{2\dot1} = a^1 - ia^2, & \zeta^{2\dot1} &= -\zeta_{1\dot2} = a^1 + ia^2.
\end{aligned}
\tag{18.2}
$$

Here

$$
\zeta_{\alpha\dot\beta}\zeta^{\alpha\dot\beta} = 2a^2. \tag{18.3}
$$

We note also that

$$
\zeta_{\alpha\dot\beta}\zeta^{\gamma\dot\beta} = \delta^\gamma_\alpha\, a^2. \tag{18.4}
$$

To see why this last relation holds, observe that the quantity $\zeta_{\alpha\dot\beta}\zeta^{\dot\beta}_{\gamma}$, viewed as a second-rank spinor in the indices α and γ, is antisymmetric and hence must be proportional to the metric spinor.

The link between $\zeta^{\alpha\dot\beta}$ and a 4-vector is actually one instance of a broader principle: any symmetric spinor of rank (k, k) can be placed in one-to-one correspondence with an irreducible symmetric 4-tensor of rank k (irreducibility meaning that contraction over every index pair gives zero).

The relationship linking a spinor to a 4-vector can be written compactly with the aid of the two-by-two Pauli matrices [2]:

$$
\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{18.5}
$$

Denoting symbolically by ζ the matrix of the quantities $\zeta^{\alpha\dot\beta}$ with upper indices (the first being undotted), formulas (18.2) take the form

$$
\zeta = \mathbf{a}\boldsymbol{\sigma} + a^0 \tag{18.6}
$$

[2] For brevity of notation, operators (matrices) that act on spin variables will be denoted by letters without carets.

(in the second term one understands, of course, the product of a^0 with the unit matrix). The inverse relations are:

$$\mathbf{a} = \tfrac{1}{2}\,\mathrm{Sp}(\zeta\boldsymbol{\sigma}), \qquad a^0 = \tfrac{1}{2}\,\mathrm{Sp}\,\zeta. \tag{18.7}$$

With the help of formulas (18.6), (18.7) one can establish the connection between the transformation laws for the 4-vector and the spinor, thereby expressing the spinor transformation law through the parameters of rotations of the 4-coordinate system.

Let us write the transformation of the spinor ξ^α in the form

$$\xi^{\alpha\prime} = (B\xi)^\alpha, \qquad B = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \tag{18.8}$$

where B is a two-by-two matrix assembled from the coefficients of the binary transformation. The transformation of the dotted spinor is then:

$$\eta^{\dot\beta\prime} = (B^*\eta)^{\dot\beta} = (\eta B^+)^{\dot\beta}, \tag{18.9}$$

and the transformation of the second-rank spinor $\zeta^{\alpha\dot\beta} \sim \xi^\alpha\eta^{\dot\beta}$ can be written symbolically as $\zeta' = B\zeta B^{+1}$). For an infinitesimally small transformation $B = 1 + \lambda$, where λ is a small matrix, and to first-order accuracy

$$\zeta' = \zeta + (\lambda\zeta + \zeta\lambda^+). \tag{18.10}$$

Consider first the Lorentz transformation to a reference frame moving with an infinitesimally small velocity $\delta\mathbf{V}$ (without altering the orientation of the spatial axes). Under this transformation the 4-vector $a^\mu = (a^0, \mathbf{a})$ changes according to

$$\mathbf{a}' = \mathbf{a} - a^0\delta\mathbf{V}, \qquad a^{0\prime} = a^0 - \mathbf{a}\delta\mathbf{V}. \tag{18.11}$$

Let us now make use of formulas (18.7). The transformation of a^0 can be expressed, on the one hand, as

$$a^{0\prime} = a^0 - \mathbf{a}\delta\mathbf{V} = a^0 - \tfrac{1}{2}\,\mathrm{Sp}(\zeta\boldsymbol{\sigma}\,\delta\mathbf{V}),$$

and on the other hand as

$$a^{0\prime} = \tfrac{1}{2}\,\mathrm{Sp}\,\zeta' = a^0 + \tfrac{1}{2}\,\mathrm{Sp}\,\zeta\,(\lambda + \lambda^+).$$

These expressions must coincide identically (i.e. for arbitrary ζ). From this we obtain the equality:

$$\lambda + \lambda^+ = -\boldsymbol{\sigma}\,\delta\mathbf{V}.$$

By the same method, examining the transformation of $\mathbf{a}$, we find

$$\boldsymbol{\sigma}\lambda + \lambda^+\boldsymbol{\sigma} = -\delta\mathbf{V}.$$

These equalities, regarded as equations for λ, yield the solution:

$$\lambda = \lambda^+ = -\tfrac{1}{2}\boldsymbol{\sigma}\,\delta\mathbf{V}.$$

Thus the infinitesimally small Lorentz transformation of the spinor ξ^α is effected by the matrix

$$B = 1 - \tfrac{1}{2}(\boldsymbol{\sigma}\mathbf{n})\,\delta V, \tag{18.12}$$

where $\mathbf{n}$ is a unit vector along the velocity $\delta\mathbf{V}$. From this it is straightforward to obtain the transformation for a finite velocity $\mathbf{V}$. To do so, recall that the Lorentz transformation amounts (geometrically) to a rotation of the 4-coordinate system in the $t\,\mathbf{n}$ plane through an angle φ related to the velocity $\mathbf{V}$ by $\operatorname{th}\varphi = V$[3]). An infinitesimally small transformation corresponds to the angle $\delta\varphi = \delta V$, while a rotation through a finite angle φ is achieved by repeating the rotation through $\delta\varphi$ a total of $\varphi/\delta\varphi$ times. Raising the operator (18.12) to the power $\varphi/\delta\varphi$ and passing to the limit $\delta\varphi \to 0$, we obtain

$$B = e^{-\frac{\varphi}{2}\,\mathbf{n}\boldsymbol{\sigma}}. \tag{18.13}$$

The mathematical meaning of this operator becomes transparent once one observes that, by the properties of the Pauli matrices, all even powers of $\mathbf{n}\boldsymbol{\sigma}$ equal 1 while all odd powers equal $\mathbf{n}\boldsymbol{\sigma}$. Noting that ch decomposes into even powers and sh into odd powers of its argument, we arrive at the final result

$$B = \operatorname{ch}\frac{\varphi}{2} - \mathbf{n}\boldsymbol{\sigma}\operatorname{sh}\frac{\varphi}{2}, \qquad \operatorname{th}\varphi = V. \tag{18.14}$$

We note that the Lorentz-transformation matrices B turn out to be Hermitian: $B = B^+$.

Consider now an infinitesimally small rotation of the spatial coordinate system. Under such a rotation the three-dimensional vector $\mathbf{a}$ transforms according to

$$\mathbf{a}' = \mathbf{a} - [\delta\theta\,\mathbf{a}], \tag{18.15}$$

where $\delta\boldsymbol{\theta}$ is the vector of the infinitesimally small rotation angle. The corresponding spinor transformation could be found in an analogous manner. There is no need for this, however, since under spatial rotations the behavior of 4-spinors coincides with that of 3-spinors, and for the latter the transformation is known beforehand from the general connection between the spin operator and the operator of an infinitesimally small rotation:

$$B = 1 + \tfrac{i}{2}\boldsymbol{\sigma}\,\delta\boldsymbol{\theta}. \tag{18.16}$$

The passage to a rotation through a finite angle θ is carried out in the same way as the passage from (18.12) to (18.14):

$$B = \exp\!\left(\frac{i\theta}{2}\,\mathbf{n}\boldsymbol{\sigma}\right) = \cos\frac{\theta}{2} + i\mathbf{n}\boldsymbol{\sigma}\sin\frac{\theta}{2}, \tag{18.17}$$

where $\mathbf{n}$ is the unit vector along the rotation axis. This matrix is unitary ($B^+ = B^{-1}$), as must be the case for a spatial rotation.

[3]We recall that in planes containing the time axis the metric is pseudo-Euclidean.

§ 19. Inversion of spinors

When developing the three-dimensional spinor theory (in Vol. III) we did not consider how spinors behave with respect to the spatial inversion operation, inasmuch as in the nonrelativistic framework this would not have produced any new physical consequences. It is worth pausing on this matter now, however, for a clearer understanding of the inversion properties of 4-spinors that will be discussed below.

The inversion operation does not alter the sign of an axial vector, and the spin vector is precisely of this type. Consequently the value of its projection s_z remains unchanged as well. It follows that under inversion each component of the spinor ψ^α can transform only into itself, i.e. one must have

$$\psi^\alpha \to P\psi^\alpha, \tag{19.1}$$

where P is a constant coefficient. Applying inversion twice returns us to the original coordinate system. For spinors, however, the return to the initial configuration can be understood in two distinct senses: as a rotation of the system by 0° or by 360°. For spinors these two interpretations are not equivalent, since ψ^α reverses sign under a 360° rotation. Two alternative definitions of inversion therefore arise: in the first,

$$P^2 = 1, \qquad P = \pm 1, \tag{19.2}$$

and in the second,

$$P^2 = -1, \qquad P = \pm i. \tag{19.3}$$

A crucial requirement is that inversion be defined in the same manner for every spinor. It would be inconsistent for distinct spinors to follow different inversion rules (one obeying (19.2), another (19.3)), for then one could not always build a scalar (or pseudoscalar) out of an arbitrary pair of spinors: if ψ^α were governed by (19.2) while φ^α were governed by (19.3), the product $\psi^\alpha\varphi_\alpha$ would pick up a factor $\pm i$ upon inversion instead of staying fixed (or simply flipping sign).

It should also be stressed that (under either definition of inversion) the assignment of one or another parity P to a spinor carries no absolute significance, because spinors change sign under a rotation through 2π, which can always be performed simultaneously with the inversion. What does possess an absolute character, however, is the “relative parity” of two spinors, defined as the parity of the scalar $\psi^\alpha\varphi_\alpha$ formed from them; since a 2π rotation reverses the sign of all spinors simultaneously, the associated ambiguity does not affect the parity of the scalar in question.

Let us turn now to four-dimensional spinors.

We note first that since inversion changes the sign of only three (x, y, z) out of the four (t, x, y, z) coordinates, it commutes with spatial rotations but does not commute with transformations that rotate the t axis. If $\widehat{L}$ denotes the Lorentz transformation to a frame moving with velocity $\mathbf{V}$, then $\widehat{PL} = \widehat{L}'\widehat{P}$, where $\widehat{L}'$ is the transformation to a frame moving with velocity $-\mathbf{V}$.

It follows from this that under inversion the components of a 4-spinor ξ^α cannot transform through themselves alone. If the inversion of the spinor $\xi^{\dot\alpha}$ were still given by the transformation (19.1) (i.e. were represented by a matrix proportional to the identity), it would commute with all Lorentz transformations whatsoever, which certainly must not be the case (since the operations $\widehat{L}$ and $\widehat{L}'$ applied to ξ^α manifestly do not coincide).

Inversion must therefore map the components of the spinor ξ^α into other quantities. These can only be the components of some other spinor $\eta_{\dot\alpha}$ whose transformation properties differ from those of ξ^α. Since inversion leaves the z-projection of the spin unchanged (as already noted above), the components ξ^1 and ξ^2 can pass under inversion only into $\eta_{\dot 1}$ and $\eta_{\dot 2}$, which correspond to the same values $s_z = 1/2$ and $s_z = -1/2$. Taking inversion to be an operation that yields 1 when applied twice, one can specify its action by the formulas

$$\xi^\alpha \to \eta_{\dot\alpha}, \qquad \eta_{\dot\alpha} \to \xi^\alpha. \tag{19.4}$$

For the covariant components ξ_α and the contravariant $\eta^{\dot\alpha}$ these transformations carry the opposite sign:

$$\xi_\alpha \to -\eta^{\dot\alpha}, \qquad \eta^{\dot\alpha} \to -\xi_\alpha, \tag{19.4a}$$

since lowering and raising one and the same index are performed with opposite signs, see (17.5) and (17.9)[4].

A certain distinction between the two definitions of inversion lies in the fact that under the second definition complex-conjugate spinors transform identically: if $\Xi_\alpha = \eta^*_{\dot\alpha}$, $H^{\dot\alpha} = \xi^{\alpha *}$, then from (19.5) one obtains $\Xi_\alpha \to -iH^{\dot\alpha}$, $H^{\dot\alpha} \to -i\Xi_\alpha$, i.e. the same rule as for $\xi_\alpha, \eta^{\dot\alpha}$. Under definition (19.4), by contrast, one would obtain the transformation $\Xi_\alpha \to H^{\dot\alpha}$, $H^{\dot\alpha} \to \Xi_\alpha$, which is the reverse in sign of the transformation of the spinors $\xi_\alpha, \eta^{\dot\alpha}$. We shall return to possible physical aspects of this distinction in § 27.

In what follows we shall everywhere adopt definition (19.5) for definiteness.

Under the rotation subgroup the spinors ξ^α and $\eta_{\dot\alpha}$ transform, as we know, in the same way. Forming from their components the combinations

$$\xi^\alpha \pm \eta_{\dot\alpha}, \tag{19.6}$$

one would obtain quantities transforming under inversion according to (19.1) with $P = \pm i$. These combinations, however, do not behave as spinors with respect to all transformations of the Lorentz group.

Incorporating inversion into the symmetry group therefore demands that one treat a pair of spinors $(\xi^\alpha, \eta_{\dot\alpha})$ jointly; such a pair goes by the name of a first-rank *bispinor*. The four components of a bispinor realize one of the irreducible representations that belong to the extended Lorentz group.

Given two bispinors $(\xi^\alpha, \eta_{\dot\alpha})$ and $(\Xi^\alpha, H_{\dot\alpha})$, their scalar product admits two distinct constructions. The expression

$$\xi^\alpha \Xi_\alpha + \eta_{\dot\alpha} H^{\dot\alpha} \tag{19.7}$$

---

[4] If instead inversion is understood in the sense that $P^2 = -1$, its action is specified by the formulas

$$\xi^\alpha \to i\eta_{\dot\alpha}, \qquad \eta_{\dot\alpha} \to i\xi^\alpha \tag{19.5}$$

or, equivalently,

$$\xi_\alpha \to -i\eta^{\dot\alpha}, \qquad \eta^{\dot\alpha} \to -i\xi_\alpha. \tag{19.5a}$$

is altogether unchanged by inversion, i.e. it constitutes a true scalar. The quantity

$$\xi^{\alpha}\Xi_{\alpha} - \eta_{\dot{\alpha}}H^{\dot{\alpha}} \tag{19.8}$$

is likewise invariant under rotations of the 4-coordinate system but reverses sign under inversion; in other words, it is a pseudoscalar.

A second-rank spinor $\zeta^{\alpha\dot{\beta}}$ can also be defined in two ways. Specifying its transformation law as

$$\zeta^{\alpha\dot{\beta}} \sim \xi^{\alpha}H^{\dot{\beta}} + \Xi^{\alpha}\eta^{\dot{\beta}}, \tag{19.9}$$

one obtains quantities that transform under inversion according to

$$\zeta^{\alpha\dot{\beta}} \to \zeta_{\dot{\alpha}\beta}. \tag{19.10}$$

The 4-vector a^{μ} to which such a spinor is equivalent undergoes, by virtue of (18.1), the mapping $(a^0, \mathbf{a}) \to (a^0, -\mathbf{a})$, so it constitutes a genuine 4-vector (while the spatial part $\mathbf{a}$ is polar).

One may, however, also define $\zeta^{\alpha\dot{\beta}}$ according to

$$\zeta^{\alpha\dot{\beta}} \sim \xi^{\alpha}H^{\dot{\beta}} - \Xi^{\alpha}\eta^{\dot{\beta}}. \tag{19.11}$$

Then [5]

$$\zeta^{\alpha\dot{\beta}} \to -\zeta_{\dot{\alpha}\beta}. \tag{19.12}$$

The 4-vector corresponding to such a spinor undergoes the inversion transformation $(a^0, \mathbf{a}) \to (-a^0, \mathbf{a})$, i.e. it is a 4-pseudovector (while the three-dimensional vector $\mathbf{a}$ is axial).

Symmetric second-rank spinors whose indices are of the same type obey the transformation laws:

$$\xi^{\alpha\beta} \sim \xi^{\alpha}\Xi^{\beta} + \xi^{\beta}\Xi^{\alpha}, \qquad \eta_{\dot{\alpha}\dot{\beta}} \sim \eta_{\dot{\alpha}}H_{\dot{\beta}} + \eta_{\dot{\beta}}H_{\dot{\alpha}}. \tag{19.13}$$

Under inversion they pass into one another:

$$\xi^{\alpha\beta} \to -\eta_{\dot{\alpha}\dot{\beta}}. \tag{19.14}$$

The pair $(\xi^{\alpha\beta}, \eta_{\dot{\alpha}\dot{\beta}})$ constitutes a second-rank bispinor. Its number of independent components equals $3 + 3 = 6$. An antisymmetric second-rank 4-tensor $a^{\mu\nu}$ possesses the same number of independent components. A definite correspondence must therefore exist between them (both furnish equivalent irreducible representations of the extended Lorentz group).

Since the spinors $\xi^{\alpha\beta}$ and $\eta_{\dot{\alpha}\dot{\beta}}$ transform independently under the proper Lorentz group, the components of the 4-tensor $a^{\mu\nu}$ can likewise be partitioned into two groups of quantities that mix only among themselves under all rotations of the 4-coordinate system. This decomposition is carried out as follows.

[5]We emphasize that the transformation laws (19.10) and (19.12), which differ by a sign on the right-hand side, are by no means equivalent, since both sides of each involve the components of one and the same spinor (cf. the footnote on p. 92).

We introduce a polar three-dimensional vector **p** and an axial three-dimensional vector **a**, related to the components of the 4-tensor $a^{\mu\nu}$ by

$$a^{\mu\nu} = \begin{pmatrix} 0 & p_x & p_y & p_z \\ -p_x & 0 & -a_z & a_y \\ -p_y & a_z & 0 & -a_x \\ -p_z & -a_y & a_x & 0 \end{pmatrix} \equiv (\mathbf{p}, \mathbf{a}) \tag{19.15}$$

(($\mathbf{p}, \mathbf{a}$) is a shorthand notation that we shall employ for listing the components of such a tensor). Here $a^{\mu\nu} = (-\mathbf{p}, \mathbf{a})$, and of the two quantities

$$\mathbf{a}^2 - \mathbf{p}^2 = \tfrac{1}{2} a_{\mu\nu} a^{\mu\nu}, \qquad \mathbf{ap} = \tfrac{1}{8} e_{\mu\nu\rho\sigma} a^{\mu\nu} a^{\rho\sigma}$$

the first is a scalar and the second a pseudoscalar; with respect to the proper Lorentz group both are equally invariant. Equally invariant alongside them are the squared moduli of the three-dimensional vectors $\mathbf{f}^{\pm} = \mathbf{p} \pm i\mathbf{a}$. This means that every rotation in 4-space amounts, for the vectors $\mathbf{f}^{\pm}$, to a "rotation" in three-dimensional space—generally speaking through complex angles (the six rotation angles in 4-space correspond to three complex "rotation angles" of the three-dimensional system). The operation of spatial inversion, on the other hand, reverses the sign of **p** (but not of **a**) and thereby interchanges the vectors $\mathbf{f}^+$ and $-\mathbf{f}^-$. The components of these vectors form precisely the two sought-after groups of quantities built from the components of the tensor $a^{\mu\nu}$.

At the same time the correspondence between the components of the 4-tensor $a^{\mu\nu}$ and those of the spinors $\xi^{\alpha\beta}$, $\eta_{\dot\alpha\dot\beta}$ becomes evident. Since spatial rotations enter the Lorentz group as a subgroup, the relations linking the spinor components to the components of the three-dimensional vectors must be the same as for three-dimensional spinors:

$$\begin{aligned} f^+{}_x &= \tfrac{1}{2}(\xi^{22} - \xi^{11}), \qquad & f^+{}_y &= \tfrac{i}{2}(\xi^{22} + \xi^{11}), \qquad & f^+{}_z &= \xi^{12}; \\ f^-{}_x &= \tfrac{1}{2}(\eta_{\dot2\dot2} - \eta_{\dot1\dot1}), \qquad & f^-{}_y &= \tfrac{i}{2}(\eta_{\dot2\dot2} + \eta_{\dot1\dot1}), \qquad & f^-{}_z &= \eta_{\dot1\dot2}. \end{aligned} \tag{19.16}$$

Problem Derive the general correspondence that connects spinors of even rank to 4-tensors.

Solution. Every spinor whose total rank $k + l$ is even realizes a single-valued (though in general reducible) representation of the extended Lorentz group; it is consequently equivalent to a 4-tensor that realizes the same representation [6].

A spinor of rank (k, k) transforms under inversion according to

$$\zeta^{\alpha\beta\ldots\dot\gamma\dot\delta\ldots} \to \pm\zeta_{\dot\alpha\dot\beta\ldots\gamma\delta\ldots}.$$

Such a spinor is equivalent to a symmetric irreducible 4-tensor of rank k—a true tensor or a pseudotensor depending on the sign in (1).

Spinors of ranks (k, l) and (l, k) making up a bispinor transform under inversion according to

$$\zeta^{\overbrace{\alpha\beta\ldots}^{k}\;\overbrace{\dot\gamma\dot\delta\ldots}^{l}} \to (-1)^{\frac{k-l}{2}} \chi_{\dot\alpha\dot\beta\ldots\gamma\delta\ldots}. \tag{2}$$

[6]Spinors whose rank is odd realize, by contrast, double-valued representations: a spatial rotation through a full turn of 360° reverses the sign of these spinors, so that each group element is associated with a pair of matrices differing in sign.

When $l = k+2$ the bispinor is equivalent to an irreducible 4-tensor $a_{[\mu\nu]\rho\sigma\ldots}$ of rank $k+2$, antisymmetric in the indices $[\mu\nu]$ and symmetric in all the rest. The irreducibility of this tensor means that it vanishes upon contraction over any pair of indices and upon dualization with respect to any triple of indices (i.e. $e^{\lambda\mu\nu\sigma}a_{[\mu\nu]\rho\sigma\ldots} = 0$); the latter condition signifies that the tensor vanishes upon antisymmetrization over any index triple.

More broadly, when $l = k+2n$ the bispinor corresponds to an irreducible 4-tensor of rank $k+2n$ that is antisymmetric in n index pairs $[\lambda\mu][\nu\rho]\ldots$ and symmetric in the remaining k indices. Tensors possessing antisymmetry of yet higher degree (in triples, quadruples, and so on of indices) are absent from this classification for a transparent reason: a completely antisymmetric tensor of third rank is dual to a pseudovector, one of fourth rank collapses to a scalar (being proportional to the Levi-Civita pseudotensor $e^{\lambda\mu\nu\rho}$), and antisymmetry in more than four indices cannot arise in a four-dimensional space at all.

§ 20. The Dirac equation in spinor representation

In its own rest frame a particle of spin 1/2 is described by a two-component wave function—a 3-spinor. From the "four-dimensional" standpoint this quantity may originate as either an undotted or a dotted 4-spinor. To describe the particle in an arbitrary frame of reference one needs both kinds of 4-spinor; let us label them ξ^α and $\eta_{\dot\alpha}$ [7].

In the case of a free particle the sole operator entering the wave equation can be (as was noted already in § 10) the 4-momentum $\widehat{p}_\mu = i\partial_\mu$. Expressed through spinor indices, this 4-vector corresponds to an operator spinor $\widehat{p}_{\alpha\dot\beta}$ with components

$$\begin{aligned} \widehat{p}^{1\dot 1} &= \widehat{p}_{2\dot 2} = \widehat{p}_0 + \widehat{p}_z, & \widehat{p}^{2\dot 2} &= \widehat{p}_{1\dot 1} = \widehat{p}_0 - \widehat{p}_z, \\ \widehat{p}^{1\dot 2} &= -\widehat{p}_{2\dot 1} = \widehat{p}_x - i\widehat{p}_y, & \widehat{p}^{2\dot 1} &= -\widehat{p}_{1\dot 2} = \widehat{p}_x + i\widehat{p}_y. \end{aligned} \tag{20.1}$$

The wave equation is a linear differential relation linking the components of the spinors through the operator $\widehat{p}_{\alpha\dot\beta}$. The demand of relativistic invariance singles out the following system of equations:

$$\widehat{p}^{\alpha\dot\beta}\eta_{\dot\beta} = m\xi^\alpha, \qquad \widehat{p}_{\dot\beta\alpha}\xi^\alpha = m\eta_{\dot\beta}, \tag{20.2}$$

where m is a dimensional constant. Introducing different constants m_1 and m_2 into these two equations (or reversing the sign in front of m) would be pointless, since by a suitable redefinition of ξ^α or $\eta_{\dot\alpha}$ the equations could always be brought back to the form given above.

Eliminating one of the two spinors from equations (20.2) by substituting $\eta_{\dot\beta}$ from the second equation into the first, we obtain:

$$\widehat{p}^{\alpha\dot\beta}\eta_{\dot\beta} = \frac{1}{m}\widehat{p}^{\alpha\dot\beta}\widehat{p}_{\gamma\dot\beta}\xi^\gamma = m\xi^\alpha.$$

[7] A three-dimensional first-rank spinor can also "descend" from 4-spinors of higher odd rank that become antisymmetric in one or several index pairs in the rest frame. Such variants, however, would lead to equations of higher order (cf. the footnote on p. 52).

But by (18.4) $\widehat{p}^{\alpha\dot{\beta}}\widehat{p}_{\gamma\dot{\beta}} = \widehat{p}^2\delta^\alpha_\gamma$, so that we arrive at

$$(\widehat{p}^2 - m^2)\,\xi^\gamma = 0, \tag{20.3}$$

from which it is clear that m is the mass of the particle.

We draw attention to the fact that the necessity of introducing a mass into the wave equation requires the simultaneous treatment of a pair of spinors (ξ^α and $\eta_{\dot{\alpha}}$): from only one of them it is impossible to compose a relativistically invariant equation that would contain any dimensional parameter. By the same token the wave equation turns out to be automatically invariant with respect to spatial inversion, provided one defines the transformation of the wave function as

$$P: \quad \xi^\alpha \to i\eta_{\dot{\alpha}}, \qquad \eta_{\dot{\alpha}} \to i\xi^\alpha. \tag{20.4}$$

One readily checks that this substitution (accompanied by the replacement $\widehat{p}^{\dot{\alpha}\beta} \to \widehat{p}_{\alpha\dot{\beta}}$, obvious from (20.1)) causes the two equations (20.2) to interchange. A pair of spinors that transform into one another under inversion make up a four-component object—a bispinor.

The relativistic wave equation embodied in the system (20.2) bears the name of the *Dirac equation* (it was derived by *Dirac* in 1928). To pursue the further study and application of this equation we shall now examine the different representations in which it can be written.

With the aid of formula (18.6) we rewrite equations (20.2) as

$$(\widehat{p}_0 + \widehat{\mathbf{p}}\boldsymbol{\sigma})\eta = m\xi, \qquad (\widehat{p}_0 - \widehat{\mathbf{p}}\boldsymbol{\sigma})\xi = m\eta. \tag{20.5}$$

Here the symbols ξ and η stand for two-component quantities—spinors

$$\xi = \begin{pmatrix}\xi^1\\ \xi^2\end{pmatrix}, \qquad \eta = \begin{pmatrix}\eta_{\dot{1}}\\ \eta_{\dot{2}}\end{pmatrix} \tag{20.6}$$

(the first carrying upper, the second lower indices), and when the matrices σ are multiplied by any two-component quantity f the ordinary rule of matrix multiplication is always understood here and below:

$$(\sigma f)_\alpha = \sigma_{\alpha\beta} f_\beta. \tag{20.7}$$

For ease of subsequent reference we write out the Pauli matrices once more:

$$\sigma_x = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}, \quad \sigma_y = \begin{pmatrix}0 & -i\\ i & 0\end{pmatrix}, \quad \sigma_z = \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix} \tag{20.8}$$

and recall their principal properties:

$$\sigma_i\sigma_k + \sigma_k\sigma_i = 2\delta_{ik}, \qquad \sigma_i\sigma_k = ie_{ikl}\sigma_l + \delta_{ik} \tag{20.9}$$

(see III, § 55).

Let us also write down the wave equation obeyed by the complex-conjugate wave function, which is built from the spinors

$$\xi^* = \left(\xi^{1*}, \xi^{2*}\right), \qquad \eta^* = \left(\eta^*_{\dot{1}}, \eta^*_{\dot{2}}\right). \tag{20.10}$$

Since every operator $\widehat{p}_\mu$ contains the factor i, one has $\widehat{p}^*_\mu = -\widehat{p}_\mu$. In taking the complex conjugate of both sides of equations (20.5) one must also take into account that, by the Hermiticity of the matrices σ ($\sigma^* = \widetilde{\sigma}$),

$$(\sigma f)^*_\alpha = \sigma^*_{\alpha\beta} f^*_\beta = f^*_\beta \sigma_{\beta\alpha} = (f^*\sigma)_\alpha,$$

and one obtains equations of the form

$$\eta^*(\widehat{p}_0 + \widehat{\mathbf{p}}\boldsymbol{\sigma}) = -m\xi^*, \qquad \xi^*(\widehat{p}_0 - \widehat{\mathbf{p}}\boldsymbol{\sigma}) = -m\eta^*. \tag{20.11}$$

In this way of writing it is tacitly assumed that the operator $\widehat{p}_\mu$ acts on the function placed to its left. The display of ξ^* and η^* as horizontal rows reflects the matrix multiplication convention adopted in these equations: the row f is contracted with the columns of the σ matrices:

$$(f^*\sigma)_\alpha = f^*_\beta \sigma_{\beta\alpha}. \tag{20.12}$$

The inversion transformation for ξ^*, η^* is defined as the complex conjugate of the transformation (20.4):

$$P: \quad \xi^{\alpha *} \to -i\eta^*_{\dot\alpha}, \qquad \eta^*_{\dot\alpha} \to -i\xi^{\alpha *}. \tag{20.13}$$

§ 21. Symmetric form of the Dirac equation

Among all ways of writing the Dirac equation the spinor form stands out as the one that displays relativistic invariance most transparently. For practical calculations, though, it is often advantageous to pass to a different representation by selecting another quartet of independent components of the wave function.

Let ψ stand for the four-component wave function (its entries being ψ_i, $i = 1, 2, 3, 4$). Written in spinor language this object is a bispinor:

$$\psi = \begin{pmatrix} \xi \\ \eta \end{pmatrix}. \tag{21.1}$$

With equal right, however, one may choose as the independent components of ψ any linear combinations of the components of the spinors ξ and η [8]. We shall restrict the admissible linear transformations solely by the requirement of unitarity; such transformations leave invariant the bilinear forms composed of ψ and ψ^* (see § 28).

For a completely general selection of the components of ψ one can cast the Dirac equation as

$$\widehat{p}_\mu \gamma^\mu_{ik} \psi_k = m\psi_i,$$

where γ^μ ($\mu = 0, 1, 2, 3$) are certain 4×4 matrices (the *Dirac matrices*). It is customary to write this equation in symbolic form, suppressing the matrix indices:

$$(\gamma\widehat{p} - m)\psi = 0, \tag{21.2}$$

[8] For brevity we shall speak of the four-component quantity ψ as a bispinor also in its non-spinor representations.

where

$$\gamma\widehat{p} \equiv \gamma^\mu \widehat{p}_\mu = \widehat{p}_0\gamma^0 - \widehat{\mathbf{p}}\boldsymbol{\gamma} = i\gamma^0\frac{\partial}{\partial t} + i\boldsymbol{\gamma}\boldsymbol{\nabla}, \quad \boldsymbol{\gamma} = (\gamma^1, \gamma^2, \gamma^3).$$

In the spinor form of the equation with the components ψ from (21.1) the corresponding matrices are[9]

$$\gamma^0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \boldsymbol{\gamma} = \begin{pmatrix} 0 & -\boldsymbol{\sigma} \\ \boldsymbol{\sigma} & 0 \end{pmatrix}, \tag{21.3}$$

as is readily seen by writing equations (20.5) in the form

$$\begin{pmatrix} 0 & \widehat{p}_0 + \widehat{\mathbf{p}}\boldsymbol{\sigma} \\ \widehat{p}_0 - \widehat{\mathbf{p}}\boldsymbol{\sigma} & 0 \end{pmatrix} \begin{pmatrix} \xi \\ \eta \end{pmatrix} = m \begin{pmatrix} \xi \\ \eta \end{pmatrix}$$

and comparing with (21.2).

In the general case the matrices γ need only satisfy the conditions ensuring that $\widehat{p}^2 = m^2$. To determine these conditions, we multiply equation (21.2) on the left by $\gamma\widehat{p}$. We have

$$(\gamma^\mu \widehat{p}_\mu)(\gamma^\nu \widehat{p}_\nu)\psi = m(\widehat{p}_\mu\gamma^\mu)\psi = m^2\psi.$$

Since $\widehat{p}_\mu\widehat{p}_\nu$ is a symmetric tensor (all operators $\widehat{p}_\mu$ commute), this equation can be rewritten as

$$\tfrac{1}{2}\widehat{p}_\mu\widehat{p}_\nu(\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu) = m^2\psi,$$

from which it is clear that one must have

$$\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2g^{\mu\nu}. \tag{21.4}$$

Thus all pairs of distinct matrices γ^μ anticommute, and the square of each is

$$\left(\gamma^1\right)^2 = \left(\gamma^2\right)^2 = \left(\gamma^3\right)^2 = -1, \qquad \left(\gamma^0\right)^2 = 1. \tag{21.5}$$

Under an arbitrary unitary transformation of the components of ψ ($\psi' = U\psi$, with U a unitary 4×4 matrix) the matrices γ transform according to

$$\gamma' = U\gamma U^{-1} = U\gamma U^{+} \tag{21.6}$$

(so that the equation $(\gamma\widehat{p} - m)\psi = 0$ goes over into $(\gamma'\widehat{p} - m)\psi' = 0$). The commutation relations (21.4), of course, remain unchanged.

The matrix γ^0 of (21.3) is Hermitian, and the matrices $\boldsymbol{\gamma}$ are anti-Hermitian. These properties are preserved under every unitary transformation (21.6), so that we shall always have[10]:

$$\boldsymbol{\gamma}^{+} = -\boldsymbol{\gamma}, \qquad \gamma^{0+} = \gamma^0. \tag{21.7}$$

[9]Here and below a compact notation is used for 4×4 matrices in terms of 2×2 blocks: each symbol in the expressions (21.3) represents a 2×2 matrix.

[10]These relations can be written together as

$$\gamma^{\lambda +} = \gamma^0\gamma^\lambda\gamma^0. \tag{21.7 a}$$

We now derive the equation satisfied by ψ^*. Complex-conjugating both sides of (21.2) and invoking (21.7) gives

$$(\widehat{p}_0\gamma^0 - \widehat{\mathbf{p}}\widetilde{\boldsymbol{\gamma}} - m)\psi^* = 0.$$

We rearrange ψ^* by $\widetilde{\gamma}^\mu\psi^* = \psi^*\gamma^\mu$ and then multiply the equation on the right by γ^0; noting that $\boldsymbol{\gamma}\gamma^0 = -\gamma^0\boldsymbol{\gamma}$ and introducing the new bispinor

$$\bar{\psi} = \psi^*\gamma^0, \qquad \psi^* = \bar{\psi}\gamma^0, \tag{21.8}$$

we obtain

$$\bar{\psi}(\gamma\widehat{p} + m) = 0. \tag{21.9}$$

Just as in (20.11), $\widehat{p}$ is understood to differentiate the function standing to its left. One calls $\bar{\psi}$ the *Dirac conjugate* (or *relativistic conjugate*) of ψ. The purpose of the γ^0 factor is to swap ξ^* and η^* (in the spinor representation), so that within $\bar{\psi} = (\eta^*, \xi^*)$ the undotted spinor occupies the first position—matching the ordering in ψ—while the dotted spinor occupies the second. This is why $\bar{\psi}$, rather than ψ^*, serves as the natural companion of ψ whenever the two enter bilinear combinations together (see § 28).

Under spatial inversion the wave function transforms according to

$$P: \quad \psi \to i\gamma^0\psi, \qquad \bar{\psi} \to -i\bar{\psi}\gamma^0. \tag{21.10}$$

In the spinor representation of ψ the matrix γ^0 interchanges, as it should under inversion, the components ξ and η. The invariance of the Dirac equation under the transformation (21.10) is in the general case obvious and can be verified directly: replacing $\widehat{\mathbf{p}} \to -\widehat{\mathbf{p}}$ and simultaneously $\psi \to i\gamma^0\psi$ in equation (21.2), we get

$$(\widehat{p}_0\gamma^0 + \widehat{\mathbf{p}}\boldsymbol{\gamma} - m)\gamma^0\psi = 0.$$

Multiplying this equation on the left by γ^0 and using the anticommutativity of γ^0 and $\boldsymbol{\gamma}$, we recover the original equation.

Left-multiplying $(\gamma\widehat{p} - m)\psi = 0$ by $\bar{\psi}$, right-multiplying $\bar{\psi}(\gamma\widehat{p} + m) = 0$ by ψ, and summing the two expressions yields

$$\bar{\psi}\gamma^\mu\left(\widehat{p}_\mu\psi\right) + \left(\widehat{p}_\mu\bar{\psi}\right)\gamma^\mu\psi = \widehat{p}_\mu\left(\bar{\psi}\gamma^\mu\psi\right) = 0,$$

where parentheses mark the function acted upon by $\widehat{p}$. Because this has the structure of a continuity equation $\partial_\mu j^\mu = 0$, the quantity

$$j^\mu = \bar{\psi}\gamma^\mu\psi = \left(\psi^*\psi, \psi^*\gamma^0\boldsymbol{\gamma}\psi\right) \tag{21.11}$$

is the particle current density 4-vector. We note that its time component $j^0 = \psi^*\psi$ is positive definite.

The Dirac equation can be written in a form explicitly solved for the time derivative:

$$i\frac{\partial\psi}{\partial t} = \widehat{H}\psi, \tag{21.12}$$

where $\widehat{H}$ is the particle Hamiltonian[11]. To this end it suffices to multiply equation (21.2) on the left by γ^0. For the Hamiltonian one obtains the expression

$$\widehat{H} = \boldsymbol{\alpha}\widehat{\mathbf{p}} + \beta m, \tag{21.13}$$

where the standard notation has been introduced for the matrices appearing here:

$$\boldsymbol{\alpha} = \gamma^0 \boldsymbol{\gamma}, \qquad \beta = \gamma^0. \tag{21.14}$$

We note that

$$\alpha_i \alpha_k + \alpha_k \alpha_i = 2\delta_{ik}, \qquad \beta\boldsymbol{\alpha} + \boldsymbol{\alpha}\beta = 0, \qquad \beta^2 = 1, \tag{21.15}$$

i.e. all the matrices $\boldsymbol{\alpha}$, β anticommute with one another and their squares are unity; all of them are Hermitian. In the spinor representation

$$\boldsymbol{\alpha} = \begin{pmatrix} \boldsymbol{\sigma} & 0 \\ 0 & -\boldsymbol{\sigma} \end{pmatrix}, \qquad \beta = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{21.16}$$

When velocities are small the particle should reduce, as in non-relativistic quantum mechanics, to a single two-component spinor. And indeed, taking $\mathbf{p} \to 0$, $\varepsilon \to m$ in (20.5) yields $\xi = \eta$: the two spinors inside the bispinor coincide. A drawback of the spinor representation shows up at this point, however—all four entries of ψ stay non-vanishing despite only two being genuinely independent. One would prefer a representation in which two of the four components of ψ go to zero in the non-relativistic limit.

Accordingly we introduce, in place of ξ and η, their linear combinations φ and χ:

$$\psi = \begin{pmatrix} \varphi \\ \chi \end{pmatrix}, \qquad \varphi = \frac{1}{\sqrt{2}}(\xi + \eta), \qquad \chi = \frac{1}{\sqrt{2}}(\xi - \eta). \tag{21.17}$$

Then for a particle at rest $\chi = 0$. This representation of ψ will be called the *standard* representation. Under inversion φ and χ transform into themselves:

$$P: \quad \varphi \to i\varphi, \qquad \chi \to -i\chi. \tag{21.18}$$

The equations for φ and χ are obtained by adding and subtracting equations (20.5):

$$\widehat{p}_0 \varphi - \widehat{\mathbf{p}}\boldsymbol{\sigma}\chi = m\varphi, \qquad -\widehat{p}_0 \chi + \widehat{\mathbf{p}}\boldsymbol{\sigma}\varphi = m\chi. \tag{21.19}$$

From these one sees that the standard representation corresponds to the matrices

$$\gamma^0 \equiv \beta = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \boldsymbol{\gamma} = \begin{pmatrix} 0 & \boldsymbol{\sigma} \\ -\boldsymbol{\sigma} & 0 \end{pmatrix}, \qquad \boldsymbol{\alpha} = \begin{pmatrix} 0 & \boldsymbol{\sigma} \\ \boldsymbol{\sigma} & 0 \end{pmatrix}. \tag{21.20}$$

Since in (21.17) the first and second components of ξ and η are combined separately, in the standard representation—as in the spinor one—the components ψ_1 and ψ_3 correspond to

[11] For a particle of spin 0 the wave equation could not be cast in this form: equation (10.5) for the scalar ψ is of second order in time, while the system (10.4) of first-order equations for the five-component quantity (ψ, ψ_k) contains time derivatives not of all of its components.

the eigenvalue +1/2 of the spin projection, while ψ_2 and ψ_4 correspond to the projection −1/2. In both representations, therefore, the matrix $\frac{1}{2}\mathbf{\Sigma}$ with

$$\mathbf{\Sigma} = \begin{pmatrix} \boldsymbol{\sigma} & 0 \\ 0 & \boldsymbol{\sigma} \end{pmatrix}, \tag{21.21}$$

is the three-dimensional spin operator: the action of $\frac{1}{2}\Sigma_z$ on a bispinor containing only the components ψ_1, ψ_3 or ψ_2, ψ_4 multiplies the bispinor by +1/2 or −1/2, respectively. In an arbitrary representation this matrix can be written as

$$\mathbf{\Sigma} = -\boldsymbol{\alpha}\gamma^5 = -\frac{i}{2}[\mathbf{aa}] \tag{21.22}$$

(for the definition of γ^5 see below, (22.14)).

Problems

1. Find the transformation formulae for the wave function under an infinitesimal Lorentz transformation and an infinitesimal spatial rotation.

   Solution. In the spinor representation of ψ, under an infinitesimal Lorentz transformation

$$\xi' = \left(1 - \frac{1}{2}\boldsymbol{\sigma}\delta\mathbf{V}\right)\xi, \qquad \eta' = \left(1 + \frac{1}{2}\boldsymbol{\sigma}\delta\mathbf{V}\right)\eta$$

   (see (18.8), (18.8 a), (18.12)). Both formulae can be written together as

$$\psi' = \left(1 - \frac{1}{2}\boldsymbol{\alpha}\delta\mathbf{V}\right)\psi. \tag{1}$$

   Similarly, the transformation law under an infinitesimal rotation is

$$\psi' = \left(1 + \frac{i}{2}\mathbf{\Sigma}\delta\boldsymbol{\theta}\right)\psi. \tag{2}$$

   In this form the formulae are valid in any representation of ψ, provided one understands $\boldsymbol{\alpha}$ and $\mathbf{\Sigma}$ as the matrices in the same representation.

   It is easy to verify that the matrices $\boldsymbol{\alpha}$ and $\mathbf{\Sigma}$ constitute the components of the antisymmetric "matrix 4-tensor"

$$\sigma^{\mu\nu} = \frac{1}{2}(\gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu) = (\boldsymbol{\alpha}, i\mathbf{\Sigma})$$

   (the listing of the components follows the rule (19.15)). Let us also introduce the infinitesimal antisymmetric tensor $\delta\varepsilon^{\mu\nu} = (\delta\mathbf{V}, \delta\boldsymbol{\theta})$. Then

$$\sigma^{\mu\nu}\delta\varepsilon_{\mu\nu} = 2i\mathbf{\Sigma}\delta\boldsymbol{\theta} - 2\boldsymbol{\gamma}\delta\mathbf{V}$$

   and both formulae (1), (2) can be written in the unified form:

$$\psi' = \left(1 + \frac{1}{4}\sigma^{\mu\nu}\delta\varepsilon_{\mu\nu}\right)\psi. \tag{3}$$

2. Write the Dirac equation in a representation that contains no imaginary coefficients (*E. Majorana*, 1937).

   Solution. In the standard representation, in the equation

$$\left(\frac{\partial}{\partial t} + \alpha_x \frac{\partial}{\partial x} + \alpha_y \frac{\partial}{\partial y} + \alpha_z \frac{\partial}{\partial z} + im\beta\right)\psi = 0$$

   the only imaginary matrices are α_y and $i\beta$. This imaginary character can be eliminated by performing a transformation $\psi' = U\psi$ that interchanges the imaginary matrix α_y with the real matrix β. To this end one sets

$$U = \frac{1}{\sqrt{2}}(\alpha_y + \beta) = U^{-1}.$$

   Then

$$\alpha'_x = U\alpha_x U = -\alpha_x, \quad \alpha'_y = \beta, \quad \alpha'_z = -\alpha_z, \quad \beta' = \alpha_y,$$

   and the Dirac equation takes the form

$$\left(\frac{\partial}{\partial t} - \alpha_x \frac{\partial}{\partial x} + \beta \frac{\partial}{\partial y} - \alpha_z \frac{\partial}{\partial z} + im\alpha_y\right)\psi' = 0,$$

   in which all the coefficients are real.

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